

A330 OE Workbook

Answered, in flashcard form

Source: 787/A321/A330 Fleets OE Workbook, Version 2, January 2026. A330 PAX items only: items marked (330), (330/321), (787/330), and the fleet-common items. Items marked (787) only, (A321) only, and the Freighter Operations group are left out.

Workbook items	336
Answered from the manuals	309
No published source, ask the check airman	27
Critical Observable Items	125
Drill cards / walkthrough items	288 / 48
Manuals used	FCOM R17 (15 MAY 26), QRH R35 (26 JUN 26), FCTM R6 (14 SEP 26), FOM 125.3 (15 SEP 26), PRC 31 AUG 26
Compiled	September 2026

How to read this document

Each question carries the answer in green, and nothing else. That green line is what you should be able to say cold. Key numbers are set larger so they land at a glance. Everything under the answer is the backup: the full answer, the manual location, a verbatim quote from the extract, and a note.

Questions are grouped by topic in roughly the order you meet them on a trip, preflight to shutdown, with the long-haul topics in the middle. The tag above each question carries the grade sheet item it came from. Walkthrough items are things you perform on the line rather than recite.

An item in amber has no published answer in the FOM, FCOM, QRH, FCTM or PRC. The card says so and tells you to ask the check airman. Nothing was guessed.

The flight deck door entry code is deliberately not recorded. It is security sensitive information and does not belong in a study document.

Numbers to Know

The facts on these cards worth knowing cold before the first leg.

Where is the passenger cabin emergency equipment location chart?	FCOM PRO-ABN-90, Detailed Cabin and Cockpit Evacuation Procedure
Where is the cockpit door described, with its operation and egress?	FCOM DSC-52-30, Cockpit Door Security System
Where are the crew rest requirements?	FOM 5.1.23, Crew Rest Facility, plus the LDMCR card in the PRC
How do you change the nav database?	On the ground only: press SECOND NAV DATABASE at 3L and confirm.
Are there high mins Captain restrictions?	Yes, relieved by Exemption 5549
B043 vs B044 flight plan?	B043 international fuel reserve, B044 planned redispach
Domestic dispatch fuel requirement?	Destination, alternate, plus 45 minutes
Flag dispatch fuel requirement?	Destination, 10 percent of trip time, alternate, 30 minutes holding
B043 special fuel requirement?	Destination, 10 percent of oceanic time, alternate, 45 minutes cruise
What is final reserve fuel?	30 minutes holding at 1500 ft, or 45 minutes under B043
When do you ask ATC about the delay?	When delay plus divert could land you under 30 minutes
When do you declare MAYDAY FUEL?	When you will land with less than 30 minutes
How far do you stand off severe cells?	20 nm laterally; clear tops by 5000 ft
No deviation clearance. What do you do?	Emergency authority, PAN-PAN three times, offset 300 ft beyond 5 nm
What do you report to Dispatch when deviating?	100 nm lateral, 4000 ft altitude, 15 minute delay, weather
Lowest visibility for CAT I?	1800 RVR
Lowest visibility for CAT II?	1000 RVR
Where are the standard customer service PAs?	Flight Operations Manual chapter 7, Standard Customer Service Announcements

Pushing back late. When is a PA required?	At scheduled departure time, then every 15 minutes
Breakaway thrust for taxi?	A little above idle, applied symmetrically; no N1 number is published
Why keep groundspeed in your scan?	To hold 30 kts max, about 10 kts congested
Recommended speed before a 90 degree turn?	Less than 10 kt for a turn of 90 degrees or more
Where are the gear geometry and turning radius figures?	FCOM DSC-20-30 Ground Handling, Taxiing, minimum turning radii
Where is the oversteer technique for turns?	FCTM PR-NP-SOP-100, Taxi Roll and Steering
Where are the single engine taxi procedures?	FCOM PRO-NOR-SOP-21B for arrival, FCTM PR-NP-SOP-102 and 280 for the considerations
Engine cooling requirement before shutdown?	At or near idle for 1 minute after high thrust operations
Where are the US holding speeds?	200 knots below 6000 ft, 230 to 14,000, 265 above
Maximum takeoff crosswind?	32 kt gust included, certified, dry; less on wet or contaminated runways
Maximum altitude with slats or flaps extended?	20 000 ft
Normal takeoff pitch target?	15 degrees, then follow the SRS bar
Pitch target after an engine failure at or after V1?	12.5 degrees, then SRS
Transition altitude and transition level?	18,000 ft MSL in the US, charted elsewhere
Minimum altitude to engage the autopilot after takeoff?	100 ft AGL and at least 5 seconds after liftoff
ETOPS alternate weather before departure?	C055 minima, earliest to latest landing time
Any Dispatch call required after takeoff, before the EEP?	Yes. 30 to 60 minutes after takeoff
What is the APU verification procedure?	Cold soak 2 hours, start within 1 hour of top of descent
Max diversion distance for ETOPS 120 and 180?	A330-200: 803 nm at 120 minutes, 1200 nm at 180 minutes
ETOPS diversion speed?	.82M/290 KIAS, the approved one-engine-inoperative cruise speed

What altitudes are RVSM airspace?	FL290 through FL410
Diverting without a clearance. What altitude do you plan?	Below FL290
Where are the ICAO oceanic weather deviation procedures?	Our Flight Operations Manual oceanic chapter, reprinted from ICAO Doc 4444
Driftdown off the route. Which way do you turn?	At least 30 degrees, left or right per listed factors
How far can you offset with SLOP?	Up to 2 nm right of centerline. Never left
Where is the aircraft Sanitation Certificate?	Not named in our manuals; the disinsection certificate is in Comply365
Where are international holding speeds?	ICAO Doc 8168 and the theater chapters
What is MedLink?	Contract ER physicians giving 24-hour inflight medical advice
How do you make a MedLink call from the aircraft?	SATCOM to (602) 282-4910, or ARINC phone patch
Medical event on an international flight. Any special notifications?	Yes. General Declaration entry; Australia notified 30 minutes before landing
When do you start the divert?	Early enough to reach the alternate with 30 minutes fuel
What must you check before accepting LAHSO?	Stopping within the ALD, using AFM distance plus 1000 ft
What altimeter setting defines a level?	Standard, 29.92 in Hg or 1013.25 hectopascals
What do you weigh before turning autothrust off for the approach?	Unsatisfactory A/THR performance, loss of autotrim, then disconnect by 1000 ft AAL
EVF communication requirements with Maintenance?	Call or ACARS Dispatch 30-60 minutes after takeoff, before EEP
Is there a QRH procedure for a tailpipe fire?	Yes. ENGINE TAILPIPE FIRE, QRH 17.09A

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Emergency Equipment (3)

1.2.3 PERFORM INTERIOR/EXTERIOR INSPECTION

What flight deck emergency equipment must you check?

Life vests, crash axe, smoke hoods, extinguisher, oxygen masks, flashlights, escape ropes, PED bag, fire gloves

Nine items on the EMER EQPT check of the Preliminary Cockpit Preparation, done by the PM or CM3. Life vests stowed, crash axe stowed, smoke hoods or portable oxygen with full-face masks stowed and ready, portable fire extinguisher lockwired with the gauge in the green, oxygen masks stowed, flashlights, escape ropes, PED Fire Containment Bag and fire gloves stowed.

FCOM PRO-NOR-SOP-04 P 13/14 Preliminary Cockpit Preparation, Emergency Equipment

"EMER EQPT..... CHECK PM-CM3 L2 Check that: ■ Life vests are stowed ■ Crash axe is stowed ■ Smoke hoods and/or portable oxygen equipment and full-face masks is stowed and ready for use ■ Portable fire extinguisher is lockwired and the pressure gauge indicator is in the green area ■ Oxygen masks are stowed ■ Flashlights are stowed ■ Escape ropes are stowed ■ PED Fire Containment Bag is stowed ■ Fire gloves are stowed"

The extinguisher check is two things: lockwire intact and gauge in the green. A missing lockwire means it may have been discharged.

Workbook wording: What flight deck emergency items are required to be checked?

Cabin Emergency Equipment Locations (Megaphone, ELT, etc.)

1.2.3 PERFORM INTERIOR/EXTERIOR INSPECTION

Where is the passenger cabin emergency equipment location chart?

FCOM PRO-ABN-90, Detailed Cabin and Cockpit Evacuation Procedure

The A330 Emergency Equipment Location figure sits in the FCOM abnormal chapter PRO-ABN-90, Detailed Cabin / Cockpit Evacuation Procedure, page 5 of 14. The systems description at DSC-25-30 covers cockpit emergency equipment and sends you to the CCOM for the cabin items and the placard symbol list.

FCOM PRO-ABN-90 P 5/14 Emergency Equipment Locations

"EMERGENCY EQUIPMENT LOCATIONS Ident.: PRO-ABN-90-90000470.9000414 / 27 DEC 23 Applicable to: ALL A330 EMERGENCY EQUIPMENT LOCATION"

Placards in the cabin also mark each item. DSC-25-30 says the symbol list is in CCOM 11-010.

Workbook wording: Where can the "Emergency Equipment Locations – Passenger Cabin" be found?

Megaphone (remove/use/replace)

MISC MISCELLANEOUS, NOT ASSOCIATED WITH GRADE SHEET TASKS WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

Remove, use and replace the megaphone

No published source located, ask the check airman

The megaphone is cabin emergency equipment. FCOM DSC-25-30 covers the cockpit items and refers the cabin equipment description to the CCOM, and the location figure is at PRO-ABN-90. No operating steps for the megaphone are in the pilot manuals.

FCOM DSC-25-30 P 2/2 Cabin Emergency Equipment

"For the description of the cabin emergency equipment, Refer to CCOM/EMERGENCY EQUIPMENT/PORTABLE EMERGENCY EQUIPMENT."

No published source located, ask the check airman. Have a flight attendant show you the stowage and the trigger.

Workbook wording: Megaphone (remove/use/replace)

Security and Doors (4)

1.2.3 PERFORM INTERIOR/EXTERIOR INSPECTION

Where is the Jumpseat and Rest Facility Briefing Card?

On the EFB, and reproduced in the Pilot Reference Cards

The FOM lists the Jumpseat Card on the electronic flight bag and marks it not required for dispatch. The Captain still owes the occupant a briefing: seat and harness, communication gear, exits, emergency equipment, sterile flight deck, and flight deck door security. The card only helps you cover it.

FOM 5.1.22.1.2.3 Flight Deck Jumpseat - Occupant Briefing

"The Jumpseat Briefing Card is not required for flight, but it may be used to assist this briefing."

The A330 Jumpseat Information Card is reproduced in the PRC, so it is also available on the flight deck without the EFB. The Captain still owes the occupant the briefing.

Workbook wording: Where is the Cockpit Jumpseat and/or Rest Facility Briefing Card located?

Crew Rest Procedures/Door Operation

1.2.3 PERFORM INTERIOR/EXTERIOR INSPECTION

What is the flight deck door entry code?

Not recorded. Security sensitive

FOM 15.2.6.3 states the keypad control device shall not be used for normal entry and that entry attempts using the keypad shall be denied by the Flight Crew. FOM 15.2.6.3 also warns that an unexpected keypad activation signal must be treated as a Level 4 hijack or attempted hijack in progress until the Flight Attendants say otherwise.

FOM 15.2.6.3 Keypad Control

"The keypad control device shall not be used for normal entry. Entry attempts using the keypad shall be denied by the Flight Crew. For A330F exception, see 15.2.6.10 – Flight Deck Access without Functioning Service Interphone."

The code is not published in the FOM, FCOM or QRH; it is SSI. The A330 keypad code is programmed by the airline as a two to seven digit code, and the keypad is for cabin crew access requests only, never for normal entry.

Workbook wording: What is the Flight Deck Door entry code?

Flight Deck Door Operation

1.2.1 PERFORM PREFLIGHT PREPARATION

Where is the cockpit door described, with its operation and egress?

FCOM DSC-52-30, Cockpit Door Security System

DSC-52-30-10 describes the door: forward-opening hinged, electrically locked latch controlled by the flight crew, cabin keypad with a two to seven digit code, a mechanical override to open it from the cockpit side, and an evacuation and decompression panel the crew kicks out if the door jams. DSC-52-30-20 covers the Cockpit Door Locking System controls and indications.

FCOM DSC-52-30-10 P 1/2 Cockpit Door Description

"A forward-opening hinge door separates the cockpit from the passenger or courier compartment. It has an electric-locking latch, controlled by the flight crew. In normal conditions, when the door is closed, it remains locked."

Two failure behaviours to know: rapid cockpit decompression unlocks the door automatically, and an electrical supply failure unlocks it but leaves it closed.

Workbook wording: Where can you find the Cockpit Door (330/321) and/or forward overhead door (787) description/operation/components/egress procedures?

Flight Deck Entry Procedures/Security

MISC MISCELLANEOUS, NOT ASSOCIATED WITH GRADE SHEET TASKS

Flight deck door procedure with no flight attendants?

Yes. Captain covers cabin duties; door closed, critical phases

FOM 16.1.11 states that when Flight Attendants are not on board a Company-operated flight, the Captain must ensure the cabin duties normally done by F/As are accomplished, and refers to fleet-specific manuals for procedures. The WARNING requires galley carts to be properly secured and the Flight Deck Door closed during critical phases of flight.

FOM 16.1.11 Pilot Duties With No Flight Attendants

"properly secured and the Flight Deck Door is closed during critical phases of flight."

FOM 15.2.6.2 sends you to Pilot Duties With No Flight Attendants at 16.1.11, which then refers to the fleet manuals for the actual procedure.

Workbook wording: Are there special Flight Deck Door Procedure on Flights with No Flight Attendants?

Crew Rest (3)

Crew Rest Procedures/Door Operation

1.2.3 PERFORM INTERIOR/EXTERIOR INSPECTION

Where are the crew rest requirements?

FOM 5.1.23, Crew Rest Facility, plus the LDMCR card in the PRC

Use is limited to Flight Crew on the Hawaiian seniority list, one occupant per bunk, no food or drink except bottled water. It cannot be occupied from pushback to 10,000 ft above field elevation, or from 45 minutes before landing to block in.

FOM 5.1.23 Crew Rest Facility

"The purpose of the crew rest facility is to provide a quiet place for the crew to rest. It is not a lounge, a meal area, or a place to stow personal carry-on items."

On the A330 the primary facility is the Lower Deck Module Crew Rest, LDMCR. Bunks 1 and 4 are the Flight Crew bunks, 2, 3, 5 and 6 belong to the Cabin Crew, and no more than 2 Flight Crew and 4 Cabin Crew may be in it at any time. The PRC reprints the LDMCR information card.

Workbook wording: Where can the Crew Rest procedures/requirements be found?

1.2.3 PERFORM INTERIOR/EXTERIOR INSPECTION

Do you have a crew rest app for long-range flying?

No published source located, ask the check airman

No FOM, FCOM, QRH, FCTM or PRC text names a crew rest app. The FOM sets the rest break structure for augmented crews: the PIC determines the length and order of pilot rest breaks, and a double crew may use a 2-4-2 or 4-4 split.

FOM 4.3.2.4.4 Augmented Crew (3 Flight Crewmembers)

"The PIC will determine the length and order of pilot rest breaks."

No published source located, ask the check airman. Bring the app you use and confirm it is the one the fleet expects.

Workbook wording: Do you have a CREW REST APP? (for Long-Range flying w/more than 2 pilots)

12.1.3.16 KNOW FOM PA POLICIES AND PERFORM EFFECTIVE COMMUNICATION FROM THE FLIGHT DECK

Which PA selection reaches the cabin but not the crew rest?

PA+ALL with the flight deck handset push to talk

FOM 5.1.23.8 tells the Flight Crew how to announce to the cabin without disturbing the LDMCR: press PA+ALL and use the push to talk button on the Flight Deck handset.

FOM 5.1.23.8 A330 Crew Rest Facility

"Flight Crew may make an announcement to the cabin without disturbing the LDMCR by pressing PA+ALL and use the push-to-talk button on the Flight Deck handset."

This matters on every augmented leg. A seat belt PA through the LDMCR wakes the relief pilot you need rested.

Workbook wording: What selection should you use to make a PA throughout the cabin, but NOT to the crew rest facilities?

EFB and FD Pro (13)

EFB Review**CRITICAL OBSERVABLE ITEM**

1.1.1 PERFORM FLIGHT PLANNING WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

EFB IOS current and updated?**Yes, iOS at or above Manual Currency minimum version**

The EFB operating system and every required app must be at or above the minimum version listed on the Manual Currency page. The CrossCheck app shows the current versions. Updates are not required or encouraged while on a trip.

FOM 2.5.1.2 EFB, App, and Document Currency

"The iOS and all required apps shall be at or above the minimum version number listed on Manual"

Check CrossCheck before you leave home, not at the gate.

Workbook wording: EFB IOS - Current/updated

CRITICAL OBSERVABLE ITEM

1.1.1 PERFORM FLIGHT PLANNING WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

FD Pro updates completed?**Yes, synchronized within 24 hours of starting a trip**

Within 24 hours of starting a trip or training event, all flight-related apps must be synchronized so charts and databases are current, and all apps must meet minimum currency. Required memos, revisions, and safety flashes must also be read and acknowledged.

FOM 2.5.1.4 EFB - Prior to Flight

"all flight-related apps must be synchronized to ensure all charts and databases are current"

Never run app or iOS updates over Flight Deck internet.

Workbook wording: FD Pro: updates completed

CRITICAL OBSERVABLE ITEM

1.1.1 PERFORM FLIGHT PLANNING WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

FD Pro: turn on AUTO UPDATE for every weather layer**No published source located, ask the check airman**

This is a Jeppesen FD Pro workflow step. The FOM sets the currency rule for the EFB and its apps but does not script the FD Pro screens. Demonstrate it on the device during OE.

FOM 2.5.1.4 EFB - Prior to Flight

No published source located, ask the check airman. The FD Pro Pubs Quick Reference guide in Comply365 covers the screens.

Workbook wording: FD Pro: "Weather" (bottom/left of screen), "Weather Layers" pane/"I" (greyed-out "info" button/s, RT side of each Weather layer)/"AUTO UPDATE" selected ON; repeat for all WEATHER layers

CRITICAL OBSERVABLE ITEM

1.1.1 PERFORM FLIGHT PLANNING WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

FD Pro: select the weather layers you want**No published source located, ask the check airman**

This is a Jeppesen FD Pro workflow step. The FOM sets the currency rule for the EFB and its apps but does not script the FD Pro screens. Demonstrate it on the device during OE.

FOM 2.5.1.4 EFB - Prior to Flight

No published source located, ask the check airman. The FD Pro Pubs Quick Reference guide in Comply365 covers the screens.

Workbook wording: Select desired layer/s on "Weather Layers" pane

CRITICAL OBSERVABLE ITEM

1.1.1 PERFORM FLIGHT PLANNING WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

FD Pro: REFRESH the Weather Layers pane**No published source located, ask the check airman**

This is a Jeppesen FD Pro workflow step. The FOM sets the currency rule for the EFB and its apps but does not script the FD Pro screens. Demonstrate it on the device during OE.

FOM 2.5.1.4 EFB - Prior to Flight

No published source located, ask the check airman. The FD Pro Pubs Quick Reference guide in Comply365 covers the screens.

Workbook wording: "REFRESH" (bottom/right) on "Weather Layers" pane

CRITICAL OBSERVABLE ITEM

1.1.1 PERFORM FLIGHT PLANNING WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

FD Pro: verify the correct aircraft at top center**No published source located, ask the check airman**

This is a Jeppesen FD Pro workflow step. The FOM sets the currency rule for the EFB and its apps but does not script the FD Pro screens. Demonstrate it on the device during OE.

FOM 2.5.1.4 EFB - Prior to Flight

No published source located, ask the check airman. The FD Pro Pubs Quick Reference guide in Comply365 covers the screens.

Workbook wording: FD Pro: Verify correct "aircraft" on top/center of screen

CRITICAL OBSERVABLE ITEM

1.1.1 PERFORM FLIGHT PLANNING WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

FD Pro: demonstrate a flight plan upload**No published source located, ask the check airman**

This is a Jeppesen FD Pro workflow step. The FOM sets the currency rule for the EFB and its apps but does not script the FD Pro screens. Demonstrate it on the device during OE.

FOM 2.5.1.4 EFB - Prior to Flight

No published source located, ask the check airman. The FD Pro Pubs Quick Reference guide in Comply365 covers the screens.

Workbook wording: FD Pro Flight plan upload demo

CRITICAL OBSERVABLE ITEM

1.1.1 PERFORM FLIGHT PLANNING WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

FD Pro: demonstrate a flight plan transfer to another EFB**No published source located, ask the check airman**

This is a Jeppesen FD Pro workflow step. The FOM sets the currency rule for the EFB and its apps but does not script the FD Pro screens. Demonstrate it on the device during OE.

FOM 2.5.1.4 EFB - Prior to Flight

No published source located, ask the check airman. The FD Pro Pubs Quick Reference guide in Comply365 covers the screens.

Workbook wording: FD Pro Flight plan transfer to another EFB demo

Taxiing

CRITICAL OBSERVABLE ITEM

2.1.1 PERFORM NORMAL TAXI-OUT (TAXI-IN) WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

Must both pilots have the FD Pro taxi chart out?

Yes, both pilots reference airport diagrams during taxi

Both pilots are required to reference the appropriate airport diagram during taxi and follow the aircraft's progress on it against the ATC clearance. The taxiing pilot should stay primarily heads-up while doing so. If a mount is broken the EFB may still be used to keep the diagram in view.

FOM 5.3.4.3 During Taxi

"Both pilots shall reference the appropriate airport diagrams during taxi."

If any doubt about the clearance, stop the aircraft and ask.

Workbook wording: Do both crewmembers must have the FD Pro TAXI chart out and available during surface movements?

FD Pro ICAO Procedures

1.1.1.16.4 KNOW INTERNATIONAL PROCEDURES (AS APPLICABLE) WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

Where in FD Pro are the ICAO country procedures?

Pubs, then the region's Airway Manual, State Rules and Procedures

Open Pubs in FD Pro, pick the region, open that region's Jeppesen Airway Manual, then State Rules and Procedures for the country. The FOM sends you there for country specific holding, and its regional chapters use the same path for Canada, Pacific, and Latin America material.

FOM 5.5.15.4 International Holding Procedures

"For country-specific holding procedures and airspeeds, see that country's State Rules and Procedures under the applicable region's Jeppesen Airway Manual in Jeppesen FD Pro on the EFB."

FOM chapter 20 lists suggested FD Pro paths for the Pacific region item by item.

Workbook wording: Where in FD Pro can you find the ICAO (country) procedures for destinations that we serve?

1.1.1.16.4 KNOW INTERNATIONAL PROCEDURES (AS APPLICABLE) WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

Where in FD Pro are the ICAO emergency procedures?

Region specific Emergency section of the Jeppesen Airway Manual

Under Pubs in FD Pro, open the region's Jeppesen Airway Manual and go to the Emergency section. The FOM points there for ICAO communication failure procedures, which differ by country. Review it for each international destination before you need it in the air.

FOM 11.1.4.2 Loss of Communication in International Airspace

"found in the Jeppesen Airway Manual, region-specific Emergency section."

Workbook wording: Where in FD Pro can you find the ICAO (country) emergency procedures for destinations that we serve?

1.1.1.16.4 KNOW INTERNATIONAL PROCEDURES (AS APPLICABLE) WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

Review the FD Pro Pubs Quick Reference guide in Comply365

No published source located, ask the check airman

Vendor guide, not manual text. The FOM sends you into FD Pro Pubs for State Rules and Procedures and the regional Emergency sections, which is what the quick reference walks you through.

FOM 5.5.15.4 International Holding Procedures

No published source located, ask the check airman. Open the guide in Comply365 before OE.

Workbook wording: Review FD Pro Pubs Quick Reference guide in Comply365

De/Anti-Ice Procedures

MISC MISCELLANEOUS, NOT ASSOCIATED WITH GRADE SHEET TASKS WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

Discuss the SureWx app**Gives liquid water equivalent based holdover times, FAA authorized**

The app reports holdover times built from liquid water equivalent, a direct measure of the water content in frozen precipitation. Those times are more precise and often longer than METAR based tables, and they shall be used unless clear visual evidence shows the data is wrong. This includes heavy snow and freezing rain.

FOM 9.3.1.1 Liquid Water Equivalent (LWE)

"If your HOT is derived from LWE, it is FAA-authorized and shall be used unless there is clear, visual evidence that the data is inaccurate"

The FOM calls it the HOTs app. The flight log deicing block says use the HOTs app; the listed backup is printed holdover times.

Workbook wording: Discuss SureWx app

Dispatch and Release (17)**Dispatch Release: Review all sections****CRITICAL OBSERVABLE ITEM****1.1.1 PERFORM FLIGHT PLANNING****What is in the dispatch release package?****Dispatch release, flight plan, weather package, and NOTAMs**

Seven items: the dispatch release including the minimum equipment list, the flight plan, the aerodrome forecast, the routine weather report, special weather and significant meteorological reports with turbulence, Class D notices, and flight data center notices. The load closeout is also required before departure.

FOM 8.5.1 Flight Information Packet/Dispatch Release Package

"Dispatch produces the Flight Information Packet/Dispatch Release Package containing required and supplemental data for flight release."

Your signature on the dispatch release says you reviewed all of it. Before pushback also confirm tail number, equipage and maintenance status match the release.

Workbook wording: What documents are included in the Dispatch/Release package? ("Lucky 7"; in order or printing)

CRITICAL OBSERVABLE ITEM**1.1.2.4 KNOW OPERATIONAL CONTROL (DISPATCH) REQUIREMENTS****Who has operational control of a flight?****The PIC and the Dispatcher, jointly**

The PIC and the Aircraft Dispatcher are jointly responsible for preflight planning, delay, and the Dispatch Release, and jointly determine the suitability of weather, field and traffic conditions, airway facilities, and the filed route. Neither can dispatch alone; both must agree.

FOM 2.1.1 Joint Responsibility

"jointly responsible for the preflight planning, delay, and Dispatch Release of a flight"

Regulatory basis 14 CFR 121.533/.535/.537/.551/.557/.627 and Ops Spec A008. Captain-side point: joint responsibility is not shared authority in flight. Once airborne the PIC retains final authority under 2.1.2.

CRITICAL OBSERVABLE ITEM**1.1.2.4 KNOW OPERATIONAL CONTROL (DISPATCH) REQUIREMENTS****What requires Dispatch notification in flight?****Any abnormality or irregularity**

Flights experiencing abnormalities or irregularities, including but not limited to those listed in 7.2.3 Pilot Reports Quick Reference, shall contact Dispatch and report the details. Notification is mandatory; the timing flexes to safety of flight. If

time does not permit contact in flight, contact shall be made after gate arrival.

FOM 11.1.2 Abnormal Operations Dispatch Notification

"Contact Dispatch" implies that while notification is mandatory, such communications should take place as safety of flight considerations permit."

Rev 125.1 revised 7.2.3 to state the table is not all-inclusive and that the Company may request safety reports for items not listed when additional information is needed. Separately 11.1.3 covers Cabin notification, and 11.2.11 gives the NTSB briefing format for the Flight Attendants.

Workbook wording: What items require Dispatch notification during a flight?

Supplemental Operations

CRITICAL OBSERVABLE ITEM

1.1.2.4 KNOW OPERATIONAL CONTROL (DISPATCH) REQUIREMENTS WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

Supplemental operations: what changes?

Nothing in fuel or minima: the flight runs under Domestic or Flag rules and the release says which

Supplemental, non-scheduled, operations are authorized within the approved areas and are conducted under Domestic or Flag rules per Ops Spec A030 or Exemption 11047. The Dispatcher notes on the release whether the flight is operated as Part 121 Domestic or Flag, and the fuel and weather rules follow from that.

FOM 8.1.1.3 Supplemental Operations

"Non-scheduled passenger or cargo operations are authorized within the areas approved for enroute operations. Online or Offline Supplemental Operations will be conducted using Domestic or Flag rules as applicable per Ops Spec A030 or Exemption 11047."

Read the release remark first: it tells you which rule set you are under for the whole leg.

Workbook wording: Supplemental Operations

1.2.2 PERFORM ACARS AND FMS INITIALIZATION AND VERIFICATION

Where are the FMS, ACARS, uplink and setup procedures?

FCOM PRO-NOR-SOP-04 and SOP-06 for the FMS; FCOM PER-ARD for the AeroData ACARS uplinks

The MCDU and FMS setup runs through the Preliminary Cockpit Preparation, PRO-NOR-SOP-04, and the Cockpit Preparation, PRO-NOR-SOP-06. The AeroData takeoff and landing performance requests, the OPS CTRL message and the load closeout uplink are described in the FCOM Performance chapter, PER-ARD.

FCOM PER-ARD-GEN P 2/4 Overview of AeroData Process

"The flight crew sends the OPS CTRL MSG (containing the Fuel on Board, Release Version, and flight crew Fitness for duty) via ACARS during the cockpit preparation."

The Final TPR uplink carries the load closeout ZFW and ZFWCG and the takeoff data into the FMS. An uplink deletes previously entered THR RED and ACCEL altitudes, so re-check them.

Workbook wording: Where are the FMS/ACARS/Uplink and Setup Procedures located?

1.2.2 PERFORM ACARS AND FMS INITIALIZATION AND VERIFICATION

How do you tell the nav database is current?

A/C STATUS page: the ACTIVE NAV DATABASE dates and HA3 name cover today

On the MCDU DATA, A/C STATUS page, verify the day, month and year of the ACTIVE NAV DATABASE. Line 2L shows the validity dates, for example 26MAR-22APR, and 2R the database name, currently HA3, whose code carries the year, month and sequence. If the active database is out of date the MCDU displays CHECK DATA BASE CYCLE.

FCOM PRO-NOR-SOP-04 P 7/14 FMS Database Validity

"FMS DATABASE VALIDITY..... CHECK CM1 Verify the Day, Month, and Year is correct for the ACTIVE NAV DATABASE:"

AIRAC cycles change at 0900 UTC in the US and Canada, 0000 UTC ICAO, 1500 UTC Japan and 1600 UTC Australia. If the new cycle becomes effective during your flight, select it before departure.

Workbook wording: How can you tell if the Nav database is current?

1.2.2 PERFORM ACARS AND FMS INITIALIZATION AND VERIFICATION

How do you change the nav database?

On the ground only: press **SECOND NAV DATABASE** at 3L and confirm.

On the A/C STATUS page, pressing the Second Database key at 3L brings up CONFIRM and CANCEL prompts. Confirming swaps the databases. Cycling erases the primary and secondary flight plans and all stored data, so the SOP cycles it twice during preliminary cockpit preparation to clear pilot-entered and uplinked data, and the FCOM says never do it in flight.

FCOM DSC-22_20-20-10-25 P 78/154 A/C STATUS page

"CAUTION Cycling the database erases the primary and secondary flight plans, as well as stored data. The flight crew must never do this in flight."

The preliminary cockpit preparation cycles the database on purpose: select the non-current one, then re-select the current one, and check STORED shows nothing.

Workbook wording: Can you change the Nav database? How?

Load Closeout/TPR Update and review procedure

1.2.2 PERFORM ACARS AND FMS INITIALIZATION AND VERIFICATION

Who reads the load closeout data?

The First Officer reads the count and jumpseat numbers

Passenger count verification is part of the load closeout procedure. The Captain checks them against the Flight Deck Report. Any discrepancy goes to Load Planning before you go.

FOM 5.2.17.1 Passenger Count Verification

"The First Officer reads the passenger count and jumpseat numbers from the Load Closeout."

The closeout passenger count drives performance. The Flight Deck Report count is what the agents and flight attendants work from, so the two must agree.

Workbook wording: Who reads the closeout data?

1.2.2 PERFORM ACARS AND FMS INITIALIZATION AND VERIFICATION

Who verifies the closeout, and when?

The Captain, before pushback

The Captain verifies the numbers match the Flight Deck Report, and as Pilot in Command must receive and review the closeout prior to pushback. If a delay or a return to the gate changes passengers or loading, get a revised closeout before takeoff.

FOM 5.2.19 Load Closeout/SABLE Weight and Balance Loadsheets

"The Flight Crew will normally receive the Load Closeout via ACARS and the Pilot-In-Command (PIC) must receive and review it prior to pushback."

On the A330 the closeout arrives by ACARS and drives the Final TPR: AeroData uses the closeout takeoff weight and uplinks the ZFW and ZFWCG and the takeoff data to the FMS with it.

Workbook wording: Who verifies it? When?

1.2.2 PERFORM ACARS AND FMS INITIALIZATION AND VERIFICATION

What closeout items must be verified?

Takeoff weight, weight and balance, passenger count, dangerous goods notice

Compare the planned takeoff weight on the dispatch release against the actual takeoff weight on the closeout. Crosscheck the weight and balance and performance numbers, including the flight management system entries, for reasonableness. Verify the passenger and jumpseat count, and confirm the dangerous goods entry is accurate.

FOM 5.2.19 Load Closeout/SABLE Weight and Balance Loadsheel

"planned Takeoff Weight on the Dispatch Release against the Actual Takeoff Weight indicated on the Load Closeout or SABLE Weight and Balance Loadsheel."

If the release and closeout takeoff weights differ, apply the dispatch release weight tolerance rules before accepting the numbers.

Workbook wording: List several items that are required to be verified.

1.2.2 PERFORM ACARS AND FMS INITIALIZATION AND VERIFICATION**No TPR available. How do you get takeoff performance?****Use the TLR on the release, or have Dispatch calculate and send an updated TLR**

AeroData takeoff data comes three ways: the TLR in the Dispatch Release, the ACARS TPR, or the Dispatcher, who can calculate an updated TLR and send it to the printer or read it to you. The hierarchy is Final TPR, Preliminary TPR, then TLR.

FCOM PER-ARD-GEN P 1/4 Takeoff Performance Data

"■ Dispatcher: An updated TLR can be calculated and sent to the aircraft printer or relayed verbally to the flight crew."

A verbal TLR from Dispatch is legitimate data. Read it back and write it down before it goes in the FMS.

Workbook wording: How do you obtain takeoff performance if a TPR is unavailable?

1.2.2 PERFORM ACARS AND FMS INITIALIZATION AND VERIFICATION**Can you use the TLR for takeoff data?****Yes. It is the tertiary source, valid for the conditions it was built on**

The TLR is prepared by the dispatcher on anticipated atmospheric conditions and planned payload and rides in the Dispatch Release. The FCOM ranks it third behind the Final and Preliminary TPR. Use it when ACARS data is not available, and confirm it is still valid before takeoff.

FCOM PER-ARD-GEN P 1/4 TLR

"A Takeoff and Landing Report (TLR) is prepared by the dispatcher based on anticipated atmospheric conditions and planned payload. The TLR is provided as part of the Dispatch Release/Flight Plan."

FOM 5.4.5 says the same thing from the policy side: takeoff data must be available and checked for each takeoff, and invalid data is replaced by ACARS or Dispatch.

Workbook wording: Can we use the TLR for takeoff information? When would we use it?

1.2.2 PERFORM ACARS AND FMS INITIALIZATION AND VERIFICATION**Concerns using the TLR: temperature and altimeter?****It was built on forecast conditions and planned weight; confirm it is still valid**

The TLR uses the environmental conditions the dispatcher considered at dispatch and the takeoff weight the dispatcher planned. The Before Pushback check has both pilots confirm wind, OAT and QNH on the report and, if using the TLR, confirm the TLR is still valid. Actual OAT above the assumed value or a lower QNH invalidates the numbers.

FCOM PRO-NOR-SOP-07 P 3/8 Final Takeoff Data

"■ Confirm it is the "FINAL" TPR (If using the TLR, confirm the TLR is still valid)"

Each version of takeoff data is valid only for the conditions and weight it was calculated on. If anything moved, request a new TPR by entering the gross weight in PTOW at 5R and SEND at 6R.

Workbook wording: Any concerns when using TLR for takeoff information? Temp and altimeter.

Cruise Altitude Considerations**CRITICAL OBSERVABLE ITEM****5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE****Is there a tropopause chart in the release package?****No chart. TROP is a Navigation Log column per waypoint**

The Navigation Log columns on the flight plan include TROP, the tropopause level at each waypoint, alongside MORA, TTK, NM, IAS, Mach, FL and TEMP (FOM 8.5.7). That per-waypoint field, not a separate chart, is the released tropopause

data. FOM 8.5.1 lists the required contents of the Flight Information Packet and does not include a tropopause chart.

FOM 8.5.7 Dispatch Release Sample (HA) - Navigation Log Legend

"MORA Minimum Off Route Altitude TROP Tropopause TTK True Track to WPT NM Nautical Miles to WPT IAS Indicated Airspeed to WPT."

FOM 8.5.1 lists everything in the Flight Information Packet (release, flight plan, TAF, METAR, SIGMETs and turbulence products, NOTAMS); no tropopause chart is among them. Graphical hazard products from the weather vendor are covered in FOM 9.8.1.

Workbook wording: Do we have a chart in our dispatch release package that shows the level(s) of the tropopause with reference to our flight plan?

SATCOM / HF Procedures

CRITICAL OBSERVABLE ITEM

5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE

What prompts a SATCOM call to Dispatch?

Pilot discretion; any required report to Dispatch

FOM 7.1.7 makes satellite voice to Dispatch or MedLink purely the pilot's call, so the circumstance is judgment driven: use it when the message is too complex, too urgent, or too consequential for ACARS text. What is not discretionary is the contact itself.

FOM 7.1.7 SATCOM (if installed)

"Satellite voice communication with Dispatch or MedLink is at the pilot's discretion."

The mandatory triggers are in FOM 7.1.12 Reports to Dispatch: 3k lb fuel deviation on the A330, 100 nm lateral, 4000 ft altitude, 15 minute arrival delay, weather, moderate or greater turbulence, safety of flight, abnormalities, diversions, mechanical irregularities and ETOPS reroutes.

Workbook wording: What circumstance precipitates a SATCOM call to Dispatch?

Analysis of Airport Conditions

6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL

Are there high mins Captain restrictions?

Yes, relieved by Exemption 5549

FOM 5.6.17 cross-references 3.16 Exemption 5549 for high minimums Captain limitations. A newly qualified Captain operates to increased minimums until the experience requirement is met, and Exemption 5549 is the FAA relief that modifies how those high-minimums restrictions apply.

FOM 5.6.17 / FOM 3.16 Exemption 5549

"See 3.16 – Exemption 5549 for high minimums Captain limitations."

Read FOM 3.16 for the exact hour threshold and the precise relief Exemption 5549 grants. This is directly relevant to you as a new Captain and is a near-certain oral question, so get the numbers from the PDF rather than from memory.

Workbook wording: Are there restrictions for High Mins Captains?

6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL

What is FAA Exemption 5549?

Relief allowing High Minimums Captains to fly published minima

Exemption 5549 lets a High Minimums Captain fly to published CAT I minima instead of the increased high-minimums values, provided the approach is flown by the Captain with an autopilot coupled to DH or missed approach (HA wording, FOM 3.16.1).

FOM 3.16 Exemption 5549

"High Minimums Captains may be dispatched using Exemption 5549 and must comply with the restrictions listed below."

A High Minimums Captain has under 100 hours PIC in type; without the exemption they add 100 ft and 1/2 sm to published minimums, but no less than 300-1 (FOM 3.17). OE flight hours do not count toward the 100 hours.

Fuel Planning (16)

Dispatch Release: Review all sections

CRITICAL OBSERVABLE ITEM

1.1.1 PERFORM FLIGHT PLANNING

Where is preflight planning and the international release?

FOM Dispatch chapter, Flight Information and Redispatch section

FOM Chapter 8 (Dispatch) holds the planning material: 8.1 authorized operations and airports, 8.2 alternates, 8.3 navigation, 8.4 weather, and 8.5 flight information documents and redispatch. FOM 8.5.1 defines the Flight Information Packet/Dispatch Release Package and makes the PIC responsible for having it prior to departure from an originating station.

FOM 8.5 Flight Information - Documents and Redispatch (8.5.1 Flight Information Packet/Dispatch Release Package)

"Dispatch produces the Flight Information Packet/Dispatch Release Package containing required and supplemental data for flight release. The PIC shall review and ensure the following information (as applicable) is in their possession prior to departure from an originating station:"

The HA international release example and its legend are FOM 8.5.7 and 8.5.7.1. Preflight duties themselves are FOM 5.2; theater-specific international items are FOM chapters 19-24.

Workbook wording: FOM Preflight Planning and International Dispatch Release

CRITICAL OBSERVABLE ITEM

1.1.1 PERFORM FLIGHT PLANNING

B043 vs B044 flight plan?

B043 international fuel reserve, B044 planned redispatch

B043 is the Ops Spec authorizing special fuel reserves in international operations, where the 10 percent applies only to planned oceanic/remote continental time and the final reserve is 45 minutes at normal cruise consumption.

FOM 8.3.2.3 and FOM 8.5.9 Redispatch Planned

"Ops Spec B043"

Both reduce carried fuel but by different mechanisms: B043 narrows the window the 10 percent applies to; redispatch (B044) shortens the leg the 10 percent is computed against. 8.5.9 lists Ops Specs A002, A004, A008, B044 and B050.

Workbook wording: What is the difference between a B044 vs B043 flight plan?

CRITICAL OBSERVABLE ITEM

1.1.1 PERFORM FLIGHT PLANNING

Standard release vs redispatch?

Redispatch runs two plans, so you carry less fuel

A standard release dispatches you to the destination on one plan. Planned redispatch is an FAA-authorized flag procedure that minimizes the fuel needed to meet the 10 percent requirement. You operate with two plans: the original dispatch to the intermediate destination, and a second plan from a planned redispatch waypoint to the intended destination.

FOM 8.5.9 Redispatch Planned

"The terms of original Dispatch Release must be adhered to until the Captain and the Dispatcher concur on the terms of the redispatch."

Terminology to have cold: Intended Destination is the ultimate destination from the redispatch waypoint; Intermediate Destination is the one on the original release; the Redispatch Point is the common point on both flight plans where the continue decision is made.

Workbook wording: Discuss the differences between a standard dispatch release vs a Re-Dispatch/Re-Release

Cruise Altitude Considerations

CRITICAL OBSERVABLE ITEM

5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE

Why does the tropopause level matter for cruise?

It sets air temperature; descending below it warms fuel

FOM 9.3.2.3 is the only place the manuals tie tropopause height to cruise altitude selection, and it does so through fuel temperature. Fuel drifts toward TAT at roughly 3 degrees C per hour, up to 12 in extreme conditions. On long cold-soak sectors the tropopause level tells you where warmer air is.

FOM 9.3.2.3 Fuel Freeze Considerations

"For instance, a 4000 ft descent below the tropopause can result in approximately a 7°C increase in TAT."

Tropopause height per waypoint (TROP) is printed on the Navigation Log. FOM 8.3.3 also notes ozone forms primarily above the tropopause, which is why ozone converters are fitted.

Workbook wording: Why is it important to consider the level of the tropopause regarding your cruise altitude?

CRITICAL OBSERVABLE ITEM

5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE

What do you expect crossing the tropopause?

Temperature stops falling; an inversion may warm you climbing

Near and through the tropopause the normal lapse rate breaks down. Temperature stops decreasing with altitude and can invert, so a climb may actually warm TAT and help a cold fuel problem. A 4000 ft descent below the tropopause is worth about 7 degrees C of TAT (FOM 9.3.2.3).

FOM 9.3.2.3 Fuel Freeze Considerations

"Occasionally, a temperature inversion at the Tropopause may occur, and, in this case, climbing will result in warmer temperatures and be beneficial for fuel warming."

The FOM only addresses the tropopause in temperature/fuel-freeze terms (9.3.2.3) and ozone terms (8.3.3). Turbulence and wind shear expectations near the tropopause are not stated in the FOM, FCOM or QRH.

Workbook wording: What can expect when cruising near, or crossing through, the tropopause?

Analyze Fuel Factors for Diversion

CRITICAL OBSERVABLE ITEM

5.1.2.6 KNOW CONTINGENCY PLAN (I.E., ALTERNATES, WEATHER MONITORING, FUEL, ETC.)

What drives normal fuel planning?

Weather, traffic delays, MEL penalties, turbulence and cruise altitude

Six factors. MEL and configuration deviation list limits and penalties. Turbulence and cruise altitude. One instrument approach with a possible missed approach at destination. Anything else that could delay landing.

FOM 8.3.1.1 Fuel Planning Factors

"Wind, severe weather, and other forecast weather conditions, and • Anticipated traffic delays, and • MEL/CDL limitations and penalties, and • Turbulence/cruise altitude, and • One instrument approach and possible missed approach at destination, and • Any other conditions that may delay landing."

If you and the Dispatcher disagree on extra fuel, you have the final call. Agree before fueling starts, and get changes to Dispatch at least 30 minutes before pushback.

Workbook wording: What items must you keep in mind when doing fuel planning (normal flight)?

CRITICAL OBSERVABLE ITEM

5.1.2.6 KNOW CONTINGENCY PLAN (I.E., ALTERNATES, WEATHER MONITORING, FUEL, ETC.)

Domestic dispatch fuel requirement?

Destination, alternate, plus 45 minutes

Flights must be dispatched with enough fuel to fly to the airport to which dispatched. Add fuel to fly to and land at the most distant alternate, plus 45 minutes at last cruise altitude consumption rate. With no alternate required, the reserve is 55

minutes at that rate instead.

FOM 8.3.2.1 Domestic Operations

"With no alternate, fly for 55 minutes at last cruise altitude consumption rate."

14 CFR 121.639. Both reserves are at LAST CRUISE ALTITUDE consumption rate, not holding. Applicable areas at 8.1.1.1.

Workbook wording: What are the dispatch fuel requirements for domestic operations?

CRITICAL OBSERVABLE ITEM

5.1.2.6 KNOW CONTINGENCY PLAN (I.E., ALTERNATES, WEATHER MONITORING, FUEL, ETC.)

Flag dispatch fuel requirement?

Destination, 10 percent of trip time, alternate, 30 minutes holding

Flag dispatch fuel is enough to reach the destination, then 10 percent of the total time from departure to landing at destination. Add fuel to the most distant alternate if one is required. Finish with 30 minutes at holding speed 1500 ft above the destination or that alternate.

FOM 8.3.2.2 Flag Operations

"Fly to and land at the most distant alternate airport, if one is required, then • Fly for 30 minutes at holding speed at 1500 ft above the destination airport or alternate if one is required."

The 10 percent here is on total flight time. That is what separates it from the B043 special reserve, where the 10 percent counts only oceanic and remote continental time.

Workbook wording: What are the dispatch fuel requirements for Flag operations?

CRITICAL OBSERVABLE ITEM

5.1.2.6 KNOW CONTINGENCY PLAN (I.E., ALTERNATES, WEATHER MONITORING, FUEL, ETC.)

Supplemental dispatch fuel requirement?

Domestic or Flag rules, whichever applies

Supplemental (non-scheduled) operations are authorized within the areas approved for enroute operations and are conducted under Domestic or Flag rules per Ops Spec A030 or Exemption 11047 (FOM 8.1.1.3). The Dispatcher must notate on the release whether the flight is being operated as Part 121 Domestic or Flag (FOM 8.5.2). Fuel then comes straight from FOM 8.3.2.1 or 8.3.2.2.

FOM 8.1.1.3 Supplemental Operations / FOM 8.5.2 Dispatch Release

"Supplemental (non-scheduled) flights shall be dispatched using either Domestic or Flag rules as applicable."

No separate supplemental fuel rule.

Workbook wording: What are the dispatch fuel requirements for Supplemental operations?

CRITICAL OBSERVABLE ITEM

5.1.2.6 KNOW CONTINGENCY PLAN (I.E., ALTERNATES, WEATHER MONITORING, FUEL, ETC.)

B043 special fuel requirement?

Destination, 10 percent of oceanic time, alternate, 45 minutes cruise

Fuel to the destination, then 10 percent of the planned time in oceanic and remote continental airspace. Add fuel to the most distant alternate if one is required. Finish with 45 minutes at normal cruise consumption, figured at the weight and altitude you would have at top of descent.

FOM 8.3.2.3 Special Fuel Reserves in International Operations

"10% of the planned time in oceanic/remote continental airspace, then • Fly to and land at the most distant alternate airport, if one is required, then • Fly for 45 minutes at normal cruising consumption."

On the release the special reserve shows as 10 percent of the oceanic and remote continental time plus 45 minutes at normal cruise consumption at the top of descent weight and altitude. A standard flag release carries 30 minutes at 1500 ft instead.

Workbook wording: What are the special fuel requirements for a B043 flight plan?

CRITICAL OBSERVABLE ITEM

5.1.2.6 KNOW CONTINGENCY PLAN (I.E., ALTERNATES, WEATHER MONITORING, FUEL, ETC.)

What is final reserve fuel?**30 minutes holding at 1500 ft, or 45 minutes under B043**

Final reserve is the fuel that must remain untouched at the end of the planned profile. For Flag operations it is 30 minutes at holding speed at 1500 ft above the destination or alternate. Under the B043 special international reserve it is 45 minutes at normal cruising consumption computed at the flight's weight and altitude at top of descent.

FOM 8.3.2.2 / 8.3.2.3

"Fly for 45 minutes at normal cruising consumption."

Tie this to 11.2.6: if it becomes apparent you will land with less than 30 minutes of fuel, declare MAYDAY MAYDAY MAYDAY FUEL. Final reserve is a planning number; the 30-minute emergency-fuel trigger is an in-flight number.

Workbook wording: What is Final Reserve Fuel?

CRITICAL OBSERVABLE ITEM

5.1.2.6 KNOW CONTINGENCY PLAN (I.E., ALTERNATES, WEATHER MONITORING, FUEL, ETC.)

When do you ask ATC about the delay?**When delay plus divert could land you under 30 minutes**

FOM 11.2.5 reinforces the same posture: crews should be proactive in assessing estimated arrival fuel and, if delays are anticipated that could lead to minimum fuel, should evaluate and inquire well in advance.

FOM 11.2.6 Emergency Fuel

"The PIC should request ATC delay information if it is anticipated that an airborne delay followed by a diversion could result in landing with less than a 30-minute fuel supply."

Inquire early. By the time you can declare minimum fuel you have already committed to an airport and given up the divert option.

Workbook wording: When should you plan to "inquire" from ATC the current delay situation when unanticipated circumstances arise?

CRITICAL OBSERVABLE ITEM

5.1.2.6 KNOW CONTINGENCY PLAN (I.E., ALTERNATES, WEATHER MONITORING, FUEL, ETC.)

When do you declare minimum fuel?**Once committed to one airport and able to accept no delay**

Minimum fuel means the fuel supply has reached a state where the flight can accept little or no delay. Declare it only after committing to land at a specific airport. Do not declare it at the destination while a divert to the alternate is still available. It is an advisory, not an emergency, and grants no priority.

FOM 11.2.5 Minimum Fuel

"should only be made when the flight has committed to land at a specific airport"

The FOM also directs crews to be proactive and inquire well in advance if delays could lead to minimum fuel.

Workbook wording: When should you declare Minimum Fuel?

CRITICAL OBSERVABLE ITEM

5.1.2.6 KNOW CONTINGENCY PLAN (I.E., ALTERNATES, WEATHER MONITORING, FUEL, ETC.)

When do you declare MAYDAY FUEL?**When you will land with less than 30 minutes**

The PIC shall declare MAYDAY MAYDAY MAYDAY FUEL when it becomes apparent that landing will occur with less than a 30-minute supply of fuel, to ensure priority handling. The 30-minute supply is based on holding fuel flow or higher, as required by configuration, at 1500 ft above field elevation.

FOM 11.2.6 Emergency Fuel

"the PIC shall declare, "MAYDAY, MAYDAY, MAYDAY, FUEL" to ensure priority handling."

Note shall, not should. Thirty minutes is a floor, not a target. Contrast with 11.2.5 minimum fuel, which is advisory and grants no priority.

Workbook wording: When should you declare "MAYDAY, MAYDAY, MAYDAY, FUEL"?

Fuel Management**6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL****Where is fuel planning in the FOM?****Dispatch chapter, under Planning Considerations and Requirements**

Fuel planning sits in the Dispatch chapter of the Flight Operations Manual, under Planning Considerations and Requirements. Fuel Planning covers the factors you must consider, departure fuel coordination with Dispatch, and tanker fuel. The next block, Fuel Requirements, carries the domestic, flag and international reserve numbers.

FOM 8.3.1 Fuel Planning

"When computing required fuel for a flight, the following factors must be considered:"

Same chapter also holds departure fuel coordination. If you and Dispatch cannot agree on extra fuel, the Captain has the final say on fuel on board.

Workbook wording: Where in the FOM is the fuel planning section?

Fuel Conservation**MISC MISCELLANEOUS, NOT ASSOCIATED WITH GRADE SHEET TASKS****Where are fuel conservation considerations?****Green Policy; green-symbol fuel efficiency items appear throughout**

FOM 1.6.3 Green Policy establishes that green-symbol guidance supports Alaska Air Group's fuel efficiency commitment, with the explicit safety is always first and fuel saving measures must never compromise safety.

FOM 1.6.3 Green Policy

"operational decisions that align with Alaska Air Group's commitment to fuel efficiency."

No section titled Fuel Conservation exists in the FOM. The guidance is marked by a green symbol wherever it applies: single engine taxi in the taxi guidance, cruise and optimum altitude, and route deviation and optimization.

Workbook wording: Where can you find the Fuel Conservation considerations in the FOM?

Weather and Minimums (17)**Dispatch Weather Minimums****1.1.2.9 KNOW WEATHER MINIMA & LIMITATIONS FOR DISPATCH, TAKEOFF, APPROACH, AND LANDING****Destination alternate weather required to depart?****At or above alternate minimums for the ETA**

A flight may not be dispatched unless the weather at the destination alternate is forecast at or above the alternate minimums for the time the aircraft would arrive there. Those minimums come from the Derived Alternate Airport IFR Weather Minimums table in 8.2.2.1 under Ops Spec C055.

FOM 8.2.2.1 Destination Alternate Airport Weather Requirements

"A flight may not be dispatched to the destination airport unless the weather conditions at a destination alternate airport are forecast to be at or above the alternate minimums"

Second alternate rules live in 8.2.2.2 and Exemption 20108 in 8.2.2.3.

Workbook wording: Weather required to be at our destination/destination alternate (Flag) to depart?

Enroute Weather Minimums

1.1.2.9 KNOW WEATHER MINIMA & LIMITATIONS FOR DISPATCH, TAKEOFF, APPROACH, AND LANDING

Destination alternate weather in flight?

Must keep meeting the minimums it was dispatched under; below them Dispatch files a new alternate, or the flight diverts

New in FOM 125.3, 8.2.2.1.1: after dispatch the destination alternate must continue to meet the weather minimum rules under which it was dispatched. If it drops below, Dispatch must file a new alternate that meets the required minimums, or remove the alternate entirely if the destination no longer requires one. If no alternate can be found the flight must divert unless, in the interest of safety, the PIC continues to the destination using emergency authority. This is different from an ETOPS alternate in flight, where FOM 6.3.5 applies the crew's operating minima (what the operating crew needs to fly the approach given their minima and any MEL), which is less restrictive than the dispatch planning criteria.

FOM 8.2.2.1.1 Destination Alternate Airport Weather Requirements After Dispatch / 6.3.5

"After dispatch, the weather at the destination alternate must continue to meet the weather minimum rules under which it was dispatched. If the destination alternate weather drops below those minimums, Dispatch must file a new alternate that does meet the required minimums."

Two standards. Destination alternate after dispatch: the dispatch minimums still apply (8.2.2.1.1, Rev 125.3). ETOPS alternate after the EEP: crew operating minima (6.3.5). Do not mix them.

Workbook wording: Weather required at destination alternate airport in flight?

Takeoff Weather Minimums

1.1.2.9 KNOW WEATHER MINIMA & LIMITATIONS FOR DISPATCH, TAKEOFF, APPROACH, AND LANDING

Takeoff weather minimums at the departure airport?

1 sm or RVR 5000 standard; lower than standard down to 500 RVR with CL and HIRL

Standard takeoff minimums are 1 sm visibility or RVR 5000. The FOM lower than standard table takes the A330 down to 1600 RVR with any of HIRL, CL, RCLM or adequate visual reference, 1200 with the day or night lighting shown, 1000 with CL or HIRL and RCLM, and 500 RVR with CL and HIRL.

FOM 5.4.4 IFR Lower than Standard Takeoff Minimum Requirements

"When RVR ≤ 1600 ft (500 m) or 1/4 sm, the Captain must perform the takeoff."

Charted SID or obstacle minimums may be more restrictive than the 10-9A. At foreign airports you are held to the higher of the 10-9A and Ops Specs.

Workbook wording: Weather required at departure airport for takeoff.

1.1.2.9 KNOW WEATHER MINIMA & LIMITATIONS FOR DISPATCH, TAKEOFF, APPROACH, AND LANDING

Who can fly a low visibility takeoff?

At or below 1600 RVR or 1/4 sm, the Captain makes the takeoff

Ops Spec C078 authorizes lower than standard takeoff minimums wherever a procedure allows standard minimums. Any time the visibility is at or below 1/4 sm or 1600 RVR, the Captain will make the takeoff. High minimums Captain takeoff requirements are in FOM chapter 3, and initial First Officer restrictions in FOM 3.6.

FOM 5.4.3 Captain Required Takeoff

"Any time the visibility is less than or equal to 1/4 sm or 1600 RVR, the Captain will make the takeoff."

The A330 floor is 500 RVR with CL and HIRL; the PRC Takeoff Limitations card tabulates the RVR, lighting and takeoff alternate requirements.

Workbook wording: Any limitations on who can perform a low visibility takeoff?

Approach/Landing Weather Minimums

1.1.2.9 KNOW WEATHER MINIMA & LIMITATIONS FOR DISPATCH, TAKEOFF, APPROACH, AND LANDING

Visibility drops on approach. Can you continue?

Yes, once the final approach segment has begun

Do not begin the final approach segment unless the latest reported RVR, visibility, or ceiling if required, is at or above the authorized minimums. However, if a weather report is obtained AFTER the final approach segment has been initiated and it indicates below minimum conditions, the pilot may continue the approach to DA/DH or MDA.

FOM 5.6.11 Visibility Requirements Approach

“the pilot may continue the approach to DA/DH or MDA.”

The trigger is the start of the FINAL APPROACH SEGMENT, not the FAF crossing or the clearance. For non-precision and CAT I: TDZ RVR is controlling, Mid and Rollout are advisory, and Mid may substitute if TDZ is unavailable. RVR above 6000 is advisory only.

Workbook wording: If the visibility drops while on the approach, can we continue the approach?

1.1.2.9 KNOW WEATHER MINIMA & LIMITATIONS FOR DISPATCH, TAKEOFF, APPROACH, AND LANDING

Different visibility rules at international airports?

Yes. Some ICAO states allow Converted Meteorological Visibility

Some ICAO countries allow use of Converted Meteorological Visibility (CMV) when RVR is not reported, for approaches that are not low-visibility. The FOM directs you to the Aerodrome Operating Minimums and Converted Meteorological Visibility references in the Jeppesen Publications for further guidance.

FOM 5.6.11 Visibility Requirements Approach

“Some ICAO countries allow for use of Converted Meteorological Visibility (CMV) when RVR is not reported for non low-vis approaches.”

Separately, at foreign airports the published 10-9A takeoff minimums may not conform to Ops Specs, and crews are restricted to the HIGHER of the 10-9A or Ops Specs (FOM 5.4.4). Theatre chapters 19 through 24 carry the country-specific items.

Workbook wording: Are there different requirements at some international airports?

Taxiing

CRITICAL OBSERVABLE ITEM

2.1.1 PERFORM NORMAL TAXI-OUT (TAXI-IN)

When do SMGCS procedures apply?

At airports with a published low visibility taxi plan

Airports may develop a low visibility taxi plan such as SMGCS and may publish low visibility taxi procedures. Canadian airports with low visibility taxi plans may produce SMGCS charts in Jeppesen format.

FOM 5.3.11 Taxi Operations Low Visibility

“If an airport is equipped with Surface Movement Guidance and Control System (SMGCS), see the AMM Low Vis feature in Jeppesen FD Pro or Jeppesen Low Visibility Taxi Route charts”

The specific RVR trigger value at which SMGCS becomes active is airport-published on the 10-9 series, not a single FOM number. Canada-specific low visibility taxi plans are at FOM 19.3.1.5.

Workbook wording: When do SMGCS procedures go into effect?

FOM and FCTM guidance on Weather avoidance

15.1.2.3.4 ANALYZE WEATHER CONDITIONS USING WEATHER RADAR SYSTEM

How far do you stand off severe cells?

20 nm laterally; clear tops by 5000 ft

FOM 9.5.5 sets 20 nm minimum lateral avoidance for any thunderstorm identified as severe or giving an intense radar echo. Overflying in the clear requires 5000 ft of clearance above the tops. With less than that, revert to the 20 nm lateral standard.

FOM 9.5.5 Thunderstorm Avoidance

"Avoid by at least 20 miles any thunderstorm identified as severe or giving an intense radar echo. When "in the clear" on top, cloud tops should be cleared by 5000 ft."

If you cannot clear the tops by 5000 ft, go back to 20 nm lateral avoidance. The FOM does not specify an upwind or downwind side for the deviation, so that preference is not manually sourced.

Workbook wording: How far (in nm's) should you avoid Severe CB's if possible? Which direction? Vertically?

15.1.2.3.4 ANALYZE WEATHER CONDITIONS USING WEATHER RADAR SYSTEM**No deviation clearance. What do you do?****Emergency authority, PAN-PAN three times, offset 300 ft beyond 5 nm**

If you cannot continue per clearance due to dangerous weather and no prior clearance can be obtained, the Captain should exercise emergency authority. Communicate the urgency signal PAN-PAN spoken three times, or a CPDLC urgency downlink, as early as possible.

FOM 5.7.4.6 Revised ATC Clearance Cannot be Obtained

"If possible, deviate away from an organized track or route system."

Weather Deviation Table 5.7.4.6(1). Flying EAST (000 to 179 magnetic): deviating LEFT descend 300 ft, RIGHT climb 300 ft. Flying WEST (180 to 359 magnetic): LEFT climb 300 ft, RIGHT descend 300 ft.

Workbook wording: If a weather deviation clearance cannot be attained, what is the procedure to deviate around weather?

15.1.2.3.4 ANALYZE WEATHER CONDITIONS USING WEATHER RADAR SYSTEM**What do you report to Dispatch when deviating?****100 nm lateral, 4000 ft altitude, 15 minute delay, weather**

Dispatch wants a call for any significant deviation from the flight plan. That means 100 nm or more off the planned route, 4000 ft or more off planned cruise altitude, anything likely to delay arrival more than 15 minutes such as holding, and enroute changes made for weather.

FOM 7.1.12 Reports to Dispatch

"Lateral deviation: 100 nm or more from the planned route. – Altitude deviation: deviation of 4000 ft or more from the planned cruise altitude."

The same section adds two you will use often. Call when actual fuel differs from planned by 3k lb on the A330, and any time you are in moderate or greater turbulence.

Workbook wording: What are the four items that are required to be reported to dispatch when deviating?

Assess Diversion and Alternate Airport Weather Conditions and Minima**CRITICAL OBSERVABLE ITEM****5.1.2.6 KNOW CONTINGENCY PLAN (I.E., ALTERNATES, WEATHER MONITORING, FUEL, ETC.)****Airborne, what weather do you need at destination, alternate and ETOPS alternates, and to start an approach?****Crew operating minima once airborne; to begin the final approach segment, reported RVR or visibility at or above minimums**

After takeoff the standard is the crew's operating minima: the weather required for the operating crew to fly the approach given their minima and any MEL. FOM 6.3.5 says that for ETOPS alternates and states it is less restrictive than the Dispatch planning criteria. Before beginning the final approach segment the latest reported RVR, visibility, and ceiling if required, must be at or above the authorized minimums; a report received after that point does not stop the approach.

FOM 6.3.5 In-Flight ETOPS Alternate Airport Weather Requirements

"Forecast weather at the ETOPS alternate must be at or above crew's operating minima (this is the weather required for the operating crew to conduct the approach, given the crew's minima and any applicable MEL). Note that these weather requirements are less restrictive than the criteria used by Dispatch in planning."

Do not apply the derived alternate minimums table to an in-flight divert. That table is for Dispatch. FOM 5.6.11 governs beginning the approach.

Workbook wording: Once airborne, what does the weather have to be at your destination, destination alternate, ETOPS alternate(s)? To start an approach at those airports?

Weather Considerations**6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL****What weather do you need to begin an approach?****Reported RVR or visibility at or above minimums**

The reported weather at the destination must be at or above the visibility minimums prescribed for the approach. Do not begin the final approach segment unless the latest reported RVR, visibility, or ceiling if required, is at or above the authorized minimums.

FOM 5.6.11 Visibility Requirements Approach

"TDZ RVR reports are controlling."

Visibility is the controlling parameter; ceiling applies only where the procedure requires it. RVR values above 6000 are advisory only. Visibility below 1/2 sm is not authorized and shall not be used for approach minimums.

Workbook wording: What must the Weather be in order to begin an approach? Ceiling and visibility?

Analysis of Airport Conditions**6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL****Lowest visibility for CAT I?****1800 RVR**

Standard CAT I is 1800 RVR. It is authorized to approved runways even without touchdown zone or centerline lights, or with those lights inoperative, provided the flight director or autopilot is used in approach mode down to decision altitude.

FOM 5.6.16.1 Reduced Landing Minima - 1800 RVR and 200 ft DH

"Authorized precision CAT I landing minima as low as 1800 RVR to approved runways without TDZ"

Reduced minima below standard CAT I are not available to you. Touchdown zone RVR is controlling, and the flight director or autopilot must be used in approach mode to decision altitude.

Workbook wording: What is the lowest visibility for a CAT I approach?

6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL**Lowest visibility for CAT II?****1000 RVR**

For CAT II the final approach segment may not begin unless the latest controlling RVR reports are at or above authorized minimums. Required airborne and ground equipment must be operational: LOC and GS, an outer marker or DME defining the FAF, HIRL, and an approach light system (FOM 5.6.17.2).

FOM 5.6.17.2 CAT II/SA CAT II Operating Limitations

"Landings and subsequent ground operations for CAT II operations to 1000 RVR must be conducted in accordance with the airport's low visibility operations plan with reference to the Low Visibility Taxi Charts or the Jeppesen Low Visibility AMM."

The controlling RVR table lives on the approach briefing cards in the PRC, not in the FOM - FOM 5.6.11 says so explicitly and FOM 5.6.14 notes CAT II/III minima are NOT tailored on the Jeppesen charts.

Workbook wording: What is the lowest visibility for a CAT II approach? (C060)

6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL

Lowest visibility for an SA CAT II approach?**No published source located, ask the check airman**

The FCOM limitation for SA CAT II gives the minimum decision height, 100 ft, with at least one autopilot in APPR mode and CAT 2 or CAT 3 on the FMA, and without a HUD the landing must be automatic. The controlling RVR sits on the approach briefing cards in the PRC, which the FOM points to and which the text extract does not carry.

FCOM LIM-AFS-20 P 1/4 Special Authorization CAT II (SA CAT II)

"Without HUD, the flight crew must perform an automatic landing."

No published source located, ask the check airman. Read the RVR off the PRC Approach and Landing Limitations card.

Workbook wording: What is the lowest visibility for a CAT II SA approach? (C060)

6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL

Lowest visibility for a CAT III approach?**75 m RVR, CAT III without DH, fail operational with CAT 3 DUAL**

FCOM Limitations: ILS Category III fail operational, dual, has an alert height of 200 ft, needs A/THR in selected or managed speed and both autopilots in APPR with CAT 3 DUAL on the FMA, and for CAT III without DH the minimum runway visual range is 75 m. Fail passive, CAT 3 SINGLE, carries a 50 ft minimum decision height.

FCOM LIM-AFS-20 P 2/4 ILS Category III Fail Operational (Dual)

"CAT III without DH: Minimum Runway Visual Range.....75 m"

Ops Spec C060 limits the crosswind to 15 kt for CAT II and III; headwind 35 kt, tailwind 10 kt. One-engine CAT III autoland is not authorized (FOM 5.6.17.3).

Workbook wording: What is the lowest visibility for a CAT III approach? (C060)

8.2.1 PERFORM CAT 1 ILS APPROACH

Inside the FAF. Minimum visibility?**Once the segment has begun, continue to DA or MDA**

You may not begin the final approach segment unless the controlling RVR is at or above the authorized minimums and both the airborne and ground equipment are operational. Ground equipment means localizer and glide slope plus the required lighting. Once the segment has begun, you may continue to decision altitude.

FOM 5.6.17.2 CAT II/SA CAT II Operating Limitations

"The Final Approach Segment (FAS) of an IAP may not begin unless the latest controlling RVR reports for the landing runway are at or above the minimums authorized"

Note visibility below 1/2 sm is not authorized and shall not be used for approach minimums (5.6.11).

Workbook wording: What about if you are inside the FAF ? Minimum visibility ?

Communications and PA ₍₁₀₎

12.1.3.16 KNOW FOM PA POLICIES AND PERFORM EFFECTIVE COMMUNICATION FROM THE FLIGHT DECK

Where are the standard customer service PAs?**Flight Operations Manual chapter 7, Standard Customer Service Announcements**

Identify yourself as the Captain or First Officer. Keep it plain: no aviation abbreviations, no jargon, no knots, broken clouds, chop or abort. Limit announcements while passengers are asleep. Drop the choice of airlines line on Southeast Alaska flights.

FOM 7.1.21 Standard Customer Service Announcements

"Pilots can use the PA system as a customer service tool to enhance safety and leave a positive impression with our customers."

Workbook wording: What section of the FOM are the Standard Customer Service Announcements (PA's) located?

12.1.3.16 KNOW FOM PA POLICIES AND PERFORM EFFECTIVE COMMUNICATION FROM THE FLIGHT DECK

Which PAs must the flight crew make?**The required PAs, starting with the seat belt sign**

Flight Crews are required to make the PAs in 7.1.19 during each flight with passengers or persons on board, acknowledging that flight conditions and priorities may occasionally preclude timely execution in the interest of safety.

FOM 7.1.19 Required Announcements

"Flight Crew may use judgment to vary the wording of these announcements except where the announcement is noted as verbatim."

Workload permitting, pilots announce any change in Seat Belt Sign condition, and the F/As must also announce it regardless of whether the pilots do. The FOM encourages Aloha from the Flight Deck and Mahalo on Hawaii flights.

Workbook wording: Which announcements are required to be made by the Flight Crew?

12.1.3.16 KNOW FOM PA POLICIES AND PERFORM EFFECTIVE COMMUNICATION FROM THE FLIGHT DECK

How do you make a PA with a headset?**Press and hold the PA transmission key on the ACP, then talk on the boom or mask mike**

From the cockpit the PA works through the ACP or the handset. With a boomset or mask, press and hold the PA transmission key on the ACP; three green lines come on. With the hand mike, hold the PA key and press the hand mike push to talk. The PA reception knob out lets you hear the PA on the loudspeaker.

FCOM DSC-23-20-40 P 1/4 Passenger Address

"PA Transmission Key Pressed and held : The flight crew may use a boom, mask, or hand mike to make an announcement. Three green lines come on."

The PA key is a hold key, not a latch. Let go and you are off the PA.

Workbook wording: How do you make a PA using a headset (which button/transmit switches)?

12.1.3.16 KNOW FOM PA POLICIES AND PERFORM EFFECTIVE COMMUNICATION FROM THE FLIGHT DECK

PA PRIO versus PA ALL on the handset?**No published source located, ask the check airman**

The FCOM describes the cockpit handset only as the pedestal handset used for PA announcements and does not define the PRIO and ALL selections. The FOM names PA+ALL as the selection that reaches the cabin without disturbing the LDMCR.

FCOM DSC-23-20-40 P 2/4 Cockpit Handset

"The cockpit handset at the bottom of the pedestal is used for PA announcements."

No published source located, ask the check airman. Have the check airman show you PRIO on the handset.

Workbook wording: What's the difference between PA PRIO and PA ALL selections?

12.1.3.16 KNOW FOM PA POLICIES AND PERFORM EFFECTIVE COMMUNICATION FROM THE FLIGHT DECK

How do you make a PA with the flight deck handset?**Pick up the pedestal handset and press its push to talk; no ACP action needed**

The cockpit handset at the bottom of the pedestal is dedicated to PA announcements. The PA from cockpit table shows handset use needs no PA transmission key and no PA reception knob action, only the push to talk on the handset.

FCOM DSC-23-20-40 P 1/4 Passenger Address

"Note: The flight crew may use a cockpit handset to make PA announcements without action on the ACPs."

FOM 5.1.23.8 adds the LDMCR trick: press PA+ALL and use the push to talk on the flight deck handset.

Workbook wording: How do you make a PA using the flight deck handset?

CRITICAL OBSERVABLE ITEM**5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE****How do you call the cabin on the headset, and on the handset?****CALLS panel pushbutton to ring the station, then CAB key on the ACP; or use the handset**

Cockpit to cabin: press the station pushbutton on the CALLS panel, FLT REST, CAB REST, PURS, FWD, MID, EXIT or AFT, or ALL. A pink light comes on at that area call panel, CAPTAIN CALL shows on the attendant indication panel and a high-low chime sounds. Then talk with the CAB transmission key pressed on the ACP, using the boom, mask or hand mike. The pedestal handset can also be used.

FCOM DSC-23-20-20 P 2/4 Cabin Call System, Call from the Cockpit

"FLT REST /CAB REST /PURS/FWD/MID/EXIT/AFT pb When pressed: ■ A steady pink lights come on, on the corresponding area call panel ■ "CAPTAIN CALL" message appears on the corresponding AIP, and a green light comes on ■ A high-low chime sounds through corresponding loudspeaker."

EMER, guarded, is the priority call: flashing pink lights, CALL PRIO CAPT on all panels and three chimes. A cabin call to you flashes ATT on the ACP with a buzzer.

Workbook wording: How do you make an interphone call to the cabin using the headset? Using the flight deck handset?

SATCOM / HF Procedures**CRITICAL OBSERVABLE ITEM****5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE****How do you place a SATCOM call?****MCDU MAIN MENU, SAT, pick the number from the DIRECTORY or MANUAL DIAL, then the SATCOM key on the ACP**

The ACPs set up and end the call, the MCDU picks the number. Select SAT on the MCDU MAIN page, then 6R for the SATCOM DIRECTORY, choose a category, press the line key of the number to dial it, or use MANUAL DIAL at 5R. Press the SATCOM transmission key on the ACP: green flashing while the call establishes, steady green connected. The MCDU status line reads DIALING, then CONNECTED.

FCOM DSC-23-30-20-20 P 1/6 SATCOM MCDU Interface

"The crew accesses this page by selecting SAT on the MCDU MAIN page."

An incoming ground call flashes the SATCOM key on the ACP and shows INCOMING CALL on the MCDU. SELCAL/CALL RESET cancels the buzzer.

Workbook wording: How do you make a SATCOM (specifically which buttons, and in what order, must be pressed in/on which components)?

CRITICAL OBSERVABLE ITEM**5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE****Where are the preprogrammed telephone numbers?****MCDU SATCOM DIRECTORY page, four priority lists: EMERGENCY, SAFETY, NON-SAFETY, PUBLIC**

From the SATCOM MAIN MENU, 6R opens the SATCOM DIRECTORY, which holds four phone number lists by priority: 1L EMERGENCY for distress numbers, 2L SAFETY for regulatory and flight safety numbers, 3L NON-SAFETY and 4L PUBLIC. Each category page lists the titles and numbers; pressing a line dials it. Protected numbers are green, unprotected blue in brackets.

FCOM DSC-23-30-20-20 P 2/6 SATCOM Directory Page

"This page provides access to 4 phone number lists, where phone numbers can be memorized, according to their priority."

Type the first three letters of a title in the scratchpad and press 5R to search a category. MedLink and Dispatch live in the SAFETY list.

Workbook wording: In the ACARS/TCP (787) where can you find a list of telephone numbers that are preprogrammed to call?

16.1.2.1.3 KNOW ACARS PROCEDURES WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

Show where the ACARS functions live**ACARS MENU: PRE-FLIGHT, IN-FLIGHT and POST-FLIGHT pages, with PERFORMANCE and DELAYS off the main menu**

The AeroData performance pages hang off the ACARS main menu: MENU PRE-FLIGHT 1/2 then PERFORMANCE for takeoff and landing requests and VIEW DATA, with the same path from IN-FLIGHT and POST-FLIGHT. The PRC ACARS Delay Guide adds that the DELAYS prompt is line select key 4R on the main menu for pre-flight, in-flight or post-flight delays.

FCOM PER-ARD-ALP P 12/18 Landing Data - Received Uplinks

"Page access : MENU PRE-FLIGHT 1/2 → PERFORMANCE → VIEW DATA MENU IN-FLIGHT 1/2 → PERFORMANCE → VIEW DATA MENU POST-FLIGHT 1/2 → PERFORMANCE → VIEW DATA"

ACARS and ATSU traffic normally goes over VHF 3 and switches to SATCOM when VHF 3 is unavailable, so do not use VHF 3 for voice unless 1 and 2 are dead.

Workbook wording: Review ACARS menus and quiz them on where to find specific functions to demonstrate proficiency.

16.1.2.1.3 KNOW ACARS PROCEDURES

Where is a directory of ACARS functions?**FCOM PER-ARD for the AeroData pages, and the PRC ACARS Delay Guide for delays**

The AeroData chapter of the FCOM, PER-ARD, documents the ACARS performance pages key by key: the PERFORMANCE page, T/O CONDITIONS, LDG CONDITIONS, RCAM CODE, received uplinks and message display. The PRC ACARS Delay Guide covers the DELAYS pages and the delay codes. There is no single ACARS function directory in the FOM.

FCOM PER-ARD-ALP P 1/18 Performance Page

"Page access : MENU PRE-FLIGHT 1/2 → PERFORMANCE Screen to access all performance related input/output pages, requested takeoff and landing runways, and load takeoff data in the FMS once available."

A password at 6R is downlinked with every performance request; it is server side, not something you type.

Workbook wording: Where can you find a list/directory of ACARS FUNCTIONS?

Pushback and Start (10)

12.1.3.16 KNOW FOM PA POLICIES AND PERFORM EFFECTIVE COMMUNICATION FROM THE FLIGHT DECK

Demonstrate a late pushback, go-around and engine failure PA**Use FOM sample PAs: delay, go-around, emergency equipment**

Late pushback: at departure time either push or talk, give the reason and an approximate length, then update every 15 minutes (FOM 7.1.21.1). Go-around: FOM 7.1.21 says a short PA should be made when workload and ATC coordination permit, and below 10,000 ft, on an extended downwind or in holding, you may state the reason and your intentions.

FOM 7.1.21 Standard Customer Service Announcements and FOM 7.1.22 Emergency Vehicles Requested Sample PA

"• Missed Approach/Go-Around – Should a go-around be necessary, a short PA announcement should be made when workload and ATC coordination permits."

No verbatim engine failure PA in the FOM. The closest scripted text is FOM 7.1.22 (emergency equipment standing by) and the NTSB brief examples at FOM 11.2.11, one of which is an engine shutdown and return.

Workbook wording: Demonstrate a late pushback PA, a go around PA, and engine failure PA

1.3.2 PERFORM FCOM PROCEDURES PRIOR TO ENGINE START

What check is called to the flight deck before start?**Walkaround Check complete, verbally from the Ground Crew**

FOM 5.2.20 requires, before movement, that the Flight Crew check all door indications to confirm closed and receive a verbal confirmation from the Ground Crew that the Walkaround Check is complete.

FOM 5.2.20 Cabin Safety Check Prior to Taxi or Pushback

"to confirm they are closed and receive a verbal confirmation from the Ground Crew that the Walkaround Check is complete. If installed, the Installed Physical Secondary Barrier (IPSB) must be verified stowed and latched."

Two separate confirmations happen here: the A Flight Attendant/FFA says 'Cabin Secure' and closes the flight deck door, and the Ground Crew verbally confirms the Walkaround Check. The crew must also check all door indications and verify the IPSB stowed and latched if installed.

Workbook wording: What "check" must be communicated to the Flt Deck prior to engine start?

1.3.2 PERFORM FCOM PROCEDURES PRIOR TO ENGINE START**Where is the departure briefing guidance?****Yes. FOM Threat-Based Departure Briefing/TPC**

FOM 5.2.7 says the PF initiates the departure briefing prior to pushback, interactive, concise and big picture. The PM identifies relevant threats first, using the Crew Briefing Reference Card threat list and the 10-7 airport threats as needed. The PF then adds any others.

FOM 5.2.7 Threat-Based Departure Briefing/TPC

"Prior to pushback, the PF initiates the departure briefing. The departure briefing should be interactive, concise, and focused on the big picture. Pilots jointly identify any relevant threats, followed by the PF presenting the departure plan and any other considerations."

TPC = Threats (PM and PF), Plan (PF), Considerations (PF). The threat list is on the Crew Briefing Reference Card plus airport threats in the 10-7. The arrival mirror image is FOM 5.5.16 Threat-Based Approach Briefing/TPC.

Workbook wording: Do we have any guidance regarding the Departure briefing? Where can it be found?

1.3.3 PERFORM AIRCRAFT PUSHBACK PROCEDURE**How do you talk to the ground crew before pushback?****Flight interphone headset, nose gear jack, standard phraseology**

Normal means is the flight interphone with standard phraseology. The ground crew plugs in at the nose and the flight deck talks on INT; the preliminary cockpit preparation has you check the INT knob is out with volume up so you can hear them. FOM 5.2.24 scripts the exact phraseology for brake release, cleared to start and cleared to disconnect, and the SOP makes the PF the one who handles ground crew communications.

FOM 5.2.23 Pushback / 5.2.24 Pushback Interphone Communication

"Interphone communication between Flight Deck and pushback crew is normally required. Standard phraseology shall be used."

FCOM PRO-NOR-SOP-01 puts ground crew communications on the PF and ATC on the PM. Before push, the N/WS DISC memo must be displayed or the tow pin is not in.

Workbook wording: How do communicate with your ground crew prior to push? (Cockpit/Ground Communications)

1.3.3 PERFORM AIRCRAFT PUSHBACK PROCEDURE**How do you alert the ground crew?****MECH pushbutton on the CALLS panel; hand signals if no interphone**

From the cockpit, the MECH pushbutton on the overhead CALLS panel lights the blue COCKPIT CALL light on the external power panel and sounds the external horn. The ground crew answers on the flight interphone. Ground to cockpit, a call flashes the MECH light on the ACP.

FCOM DSC-23-20-30 P 3/4 Service Interphone System, MECH pb

"COCKPIT CALL lights up blue on the external power panel. held) An external horn sounds. Released : COCKPIT CALL remains lighted."

The horn keeps the COCKPIT CALL light on until the mechanic presses HORN RESET on the external power panel. FOM 5.3.12 gives the CONNECT INTERPHONE hand signal for the no-headset case.

Workbook wording: How do you alert/call the ground crew?

1.3.3 PERFORM AIRCRAFT PUSHBACK PROCEDURE**No interphone for pushback. What are the hand signals?****Set brakes, release brakes, emergency stop, all clear**

No interphone means you brief it before anyone moves. Cover the pushback plan and the signals you will use, and be specific about how the ground crew tells you to stop. The signal set itself is the standard marshalling one: set brakes, release brakes, emergency stop, all clear.

FOM 5.2.23.1 Pushback - No Interphone

"In the event that interphone communication is not available, the Captain will brief the Ground Crew on expected pushback procedures and review hand signals."

Start engines doubles as the walkaround complete signal when headset communication is not available.

Workbook wording: If pushing back without an interphone, what are the hand signals?

1.3.3 PERFORM AIRCRAFT PUSHBACK PROCEDURE

Can you be pulled forward with two engines running?

No. Prohibited with both engines running

FOM 5.2.23 states the ground crew policy currently prohibits forward tow operations with both engines running. Shut one down first, or use a single engine taxi procedure if the tow is really a repositioning.

FOM 5.2.23 Pushback

"The Ground Crew policy currently prohibits forward tow operations with both engines running."

This is a ground crew policy limit, not an aircraft limitation. Expect the crew chief to refuse the pull rather than the airplane to stop you.

Workbook wording: Can you have the aircraft pulled forward with two engines running? (787/330 only)

Non-Normal Start Operations

1.3.4.1 COMPLETE NORMAL ENGINE START FCOM PROCEDURE

Which ECAM alerts can you get during engine start?

ENG 1(2) START FAULT: starter time, stall, EGT overlimit, no light-up, low N1, levers not idle.

The A330 start is automatic. The EEC aborts and the ECAM presents ENG 1(2) START FAULT when the starter time is exceeded, on a stall, an EGT overlimit, no light up, low N1, or thrust levers not at idle. Ignition and start valve faults have their own alerts, ENG 1(2) IGN A(B) FAULT and ENG 1(2) IGN A+B FAULT.

FCOM PRO-ABN-ENG P 91/114 ENG 1(2) START FAULT

"This alert triggers when one of the following conditions is detected: ■ Starter time exceeded, or ■ Stall, or ■ EGT overlimit, or ■ No light up, or ■ Low N1, or ■ THR levers not at idle."

On a hot engine the EEC dry cranks before it introduces fuel: EGT above 100 C on the first automatic start, above 150 C on the second (PRO-NOR-SOP-08).

Workbook wording: Which Quick Action Items (787), or ECAM action items (330/321), may be required during ENG START e.g., Aborted Engine Start, Hot Start, Hung Start?

1.3.4.1 COMPLETE NORMAL ENGINE START FCOM PROCEDURE

What is the hand signal for a tailpipe fire?

Horizontal figure eight; other hand points to fire

FOM 5.3.12 defines FIRE as a horizontal figure 8 made by hand or wand, with the other hand or wand pointing at the location of the fire.

FOM 5.3.12 Hand Signals

"FIRE; signifies a fire and it's location to the Flight Crew by indicating a horizontal figure 8, by hand or wand."

Same FIRE signal is used for any fire location, not just a tailpipe fire. Pair it with the QRH 17.09A ENGINE TAILPIPE FIRE procedure: master off, bleed pressure, crank.

Workbook wording: What is the hand signal for Tailpipe Fire?

1.3.5 PERFORM FCOM PROCEDURES FOLLOWING ENGINE START/BEFORE TAXI

What must you consider before pushback and taxi?

NOTAMs, ATIS, closures or construction, and the taxi route

Two things before you move. Review NOTAMs and current ATIS for taxiway or runway closures, construction, or any other airport risk that touches your taxi route. Then confirm both pilots understand the expected route. If either of you is unsure, stop and ask the controller.

FOM 5.3.4.2 Prior to Taxi

“Review NOTAMs and current ATIS for any taxiway or runway closures, construction activity, or other airport risks that could affect the taxi route.”

Complex taxi instructions have to be written down. Dual acknowledgment is required for runway crossing, hold short, line up and wait, cleared for takeoff and cleared to land.

Workbook wording: What items must be considered prior to pushback/taxi?

Taxi and Ground (25)

12.1.3.16 KNOW FOM PA POLICIES AND PERFORM EFFECTIVE COMMUNICATION FROM THE FLIGHT DECK

Pushing back late. When is a PA required?

At scheduled departure time, then every 15 minutes

FOM 7.1.21.1 states that at the departure time the Captain will either authorize pushback/engine start or notify Operations and the passengers of the delay, giving the reason if known.

FOM 7.1.21.1 General - Departure Time Delay Announcement ('push or talk')

“Departure Time Delay Announcement (“push or talk”) – At the departure time, the Captain will either authorize pushback/starting engines procedures or notify Operations and the passengers of the delay.”

The 'push or talk' rule is the trigger: at departure time you either authorize pushback or you tell Operations and the passengers.

Workbook wording: When should you need to make a PA if you are pushing off the gate late? Minutes?

Taxiing

CRITICAL OBSERVABLE ITEM

2.1.1 PERFORM NORMAL TAXI-OUT (TAXI-IN)

Where is taxi guidance in the FOM?

Flight Operations chapter, Taxi section, right after pushback and engine start

Taxi sits in the Flight Operations chapter between pushback and takeoff. It covers when taxi is prohibited, taxi thrust, taxi guidance and speeds, runway incursion prevention, low visibility taxi, hand signals and tarmac delays. It then points you to the fleet manuals for airplane specific technique.

FOM 5.3 Taxi

“While taxiing in the ramp/gate area, any congested area, where taxi instructions are complex, or during periods of reduced visibility, the Flight Crew shall devote their full attention to taxiing the aircraft.”

Taxi is prohibited when braking action is reported NIL by the airport or another aircraft. That is a company rule, not a technique.

Workbook wording: Where in the FOM can you find information on taxiing?

CRITICAL OBSERVABLE ITEM

2.1.1 PERFORM NORMAL TAXI-OUT (TAXI-IN)

Breakaway thrust for taxi?

A little above idle, applied symmetrically; no N1 number is published

The FCTM says the crew needs a little power above idle to move the aircraft, that excessive thrust risks jet blast damage and FOD, and that thrust should normally be used symmetrically. No breakaway N1 figure is published for the A330.

FCTM PR-NP-SOP-100 P 3/8 Taxi Roll and Steering, Thrust Use

"The flight crew will need a little power above idle thrust to move the aircraft. Excessive thrust application can result in exhaust-blast damage or Foreign Object Damage (FOD). Thrust should normally be used symmetrically."

Avoid stopping in a turn, because it takes excessive thrust to get moving again.

Workbook wording: Thrust Use – Max breakaway thrust, N1?

CRITICAL OBSERVABLE ITEM

2.1.1 PERFORM NORMAL TAXI-OUT (TAXI-IN)

Max taxi speed?**30 kt straight, one smooth application back to 10 kt; under 10 kt in a 90 degree turn**

FCTM: on long straight taxiways with no constraints, let the aircraft accelerate to 30 kt, then use one smooth brake application to decelerate to 10 kt, avoiding continuous braking. Use the ND ground speed. For turns of 90 degrees or more, be below 10 kt. FOM 5.3.3 sets the same 30 kt cap and about 10 kt in congested areas.

FCTM PR-NP-SOP-100 P 1/8 Taxi Speed and Braking

"the PF should allow the aircraft to accelerate to 30 kt, and should then use one smooth brake application to decelerate to 10 kt. The PF should avoid continuous brake applications. The GS indication on the ND should be used to assess taxi speed."

Carbon brake wear depends on the number of applications, not the pressure or duration, so fewer firmer applications are the technique.

Workbook wording: Max Taxi Speed/s, and Braking technique?

CRITICAL OBSERVABLE ITEM

2.1.1 PERFORM NORMAL TAXI-OUT (TAXI-IN)

Why keep groundspeed in your scan?**To hold 30 kts max, about 10 kts congested**

FOM 5.3.3 caps taxi speed at 30 kts, only on straight smooth taxiways, and approximately 10 kts in congested areas, and requires full crew attention while taxiing in the ramp, congested areas, complex taxi instructions, or reduced visibility.

FOM 5.3.3 Taxi Guidance

"Speed shall be limited to not more than 30 kts, achieved only on straight, smooth taxiways, and approximately 10 kts in congested areas."

The FCTM says to use the GS indication on the ND to assess taxi speed. There is no taxi speed readout anywhere else.

Workbook wording: Emphasize importance of keeping groundspeed in scan

CRITICAL OBSERVABLE ITEM

2.1.1 PERFORM NORMAL TAXI-OUT (TAXI-IN)

Recommended speed before a 90 degree turn?**Less than 10 kt for a turn of 90 degrees or more**

The FCTM taxi chapter sets the number: for turns of 90 degrees or more the aircraft speed should be less than 10 kt. With a deflated main gear tire it is 7 kt with one tire and 3 kt with two, with steering limited to 30 degrees.

FCTM PR-NP-SOP-100 P 5/8 Taxi Roll and Steering

"For turns of 90 ° or more, the aircraft speed should be less than 10 kt."

The inside main gear cuts the corner and tracks inside the nosewheel, so oversteer and slow down before the turn, not in it.

Workbook wording: Review recommended speed prior to entering a 90° turn

CRITICAL OBSERVABLE ITEM

2.1.1 PERFORM NORMAL TAXI-OUT (TAXI-IN) WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

Demonstrate differential braking when one side's brakes are much hotter**No published source located, ask the check airman**

No published A330 procedure calls for differential braking to balance brake temperatures. The FCTM notes differential braking stays available with antiskid off and warns against using it to stop one gear for a pivot turn; the FCOM BRAKES HOT procedure says delay takeoff until below 300 C, or 150 C with the fans on.

FCOM PRO-ABN-BRAKES P 7/16 BRAKES HOT

"Delay takeoff until the brake temperature is below 300 °C (or 150 °C with the brake fans on)."

No published source located, ask the check airman. The green arc marks the hottest wheel above 100 C, amber above 300 C.

Workbook wording: Demonstrate use of differential braking if brake temperature on one side is significantly hotter than the other side

CRITICAL OBSERVABLE ITEM

2.1.1 PERFORM NORMAL TAXI-OUT (TAXI-IN)

Where are the gear geometry and turning radius figures?**FCOM DSC-20-30 Ground Handling, Taxiing, minimum turning radii**

The Taxiing page for the A330 PAX fleet tabulates the minimum turning radii at the 72 degree nosewheel steering limit: Y 12 m, A 44 m, R3 25 m, R4 43 m, R5 31 m and R6 37 m, assuming symmetric thrust and no differential braking. The cockpit cut-off angle is 20 degrees with a 53 ft obscured segment.

FCOM DSC-20-30 P 1/2 Taxiing - A330 PAX Fleet, Minimum Turning Radii

"The above figure assume symmetric thrust and no differential braking."

The FCTM adds the visual rule: start the turn before an obstacle reaches the 53 ft obscured segment, for wing and tail clearance.

Workbook wording: Location of gear/turning Radius

CRITICAL OBSERVABLE ITEM

2.1.1 PERFORM NORMAL TAXI-OUT (TAXI-IN)

Where is the oversteer technique for turns?**FCTM PR-NP-SOP-100, Taxi Roll and Steering**

The FCTM explains that the main gear on the inside of a turn always cuts the corner and tracks inside the nosewheel track, so over-steer must be used. Anticipate the steer out and let the aircraft roll forward a short distance after a tight turn to relieve the main gear.

FCTM PR-NP-SOP-100 P 5/8 Taxi Roll and Steering

"The flight crew should be aware that the main gear on the inside of a turn will always cut the corner and track inside of the nosewheel track. For this reason, over-steer must be used."

Nosewheel skidding in a turn means slow down or open the radius.

Workbook wording: Where can you find taxi guidance/technique regarding oversteering in turns?

CRITICAL OBSERVABLE ITEM

2.1.1 PERFORM NORMAL TAXI-OUT (TAXI-IN)

Normal 180 degree turn width on the runway?**About 41 m/133 ft for the A330-200, 48 m/156 ft for the A330-300, dry, no margin.**

With the recommended 180 degree turn technique on a dry runway and 72 degrees of nosewheel steering, the approximate turn width is 41 m for the A330-200 and 48 m for the A330-300, without margin. Add margin when the runway is wet or contaminated. The technique itself is in FCTM PR-NP-SOP-100.

FCOM DSC-20-30 P 2/2 180 Degrees Turn on Runway

"With the recommended 180 ° turn technique, on dry runway, with the maximum nosewheel steering of 72 °, the approximate turn width is: ■ For the A330-200, 41 m (133 ft) without margin ■ For the A330-300, 48 m (156 ft) without margin."

Technique: taxi on the far side, 5 to 10 kt, diverge 20 degrees, and when you are physically over the runway edge turn with full tiller, using asymmetric thrust or differential braking only to hold a constant speed.

Workbook wording: What is the normal 180 degree turning radius?

CRITICAL OBSERVABLE ITEM

2.1.1 PERFORM NORMAL TAXI-OUT (TAXI-IN)

Pivot turn radius for a 180?**Not published. The FCTM says avoid braked pivot turns**

Airbus publishes no pivot turn radius. The FCTM 180 degree turn note says to avoid using differential braking to fully stop one main landing gear, the braked pivot turn technique, because of stress and fatigue in the main gear.

FCTM PR-NP-SOP-100 P 6/8 180 Degrees Turn on Runway

"In order to avoid stress and fatigue in the main landing gear, the flight crew should avoid using differential braking to fully stop one main landing gear (braked pivot turn technique)"

If a runway is too narrow for the published width, you back-taxi to a turn pad or ask for a tow, not pivot.

Workbook wording: What is the non-normal (pivot) 180 degree turning radius?

Airport Ops**2.1.3 PERFORM FCOM PRE-TAKEOFF PROCEDURES****Runway change during taxi. What do you do?****Recompute the takeoff data, revise and crosscheck the FMS, flaps, FCU ALT, T.O CONFIG, re-brief, Departure Change checklist**

The FCOM Departure Change procedure: the PM checks the AeroData inputs and sends for a new TPR if required, revises the FMS takeoff data, runway and SID, and the PF crosschecks the revised FMS against the TPR or TLR. Then flaps set, FCU altitude set, T.O CONFIG test, a re-briefing by both, and the Departure Change checklist.

FCOM PRO-NOR-SOP-10B P 1/2 Departure Change

"FINAL TAKEOFF PERF DATA.....RECOMPUTE PM L2 The PM ensures the Aerodata inputs (runway and environmental) are correct and if required, sends for a new TPR."

Watch the slat and flap configuration on the new data. A different runway can mean a different CONF, and the T.O CONFIG test only proves the levers agree with the FMS.

Workbook wording: If given a change of departure runway during the taxi, how is this accomplished prior to takeoff and what items need to be accomplished/reviewed?

Passenger Ops**2.1.3 PERFORM FCOM PRE-TAKEOFF PROCEDURES****Must you stop taxiing if a guest stands up?****No**

Depending on the circumstances, it may be safer to continue to taxi.

FOM 5.3.9 Passengers Standing While Taxiing

"There is no regulatory requirement that the aircraft stop if a passenger gets out of their seat."

Captain judgment call. Stopping on an active taxiway can create a bigger hazard than the standing passenger. Coordinate with the Cabin Crew rather than reflexively braking.

Workbook wording: Are we required to stop the aircraft during taxi if a guest stands up?

STAR/Airport/Arrival Review**6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL****What do you brief off the STAR, approach and taxi charts?****Route and constraints, minimums, missed approach, landing runway and taxi**

The briefing runs threats, plan, then considerations. Off the charts you cover the arrival, approach and approach mode, altitude and speed constraints, minimums, fuel and route to the alternate, the missed approach, the landing runway and its assessment, the exit and taxi, autobrakes, flaps and approach speed.

FOM 5.5.16 Threat-Based Approach Briefing/TPC

"Prior to beginning the descent, the PF initiates the approach briefing. The approach briefing should be interactive, concise, and focused on the big picture."

The pilot monitoring calls threats first, off the Crew Briefing Reference Card and the 10-7 page. Keep it interactive and big picture, not a recital.

Workbook wording: What items should be briefed on the STAR/Approach/Taxi charts?

Diversion Considerations

6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL

What drives a diversion?

Anything degrading a safe landing at destination

If an engine fails or is shut down in flight, land at the nearest suitable airport based on time considerations where a safe landing can be made. Suitability factors include aircraft configuration, weight, systems status and fuel remaining, and wind and weather conditions enroute at the diversion altitude.

FOM 11.2.7 / 11.2.7.1 Engine Shutdown in Flight, Nearest Suitable Airport

"land the aircraft at the nearest suitable airport, based on time considerations, where a safe landing can be made."

Nearest Suitable Airport is FAA-approval-required text, so it is regulatory, not company preference. Nothing justifies flying past the nearest suitable airport after an engine shutdown.

Workbook wording: What factors may precipitate a diversion?

2.1.1.10 KNOW SINGLE ENGINE TAXI PROCEDURES

Where are the single engine taxi procedures?

FCOM PRO-NOR-SOP-21B for arrival, FCTM PR-NP-SOP-102 and 280 for the considerations

FOM 5.3.3 makes single engine taxi the standard procedure when feasible and defers the procedure to the fleet manual. The A330 arrival procedure is FCOM PRO-NOR-SOP-21B One Engine Taxi at Arrival, and the FCTM carries the considerations at PR-NP-SOP-102 for departure and PR-NP-SOP-280 for arrival. The FCTM revision added the departure unit and lets the PM act on the engine master lever.

FCOM PRO-NOR-SOP-21B P 1/2 One Engine Taxi at Arrival

"When taxiing in a straight line: ENG 2 SHUTDOWN..... ORDER PF"

Taxi on engine 1 so the blue system keeps the brake accumulator up; on engine 2 check ACCU PRESS. Open the crossbleed for one engine taxi at departure so both packs run.

Workbook wording: Where can you find the single engine taxi procedures? Review any differences in SOP.

2.1.1.10 KNOW SINGLE ENGINE TAXI PROCEDURES

Engine cooling requirement before shutdown?

At or near idle for 1 minute after high thrust operations

Operate the engines at or near idle for a cooling period of 1 minute before shutdown to thermally stabilize them. Idle reverse and normal taxi thrust do not count as high thrust, so with idle reverse the clock starts at the flare when the levers are retarded, and with max reverse it starts when the levers come back to idle reverse on the rollout. The PM announces COOLING TIME ELAPSED.

FCOM PRO-NOR-SOP-21B P 1/2 One Engine Taxi at Arrival

"The flight crew should operate the engines at or near idle thrust for a cooling period of 1 min before engine shutdown, in order to thermally stabilize the engines."

Routine short cooling degrades the engine. The same 1 minute rule applies to both engines at the gate in the Parking SOP.

Workbook wording: What is the engine cool requirements before shutting down an engine?

2.1.1.10 KNOW SINGLE ENGINE TAXI PROCEDURES

What do you weigh before a single engine taxi?

Systems redundancy, jet blast, fuel imbalance, icing at +1 C or below, active runway crossings, slope, slippery surface, weight

One engine taxi is always the crew's decision. Consider the reduced systems redundancy and the aircraft status, the higher thrust needed and the FOD and jet blast risk, fuel imbalance against the takeoff limit, that it is not permitted in icing conditions at OAT +1 C or below, not recommended across an active runway, and that uphill slope, a slippery taxiway or high gross weight may make it impractical. Taxi on engine 1 for the brake accumulator.

FCTM PR-NP-SOP-102 P 1/2 One Engine Taxi at Departure

“■ One engine taxi is not permitted in icing conditions with OAT at +1 °C (34 °F) or below (due to ice accretion and the required engine acceleration during the ice shedding procedure). ■ One engine taxi is not recommended in the case of crossing an active runway.”

Green policy is not a mandate. If any of these bite, taxi on two engines and say why.

Workbook wording: What factors should you consider before conducting a single engine taxi?

10.1.1.1 COMPLETE PARKING FCOM PROCEDURE WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL**What are the gate arrival requirements?****Two wing walkers plus marshaller or VDGS operator; correct type on VDGS; stop on WAIT or STOP; marshaller overrides VDGS.**

For the A330 the required personnel are two wing walkers and one marshaller or VDGS operator, with station exceptions on the 10-7. Under manual marshalling the signals must stay visible to the Captain the whole way. With a VDGS confirm the correct aircraft type before entering the gate area, see the centerline guidance before the nose passes the jet bridge, and stop on WAIT or STOP. If the marshaller and the VDGS give conflicting guidance, follow the marshaller (Rev 125.3 wording).

FOM 5.6.33 Parking Requirements (manual or VDGS)

“• (787/A330) Two Wing Walkers and One Marshaller/VDGS Operator”

You may stop and ask for wing walkers at any point if clearance is in question. Poor lead-in line visibility at night or in weather justifies requesting manual marshalling.

Workbook wording: Gate Arrival Procedures

10.1.1.1 COMPLETE PARKING FCOM PROCEDURE**What do you check before taxiing into the gate area?****Correct aircraft type displayed on the VDGS**

FOM 5.6.33 gives the VDGS sequence: correct aircraft type displayed before entering the gate area, guidance screen with centerline indicator displayed before the nose passes the jet bridge, and stop on WAIT or STOP. If lead-in line visibility is poor at night or in weather, the crew may request manual marshalling from Station Ops.

FOM 5.6.33 Parking Requirements (manual or VDGS)

“Ensure the correct aircraft type is displayed prior to entering the gate area.”

Also confirm the required personnel are in place, two wing walkers plus the marshaller or VDGS operator for the A330, that the Circle of Safety is clear, and that the 1 minute engine cooling period will be satisfied.

Workbook wording: What should be checked prior to taxiing into the gate area?

10.1.1.1 COMPLETE PARKING FCOM PROCEDURE**No marshaller. Where do you stop?****Stop; marshalling signals must be visible throughout**

FOM 5.6.33 lists one marshaller or VDGS operator and two wing walkers as required personnel for the A330, and under manual procedures requires the marshalling signals to be clearly visible to the Captain at all times. If nobody is marshalling, stop and ask for guidance; the FOM does not name a stop point.

FOM 5.6.33 Parking Requirements (manual or VDGS)

“Manual – for standard manual procedures, the marshalling signals shall be clearly visible to the Captain at all times (see 5.3.12 – Hand Signals).”

The FOM does not name a specific stop point for a missing marshaller.

Workbook wording: If there is no marshaller or if they are not marshalling you in, where should you stop the aircraft?

10.1.2 PERFORM POSTFLIGHT WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL**Perform or discuss the securing procedure****Parking flow first, then the Securing the Aircraft checklist: ADIRS off, external power, lights, batteries**

The Parking SOP runs the shutdown: ACCU pressure, parking brake per brake temperature, anti-ice off, APU bleed, engines off after the 1 minute cool, beacon off when spooled down, fuel pumps off, transponder standby, IRS performance and fuel quantity checks, slides disarmed on the DOOR page, chocks by hand signal, logbook from the STATUS page, then the Parking checklist. Securing the Aircraft follows when the aircraft is left, per the PRO-NOR-SUP-SEC procedure and its checklist.

FCOM PRO-NOR-SOP-22 P 6/6 Parking, Logbook

"STATUS page..... CHECK CM1 L2 Press the STS pb (ECAM control panel). L1 LOGBOOK.....COMPLETE CM1 L2 Complete the logbook according to the STATUS page."

If one brake is above 300 C, 150 C with fans, release the parking brake once chocks are in. Wing lights off before the jet bridge comes on; they can heat-damage it.

Workbook wording: Perform Securing Procedure (Optional/If not performed, discuss)

10.1.2 PERFORM POSTFLIGHT

Who starts the post-flight debrief?

PF initiates it and asks PM for feedback

FOM 5.6.39 says the postflight debriefing should be interactive, concise and focused on relevant items, to learn from experiences and facilitate mentoring. The PF initiates it by asking the PM for positive and/or constructive feedback, referencing the Crew Briefing Reference Card.

FOM 5.6.39 Debrief

"After the completion of each flight, as workload allows, the PF should initiate the postflight debriefing."

The PF initiates regardless of seat, so as Captain you may be debriefed by your FO after their leg. Use the Crew Briefing Reference Card. Separate from FOM 11.1.12 Postflight Debriefing/Flight Irregularities, which is the irregularity reporting path.

Workbook wording: Debrief

11.1.10.4 KNOW AIRPLANE SHUTDOWN AND SECURING PROCEDURES WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

What is the Parking SOP sequence at the gate?

1 minute cool, APU bleed, masters off, beacon off when spooled down, pumps off, slides disarmed, chocks, logbook from STATUS.

Parking: ACCU pressure in the green or chocks before shutdown, parking brake on unless a brake is above 500 C, anti-ice off, APU bleed on and wait for the memo, engines off no less than 1 minute after high thrust, wing lights off before the jet bridge, beacon off once spooled down, fuel pumps off, transponder standby, IRS performance and fuel quantity checks, slides disarmed on the DOOR/OXY page, chocks confirmed by hand signal, logbook completed from the ECAM STATUS page, Parking checklist. Securing the Aircraft follows per PRO-NOR-SUP-SEC when the aircraft is left.

FCOM PRO-NOR-SOP-22 P 3/6 Parking, Exterior Lights

"Turn off the beacon lights when all engines spooled down."

With a hot brake above 300 C, 150 C with fans, release the parking brake after chocks are in. The fuel used plus fuel on board must reconcile with departure fuel or maintenance action is due.

Workbook wording: Know Airplane Shutdown and Securing Procedures

Holding (Procedures, speeds)

MISC MISCELLANEOUS, NOT ASSOCIATED WITH GRADE SHEET TASKS

Where are the US holding speeds?

200 knots below 6000 ft, 230 to 14,000, 265 above

At or below 6000 ft hold at 200 knots or less. Above 6000 ft through 14,000 ft the limit is 230 knots. Above 14,000 ft it is 265 knots. Inbound leg is one minute at or below 14,000 ft and one and a half minutes above.

FOM 5.5.15 Holding

"Altitude Maximum Airspeed Time (Inbound Leg) 6000 ft or below 200 KIAS Above 6000 ft 230 KIAS 1 minute Through 14,000 ft (210 KIAS) Above 14,000 ft 265 KIAS 1 1/2 minutes"*

Charts marked 210K in Alaska or New York hold you to 210 knots above 6000 through 14,000 ft, except the Anchorage and Fairbanks approach control areas. Pattern protection assumes 280 knots or Mach 0.8, whichever is lower.

Workbook wording: Where can you find the holding speeds (FAR) in USA?

Cold Weather and Runway Condition ⁽¹⁵⁾

Cold Weather Ops**2.1.3 PERFORM FCOM PRE-TAKEOFF PROCEDURES**

Active precipitation or contaminated surface: what check before departure regardless of holdover?

A Pre-Takeoff Check from the seat, always, whenever icing conditions exist

If takeoff will occur within the holdover time, a Pre-Takeoff Check must always be done when icing conditions exist. It is done seated in the cockpit and has two parts: a visual check of the representative surfaces, the portion of the wing visible from the flight deck window, and a judgment that the holdover time is still adequate given changes in precipitation type, intensity, temperature and wind.

FCOM PRO-NOR-SUP-ADVWXR P 4/24 Pre-Takeoff Check

"A Pre-Takeoff Check must always be accomplished when icing conditions exist. It is accomplished by the flight crew from the seated position inside the cockpit, and consists of two parts."

The outcome is one of three: go, get deiced and anti-iced again, or perform a Contamination Check.

Workbook wording: While using SureWx app, if/when there is active precipitation or during contaminated surface operations, regardless of the holdover time, what check to we always do prior to departure?

2.1.3 PERFORM FCOM PRE-TAKEOFF PROCEDURES

Holdover time expired. What check before takeoff?

A Contamination Check from the cabin within five minutes of takeoff

If takeoff will occur after the holdover time expires, or in heavy snow, a Contamination Check must be done within five minutes before beginning the takeoff. A pilot assigned to the flight goes to the cabin and visually checks both wings and the engine nacelles for any ice or snow. It may be repeated to meet the five minute window.

FCOM PRO-NOR-SUP-ADVWXR P 4/24 Contamination Check

"A Contamination Check must be accomplished within five minutes prior to beginning takeoff if the holdover time has expired or if heavy snow is occurring."

The FODM Contamination Check Requirements section carries the detail. The PRC Cold Weather Card lists the same two checks.

Workbook wording: While using SureWx app and the holdover time has expired, what check is required prior to takeoff?

2.1.3 PERFORM FCOM PRE-TAKEOFF PROCEDURES

Which manual holds the cold weather and de-icing material?

The FODM, plus fleet-specific manuals

FOM 9.3.1 sends you to the fleet-specific manuals and the FODM for cold weather procedures, and FOM 9.3.1.1 sends you to the FODM again for Liquid Water Equivalent holdover times. FOM 16.2.9 names the ASA Deicing Procedures Manual alongside the FODM.

FOM 9.3.1 Cold Weather Preflight Requirements

"Comply with the cold weather procedures in the fleet-specific manuals and FODM."

Aircraft-side cold weather procedures are in FCOM PRO-NOR-SUP-ADVWXR, Supplementary Procedures, Adverse Weather; FOM 9.3 is the policy layer only.

Workbook wording: In which manual is most of the cold weather/de-icing information found?

2.1.3 PERFORM FCOM PRE-TAKEOFF PROCEDURES

What are your cold weather ops resources?

Deicing manual (FODM), holdover times app, and fleet-specific manuals

Three sources. The deicing manual, FODM, carries the deice and anti-ice procedures and the printed holdover tables. The holdover times app on the electronic flight bag gives the live numbers. The FCOM adverse weather supplementary procedures carry the airplane specific cold weather steps.

FOM 9.3.1 Cold Weather Preflight Requirements

"Comply with the cold weather procedures in the fleet-specific manuals and FODM."

Holdover times built on liquid water equivalent data are FAA authorized and shall be used over METAR or ATIS derived times unless you can see the data is wrong.

Workbook wording: What are your cold weather ops resources? (Manuals, cards etc.)

Contaminated Runway Ops

2.1.3 PERFORM FCOM PRE-TAKEOFF PROCEDURES

Wet vs contaminated runway?

Wet is a water film. Contaminated is beyond that

The FOM carries the operational consequence rather than a bare definition: for a Landing Distance Available of less than 7000 ft, do not use WET data. The wet versus contaminated distinction drives which performance table you are allowed to use, which is why the LDA threshold matters more on the line than the textbook definition.

FOM 9.1.12.3 Wet vs. Contaminated Data

"For LDA less than 7000 ft, do not use WET data."

The full runway state definitions and the wet and contaminated crosswind table for the A330 are in FCOM LIM-AG-OPS, keyed to RWYCC. Related: FOM 9.1.13 Degraded Braking uses the Can U StoP mnemonic.

Workbook wording: What is the difference between a wet vs contaminated runway?

2.1.3 PERFORM FCOM PRE-TAKEOFF PROCEDURES

Braking action report vs RCAM?

PIREP is pilot-reported deceleration; RCAM is an assessment tool

A braking action report is a pilot qualitative PIREP (GOOD, GOOD to MEDIUM, MEDIUM, MEDIUM to POOR, POOR, NIL) and airport personnel cannot issue one. The RCAM is a published FAA or ICAO matrix used primarily by airports to assess conditions and generate a Runway Condition Code, 0 (NIL) through 6 (dry), and correlate it with those same descriptive terms.

FOM 9.1.2 Braking Action Report / FOM 9.1.6 Runway Condition Assessment Matrix (RCAM)

"A braking action report (PIREP) is a qualitative description of deceleration by pilots such as GOOD, GOOD to MEDIUM, MEDIUM, MEDIUM to POOR, POOR, and NIL."

Key operational rule in FOM 9.1.2: a PIREP may downgrade expected braking action from the RCAM or FICON, but must never be used to upgrade it.

Workbook wording: What's the difference between a braking action report vs RCAM?

2.1.3 PERFORM FCOM PRE-TAKEOFF PROCEDURES

What is a FICON NOTAM?

NOTAM giving runway condition codes by runway third

A FICON reports condition for touchdown, midpoint, and rollout in the direction of takeoff and landing, listed as runway thirds such as 3/4/2, with an expected braking action expressed as a Runway Condition Code. Where a single report gives multiple conditions along the runway, assume the most restrictive for the whole runway.

FOM 9.1.3 Field Condition (FICON) NOTAM

"A FICON is a specifically formatted NOTAM reporting runway condition for each third of the runway, and giving an expected braking action as a Runway Condition Code (RWYCC)."

Absence of a FICON does not assure the runway is dry; airports are encouraged but not required to issue one for a wet runway.

Workbook wording: What is a FICON (Field Condition) NOTAM?

2.1.3 PERFORM FCOM PRE-TAKEOFF PROCEDURES**What is a runway CC (Runway Condition Code)?****A 0 to 6 code for expected braking action, 0 NIL through 6 dry**

Runway Condition Code values run from 0, NIL, to 6, dry. The RCAM ties each code to the descriptive terms GOOD through NIL. Airport operators use the RCAM to generate the RWYCC for the runways in use, and the FICON reports it per runway third.

FOM 9.1.6.1 Runway Condition Code (RWYCC)

"Runway Condition Code values range from 0 (NIL) to 6 (dry)."

For the A330 the FOM sends you to the FCOM In-Flight Performance chapter for the RCAM in the fleet format, including the wind limits that depend on runway condition.

Cold Temperature Altimeter Correction/VNAV Temperature Limitation**6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL****When and why do you apply cold temperature corrections on approach?****When OAT is well below ISA, because true altitude is lower than indicated; charts flag when required**

Below ISA the true altitude is lower than indicated, and the error grows with cold and matters most where terrain separation is minimum. In the US the chart briefing strip carries a note when cold temperature corrections apply. The FOM sends you to the fleet manual for the procedure. The A330 SOP: consider altitude corrections for low temperature during descent preparation, and note that managed vertical guidance must not be used below the chart minimum temperature and FINAL APP may not engage when corrections are required.

FOM 9.3.3.1 Cold Temperature Altimeter Corrections/Cold Temperature Restricted Airports

"When the temperature is lower than International Standard Atmosphere (ISA), true altitude is lower than indicated altitude. This temperature effect on indicated altitude becomes extremely important with very cold temperatures and minimum terrain separation."

For an RNAV(GNSS) approach with LNAV/VNAV minima the FMS profile ignores low OAT: vertical managed guidance must not be used below the chart minimum temperature (PRO-NOR-SOP-18-C).

Workbook wording: When/why do you make cold weather corrections while on approach?

De/Anti-Ice Procedures**MISC MISCELLANEOUS, NOT ASSOCIATED WITH GRADE SHEET TASKS****Definition of icing conditions?****Ground OAT or in-flight TAT 10 C or less with visible moisture; or ground contamination at 10 C or less.**

Icing conditions exist when the OAT on the ground or after takeoff, or the TAT in flight, is at or below 10 C and visible moisture in any form is present: clouds, fog with visibility 1 sm or less, rain, snow, sleet or ice crystals. They also exist when the OAT on the ground and for takeoff is at or below 10 C and the aircraft operates on surfaces where snow, standing water or slush may be ingested or freeze on the engines, nacelles or probes.

FCOM LIM-ICE_RAIN P 1/2 Definition of Icing Conditions

"Icing conditions exist when the OAT (on ground or after takeoff) or the TAT (in flight) is at or below 10 °C and visible moisture in any form is present (such as clouds, fog with visibility of 1 sm (1 600 m) or less, rain, snow, sleet or ice crystals)."

Severe ice accretion is about 5 mm, 0.2 in, on the airframe. Thin hoarfrost is thin enough to read markings through.

Workbook wording: What is definition of icing conditions?

MISC MISCELLANEOUS, NOT ASSOCIATED WITH GRADE SHEET TASKS

Where are the aircraft de-ice and anti-ice procedures?**FCOM PRO-NOR-SUP-ADWXR, Adverse Weather, with the FODM for the ground deicing rules**

The A330 supplementary procedures for adverse weather carry the cold weather work: the ground deicing and anti-icing procedure, the pre-takeoff check, the contamination check, engine ice shedding on ground, and the cold soak items. The FCOM refers to the Flight Operations Deicing Manual for the pre-takeoff check and contamination check requirements. The PRC Cold Weather and Deicing cards summarize both.

FCOM PRO-NOR-SUP-ADWXR P 4/24 Pre-Takeoff Check

"For more information, refer to Flight Operations Deicing Manual (FODM) "Pre-Takeoff Check Requirements."

The FOM policy layer is chapter 9.3; the airplane steps are in the FCOM supplementary procedures, not the FOM.

Workbook wording: Where do you find the aircraft de/anti-Ice procedures?

MISC MISCELLANEOUS, NOT ASSOCIATED WITH GRADE SHEET TASKS

Where are airport specific deice procedures?**FODM and the ASA Deicing Procedures Manual**

FOM 16.2.9 states plainly that all deice procedures are found in the ASA Deicing Procedures Manual and the FODM, with the Deicing Procedures Manual available through Maintenance Control.

FOM 16.2.9 Deice - Charters

"All deice procedures are found in the ASA Deicing Procedures Manual and the FODM ."

Verified only to this level: the FOM contains no airport-specific deice pages, and states all deice procedures live in the FODM (on the EFB) and the ASA Deicing Procedures Manual, which is obtained or reviewed through Maintenance Control.

Workbook wording: Where do you find specific airport deice procedures?

MISC MISCELLANEOUS, NOT ASSOCIATED WITH GRADE SHEET TASKS

When must engine anti-ice be on?**Whenever icing conditions exist or are anticipated, except in climb and cruise with SAT below -40 C**

Engine anti-ice must be ON during all ground operations when icing conditions exist or are anticipated, and in flight when icing conditions exist or are anticipated, except during climb and cruise when the SAT is below -40 C. The caution in every phase: turn it on in icing conditions and do not wait to see ice build up.

FCOM PRO-NOR-SOP-13 P 1/2 After Takeoff, Anti Ice

"Engine anti-ice must be set to ON when icing conditions exist or are anticipated, except during climb and cruise when the SAT is below -40 °C (-40 °F)."

On the ground engine anti-ice raises ground idle, so watch taxi speed on wet or slippery surfaces. Descent and approach have no -40 exemption.

Workbook wording: When are you required to turn the engine anti-ice on?

MISC MISCELLANEOUS, NOT ASSOCIATED WITH GRADE SHEET TASKS

When must wing anti-ice be on?**With any evidence of ice accretion: visual indicator, wipers, or SEVERE ICE DETECTED; may be used earlier to prevent it.**

In icing conditions the crew may turn on wing anti-ice to prevent ice on the wing leading edge, and must turn it on when there is evidence of accretion such as ice on the visual indicator or the wipers or the SEVERE ICE DETECTED alert, to remove the accumulation. On approach it goes OFF for landing unless severe icing.

FCOM PRO-NOR-SOP-13 P 1/2 After Takeoff, Anti Ice

“The flight crew must turn on the wing anti-ice if there is evidence of ice accretion, such as ice on the visual indicator, or on the wipers, or with the SEVERE ICE DETECTED alert. This is to remove any ice accumulation from the wing leading edge.”

ICE NOT DET appears in green if no ice is detected for 130 s after you switch it on. Wing anti-ice inhibits the bleed temperature reduction and can trigger BLEED LO TEMP.

Workbook wording: When are you required to turn on the wing de/anti-ice

MISC MISCELLANEOUS, NOT ASSOCIATED WITH GRADE SHEET TASKS

Where are cold weather procedures?**Weather chapter, the fleet manuals, and the FODM deicing manual**

The FOM does not list cold weather steps. It sends you to the fleet books and to the FODM, which carries the deicing procedures, holdover times and the post-deicing check. For the A330 the procedures are in FCOM PRO-NOR-SUP-ADVWXR, Supplementary Procedures, Adverse Weather, summarized on the PRC Cold Weather and Deicing cards.

FOM 9.3 Cold Weather Operations

“Comply with the cold weather procedures in the fleet-specific manuals and FODM.”

The Weather chapter still owns the policy side: icing intensity definitions, adverse weather restrictions and liquid water equivalent holdover times. Cold weather approach restrictions and altitude corrections point back to the fleet manuals.

Workbook wording: Where can you find the cold weather procedures in the FOM? Any other manual?

Takeoff and Departure (26)

1.3.2 PERFORM FCOM PROCEDURES PRIOR TO ENGINE START WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

Review the engine failure after V1 procedure**Continue, rotate 3 degrees per second to 12.5 degrees, fly the SRS and beta target, secure the engine, then accelerate**

If an engine fails after V1 the takeoff continues. Rudder conventionally on the ground, rotate at about 3 degrees per second to 12.5 degrees, then follow SRS which targets the speed at failure, between V2 and V2+15. Center the beta target with rudder, trim it out, engage the AP. Fly the trajectory first, delay the acceleration only to secure the engine, then level at the engine-out acceleration altitude and accelerate.

FCTM PR-AEP-ENG P 10/26 Engine Failure After V1

“At VR, the flight crew should rotate the aircraft using a continuous pitch rate of approximately 3 °/s towards an initial pitch attitude of 12.5 °. The combination of high FLEX temperatures and low VR speeds requires precise handling during the rotation and liftoff.”

Secure the engine means run the ECAM to ENG MASTER OFF, or AGENT 1 DISCH with damage, or fire out or AGENT 2 DISCH for a fire, before you accelerate. A derated takeoff carries a warning: TOGA below F speed can cause loss of control.

Workbook wording: Review Engine failure procedure

1.3.2 PERFORM FCOM PROCEDURES PRIOR TO ENGINE START

Where is the engine failure procedure for a specific runway?**On the TPR or TLR, as the Engine Out Departure Procedure; check the EOSID on the ND**

The Final Takeoff Data check has both pilots confirm the Engine Out Departure Procedure printed on the TPR or TLR. The FMS side is the EOSID, shown as the yellow line on the ND in plan mode, and it is checked during the flight plan check.

FCOM PRO-NOR-SOP-07 P 3/8 Final Takeoff Data

“Takeoff Runway / Intersection ■ Engine Out Departure Procedure ■ Air Conditioning on/off ■ V speeds”

To review the EOSID detail, select it as a temporary flight plan, read it, then erase it (PRO-NOR-SRP-01-10 flight plan check).

Workbook wording: Where can you find the EFP for a specific runway? Anywhere else?

1.3.1 PERFORM FOM TAKEOFF AND DEPARTURE BRIEFING INCLUDING IDENTIFICATION OF OPERATIONAL THREATS THREAT BASED BRIEFING/TPC

What is the main concept of the TPC briefing?

Threats jointly, then the PF plan, then considerations

TPC replaces a read-the-plate monologue with a threat-led, interactive exchange. The PM opens by naming relevant threats, the PF adds any others, then the PF briefs the plan and the considerations. It should be interactive, concise and focused on the big picture, with management strategies discussed as threats surface.

FOM 5.5.16 Threat-Based Approach Briefing/TPC

"The approach briefing should be interactive, concise, and focused on the big picture."

Captain-side: an interactive briefing is where you set flight-deck tone. If the FO never speaks during your brief, the T was skipped and you have lost the main safety value of the format.

Workbook wording: Discuss the main concept of the TPC briefing

1.3.1 PERFORM FOM TAKEOFF AND DEPARTURE BRIEFING INCLUDING IDENTIFICATION OF OPERATIONAL THREATS THREAT BASED BRIEFING/TPC

Who starts the briefing?

The PF, but the PM speaks first on threats

Within the format the PM identifies relevant threats first, then the PF briefs any additional threats followed by the plan and considerations.

FOM 5.5.16 Threat-Based Approach Briefing/TPC

"Prior to beginning the descent, the PF initiates the approach briefing."

1.3.1 PERFORM FOM TAKEOFF AND DEPARTURE BRIEFING INCLUDING IDENTIFICATION OF OPERATIONAL THREATS THREAT BASED BRIEFING/TPC

Where is the briefing guidance?

Yes. Threats, Plan, Considerations departure briefing before pushback

The pilot flying starts it before pushback, and it runs Threats, Plan, Considerations. The pilot monitoring names the relevant threats first, using the Crew Briefing Reference Card and the airport 10-7 page. Then the pilot flying briefs taxi and departure runway, performance and configuration, route, and the return plan.

FOM 5.2.7 Threat-Based Departure Briefing

"Prior to pushback, the PF initiates the departure briefing. The departure briefing should be interactive, concise, and focused on the big picture."

Considerations is where you assign specific pilot monitoring duties tied to the threats you just named. Recap the critical threats at the end.

Workbook wording: Do we have any guidance regarding this briefing? Where can it be found?

Airport Ops

2.1.3 PERFORM FCOM PRE-TAKEOFF PROCEDURES

Can you depart at night with runway edge lights out?

Yes. Only with visibility at least 1/4 sm or RVR 1600 and FODO permission.

If ALL runway lights are inoperative, takeoff is not authorized during darkness unless both conditions are met: reported visibility greater than or equal to 1/4 sm or RVR 1600, and permission of the Flight Operations Duty Officer.

FOM 5.4.9 Runway Lights Inoperative

"Permission of the Flight Operations Duty Officer (FODO)"

Two different tests depending on all versus partial. The FODO call is only required for ALL lights inoperative. This is a Captain decision item that people commonly get wrong by assuming any light outage grounds the departure.

Workbook wording: Can we depart at night from a runway where the edge lights are inoperative?

Crosswind Control**CRITICAL OBSERVABLE ITEM****3.1.1 PERFORM TAKEOFF****Maximum takeoff crosswind?****32 kt gust included, certified, dry; less on wet or contaminated runways**

The maximum certified crosswind for takeoff on the A330 PAX fleet is 32 kt gust included, an AFM engine limitation. On wet and contaminated runways the FCOM table drops it by runway condition code: 32 kt at RWYCC 5, 27 kt at 4, 20 kt at 3 and 2, and 15 kt on ice at RWYCC 1. Maximum tailwind for takeoff is 15 kt.

FCOM LIM-AG-OPS P 2/4 Crosswind Takeoff and Landing - A330 PAX Fleet

"Maximum certified crosswind for takeoff..... 32 kt (gust included)"

The landing figure is different: 45 kt demonstrated, not a limitation. The wet and contaminated table applies to both, and it is what AeroData uses.

Workbook wording: Max takeoff crosswind limitations?

Gear/Flap Management**CRITICAL OBSERVABLE ITEM****3.1.1 PERFORM TAKEOFF****Maximum altitude with slats or flaps extended?****20 000 ft**

FCOM Limitations, Flight Controls: the maximum operating altitude with slats and or flaps extended is 20 000 ft.

FCOM LIM-F_CTL P 1/2 Maximum Operating Altitude with Slats and/or Flaps Extended

"Maximum operating altitude with slats and/or flaps extended.....20 000 ft"

Holding with CONF 1 is limited to 240 kt, and the FCTM cold weather card says hold clean when able.

Workbook wording: Max altitude with flaps extended?

Rotation Rate**CRITICAL OBSERVABLE ITEM****3.1.1 PERFORM TAKEOFF****Normal rotation rate?****About 3 degrees per second, liftoff near 10 degrees after 4 to 5 seconds**

A positive backward stick input starts the rotation and the crew holds a rate of approximately 3 degrees per second for a continuous pitch increase. Liftoff occurs at about 10 degrees, typically 4 to 5 seconds after starting, then the PF targets the required pitch. A slow rotation or under-rotation lengthens the takeoff run and reduces obstacle clearance.

FCTM PR-NP-SOP-120 P 2/14 Rotation Technique

"the flight crew achieves a rotation rate of approximately 3 °/s resulting in a continuous pitch increase. During the rotation, the aircraft liftoff occurs at approximately 10 ° of pitch, typically around 4 to 5 s after the initiation of the rotation."

Monitor the rotation with outside references, then fly the pitch target on the PFD once airborne. An abrupt rate increase near liftoff is the tail strike setup.

Workbook wording: Normal rotation rate? (degrees/sec)

Pitch Attitude**CRITICAL OBSERVABLE ITEM****3.1.1 PERFORM TAKEOFF****Normal takeoff pitch target?****15 degrees, then follow the SRS bar**

At VR, initiate the rotation with a positive sidestick input to a continuous rate of about 3 degrees per second towards 15 degrees of pitch, 12.5 degrees with an engine failed. Minimize lateral inputs to avoid spoiler extension, and after liftoff follow the SRS pitch command bar.

FCOM PRO-NOR-SOP-12 P 3/6 Takeoff, Rotation

“■ At VR, initiate the rotation with a positive sidestick input to achieve a continuous rotation rate of about 3 °/s, towards a pitch attitude of 15 ° (12.5 °, if one engine is failed)”

Deviation callout on takeoff climb: PITCH if greater than 20 degrees up or less than 10 degrees up (PRO-NOR-SCO).

Workbook wording: What is the normal takeoff rotation target pitch?

CRITICAL OBSERVABLE ITEM

3.1.1 PERFORM TAKEOFF

Pitch target after an engine failure at or after V1?

12.5 degrees, then SRS

With one engine failed the rotation target is 12.5 degrees at about 3 degrees per second. The FCTM explains the 12.5 degree target ensures the aircraft becomes airborne despite high FLEX temperatures and low VR speeds. Once safely airborne follow SRS, which targets the speed at failure, no lower than V2.

FCTM PR-AEP-ENG P 10/26 Engine Failure After V1

“The 12.5 ° pitch target will ensure the aircraft becomes airborne.”

If the failure occurs above V2, maintain the SRS commanded attitude; minimum speed is always V2 (FCTM engine failure during initial climb).

Workbook wording: What is the takeoff rotation target pitch attitude after an engine failure at/after V1 and prior to liftoff?

Callouts

CRITICAL OBSERVABLE ITEM

3.1.1 PERFORM TAKEOFF

What are the callouts during a rejected takeoff?

Captain calls STOP, or GO to continue; STOP also transfers control

The decision and the stop action belong to the Captain and must come before V1, hand on the thrust levers until V1 whether PF or PM. For a malfunction before V1 the Captain calls GO to continue, or STOP to reject. STOP both confirms the rejection and states the Captain now has control; it is the only handover not accompanied by I have control. The PM rollout checks follow the FCTM RTO task-sharing figure.

FCTM PR-AEP-MISC P 25/38 Rejected Takeoff, Decision Callouts

“■ If a decision is made to reject the takeoff, the Captain calls “STOP”. This call both confirms the decision to reject the takeoff and also states that the Captain now has control.”

Above 100 kt be go-minded: fire or severe damage, sudden thrust loss, unambiguous evidence the aircraft will not fly, or any ECAM alert. EGT red line or nose gear vibration are not reasons above 100 kt.

Workbook wording: What are the PM callouts during an RTO?

Noise Abatement Departure Procedures (NADP)

4.1.1 PERFORM DEPARTURE AND CLIMB

What is NADP1?

The close-in noise abatement profile

NADP procedures are governed by Ops Spec C068. Specific requirements, including gradients and altitudes for configuration changes, may be defined in aircraft-specific manuals, the 10-7, or the Jeppesen Airway Manual Flight Procedures Noise Abatement Procedures.

FOM 5.4.7 Noise Abatement Departure Procedure (NADP)

"Normal takeoff procedures satisfy either close-in or distant NADP profile requirements."

That quote is (AS) bannered. This matters for the JCAB ramp-check question in section 1.2.3 about whether HA may perform NADP 1: the OpSpec is C068. confirm the HA-specific position and the OpSpec paperwork the JCAB inspector wants to see.

Workbook wording: What is NADP1? Describe associated procedures

4.1.1 PERFORM DEPARTURE AND CLIMB**What is NADP2?****The distant noise abatement profile**

NADP2 is the distant profile. Per FOM 5.4.7, normal takeoff procedures satisfy either close-in or distant NADP profile requirements, so no separate profile is flown unless a specific published procedure requires it. Gradients and configuration-change altitudes come from the aircraft-specific manuals, the 10-7, or the Jeppesen Airway Manual.

FOM 5.4.7 Noise Abatement Departure Procedure (NADP)

"Flight Crews must comply with published noise abatement procedures when operating from noise-sensitive airports"

Where thrust reduction and acceleration occur relative to the airport is what separates the two profiles. The A330 flies the THR RED and ACC ALT set on the PERF TAKEOFF page, defaulted from the takeoff data.

Workbook wording: What is NADP2? Describe associated procedures

Climb Airspeed / Mach Control**4.1.1 PERFORM DEPARTURE AND CLIMB****How do you find max rate of climb speed?****Between ECON climb and green dot; rule of thumb, use turbulence speed**

The FCTM says the speed for maximum rate of climb, to reach a given altitude in the shortest time, lies between ECON climb speed and green dot. It is not shown on the PFD, so the rule of thumb is to use the turbulence penetration speed. Select it on the FCU when airborne, or pre-select it on the PERF CLB page on the ground.

FCTM PR-NP-SOP-140 P 3/4 Climb, Selected

"The speed to achieve the maximum rate of climb, i.e. to reach a given altitude in the shortest time, lies between ECON climb speed and green dot. As there is no indication of this speed on the PFD, a good rule of thumb is to use turbulence speed to achieve maximum rate."

Selected speed spoils the optimum profile, so use it only when ATC or weather requires it.

Workbook wording: How do you determine max rate of climb speed?

4.1.1 PERFORM DEPARTURE AND CLIMB**Severe turbulence penetration speed?****0.80 Mach at FL370 and above, 260 kt FL250 to 350, 240 kt at FL200 and below**

The QRH SEVERE TURBULENCE table gives the speed and N1 by flight level and weight: Mach 0.8 from FL370 up, 260 kt from FL250 to FL350, and 240 kt at FL200 and below. Seat belts on, keep the autopilot on, disconnect A/THR only for excessive thrust variations, and consider descending to or below optimum.

QRH 22.08A SEVERE TURBULENCE

"SEVERE TURBULENCE Ident.: ABN-22-00012057.0030001 / 16 MAY 23 Applicable to: ALL SEAT BELTS.....ON SPEED and THRUST..... ADJUST"

For the approach the QRH wants A/THR on and managed speed. The FCTM also uses turbulence speed as the max rate of climb rule of thumb.

Workbook wording: What is severe turbulence penetration speed?

4.1.1 PERFORM DEPARTURE AND CLIMB**Where is the max angle of climb speed?****Green dot, shown on the PFD; PERF CLB page gives time and distance at it**

The speed for the maximum gradient of climb, altitude in the shortest distance, is green dot. The MCDU PERF CLB page displays the time and distance to reach the selected altitude climbing at green dot. Avoid reducing to green dot at high altitude and heavy weight because the reacceleration to ECON Mach takes a long time.

FCTM PR-NP-SOP-140 P 3/4 Climb, Selected

"The speed to achieve the maximum gradient of climb, i.e. to reach a given altitude in a shortest distance, is green dot."

A speed below green dot can be selected but gives no operational benefit.

Workbook wording: Where can you find max angle climb speed?

SID Departure

4.1.1 PERFORM DEPARTURE AND CLIMB

What else gets briefed on a SID beyond legs and restrictions?

Taxi and runway, takeoff performance, route, return plan

Before pushback the pilot flying briefs the plan: taxi and departure runway, takeoff performance and configuration, route with the ATC clearance compared against the programmed route, and the return plan for an emergency or takeoff alternate. Pilot monitoring calls the threats first, and considerations close it out.

FOM 5.2.7 Threat-Based Departure Briefing

"Taxi, departure runway – Takeoff performance/configuration – Route (ATC clearance vs. programmed route)"

Keep it interactive and big picture. Threats come from the pilot monitoring using the Crew Briefing Reference Card and any airport specific items on the 10-7 page.

Workbook wording: Though the SID legs/restrictions get briefed from the plan view vs the CDU/FMS, what other items should be briefed from the SID plate?

4.1.1 PERFORM DEPARTURE AND CLIMB

Is a SID for specific runways only?

No published source located, ask the check airman

Runway applicability is a Jeppesen chart property. Neither the FOM nor the A330 manuals define it. The FCOM cockpit preparation does require the PM to check each waypoint and constraint against the Jeppesen SID, the release and the ATC clearance.

FCOM PRO-NOR-SOP-06 P 26/28 F-PLN check

"Follow along with the PF by checking each waypoint and constraint against the Jeppesen SID, Dispatch release, and current ATC clearance."

No published source located, ask the check airman. Read the runway box on the SID plate.

Workbook wording: Is the SID for a specific runway(s) only?

4.1.1 PERFORM DEPARTURE AND CLIMB

Obstruction climb gradient restrictions?

From runway analysis data, checked before every takeoff

Climb gradient and obstacle numbers come off the runway analysis, which has to be available and checked for every takeoff. It rides in the Flight Information Packet or on the Dispatch Release. If the data is invalid, get new takeoff data by datalink or from Dispatch.

FOM 5.4.5 Runway Analysis

"Runway analysis data must be available and checked for each takeoff."

If an engine fails on takeoff from a runway that has a published engine failure procedure, fly that procedure. That is what guarantees the obstacle clearance, not raw climb performance.

Workbook wording: Obstruction climb gradient restrictions.

4.1.1 PERFORM DEPARTURE AND CLIMB

Red boxed notes on the SID, such as speed restrictions?

No published source located, ask the check airman

Chart symbology is Jeppesen material, not manual text. The FOM departure briefing framework is where the notes get covered: the PM names the threats first from the Crew Briefing Reference Card and the airport 10-7, then the PF briefs the plan.

FOM 5.2.7 Threat-Based Departure Briefing/TPC

No published source located, ask the check airman. Jeppesen chart legend in FD Pro covers the boxed notes.

Workbook wording: Red Boxed notes e.g., speed restriction until further advised, etc.

4.1.1 PERFORM DEPARTURE AND CLIMB

SID top altitude on a climb via clearance?

No published source located, ask the check airman

The FOM uses climb via only in the Mexico theater, where the controller expects the pilot to fly the published lateral routing and altitudes, and it notes Central America does not use the phrase at all. The US definition of top altitude is AIM material, not FOM.

FOM 21.3.3.5 Mexico Phraseology for Climbs and Descents

"Typically, the controller will issue a "Descend Via" or "Climb Via" clearance to keep the radio transmission short and concise."

No published source located, ask the check airman. Know the top altitude on every SID you brief.

Workbook wording: SID top altitude (if given "CLIMB VIA" from ATC)?

4.1.1 PERFORM DEPARTURE AND CLIMB

ATC says climb and maintain. Do SID restrictions still apply?

Yes, until ATC explicitly cancels them

Published restrictions stay in force. A controller cancels them with explicit words like cancel level restriction, climb unrestricted, or disregard SID altitude restrictions. Without that phrase, keep flying the published vertical profile up to the newly assigned altitude. If the clearance is ambiguous, ask.

FOM 23.16.2 SID and STAR Altitude Restrictions

"European controllers explicitly cancel intermediate altitude restrictions using standardized phraseology such as "Cancel level restriction," "Climb unrestricted to [level]," or "Climb to [level], disregard SID/STAR altitude restrictions" when these restrictions on a SID or STAR are no longer necessary."

The FOM says the same for Central America, where an altitude clearance still requires adherence to charted altitude restrictions unless you are specifically relieved.

Workbook wording: While on the SID, if ATC issues a "Climb and Maintain XXX" clearance, does the flight crew still need to comply with altitude constraints on the SID?

4.1.1 PERFORM DEPARTURE AND CLIMB

Transition altitude and transition level?

18,000 ft MSL in the US, charted elsewhere

Below the transition altitude you set QNH and state altitudes in thousands and hundreds. At and above it you set 29.92 and say flight level. Outside the US the transition altitude is charted and the transition level comes from the ATIS or the approach controller.

FOM 7.1.2.3 Flight Altitudes

"Up to, but not including the transition altitude (18,000 ft MSL in the US and some other areas), state the separate digits of the thousands plus the hundreds, (e.)"

Central America runs transition level FL200 and transition altitude 19,000 ft MSL unless charted otherwise. Europe is generally lower than the US, so read the chart every time.

Workbook wording: Transition Altitude/LVL?

Manual Aircraft Control

4.1.1 PERFORM DEPARTURE AND CLIMB

Minimum altitude to engage the autopilot after takeoff?

100 ft AGL and at least 5 seconds after liftoff

FCOM Limitations, Autopilot Function: the autopilot can be used at takeoff from 100 ft AGL and at least 5 s after liftoff. After a manual go-around it is 100 ft AGL. In all other phases the minimum is 500 ft AGL. The takeoff SOP adds that AP must be engaged for RNP AR below 0.3 NM.

FCOM LIM-AFS-10 P 2/8 Autopilot Function

"At takeoff 100 ft AGL and at least 5 s after liftoff"

Trim first. With an engine failed the FCTM wants the aircraft trimmed with the beta target centered before you engage the AP, and pedal pressure held after engagement can disengage it.

Workbook wording: Minimum Altitude for autopilot engagement?

4.1.1 PERFORM DEPARTURE AND CLIMB

Is the autopilot required on an RNAV SID?

No

FOM 5.1.4.1 is the only autopilot requirement tied to RNAV/RNP procedures. It applies to RNP AR with RNP less than 0.3 or Radius to Fix legs, where autopilot or flight director driven by the RNAV system shall be used. RNP AR departure capability must be authorized by the fleet-specific manuals (FOM 5.1.4.3).

FOM 5.1.4.1 RNP Authorization Required (RNP AR) Procedures

"RNP AR procedures with RNP values less than 0.3 or with Radius to Fix (RF) legs shall use autopilot or Flight Director driven by the RNAV system."

Only RNP AR procedures with RNP below 0.3 or RF legs require autopilot or flight director driven by the RNAV system. Nothing in the FOM, FCOM, or QRH requires the autopilot on a standard RNAV SID.

Workbook wording: Is the autopilot required for an RNAV SID?

Cruise and Diversion (23)

CRITICAL OBSERVABLE ITEM

5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE

Review PA usage

No PA in critical phases except emergency announcements

FOM 7.1.18 keeps the PA out of critical air and ground phases except for emergencies, because scanning and flight deck duties come first; delay routine announcements until established in cruise. FOM 7.1.19 lists the announcements the crew is required to make, and crews may vary the wording except where the manual says verbatim.

FOM 7.1.18 Public Address System

"Flight Crewmembers should not use the Public Address System during any critical air or ground phase of flight for other than emergency announcements."

Required PAs are listed in FOM 7.1.19 (seat belt sign on/off, anticipated turbulence, turbulence, stopping short of the jetway).

Customer service PA guidance and sample scripts are in FOM 7.1.21.

Workbook wording: Review PA usage (see 12.1.3.16 Know FOM PA policies and Perform Effective Communication from the flight deck)

CRITICAL OBSERVABLE ITEM

5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE

Where in Comply365 is the AIREP form?

No published source located, ask the check airman

No FOM, FCOM or PRC text names an AIREP form or its Comply365 location. The PRC ORCN / ETOPS Reference Card carries the ten-item non-CPDLC position report format, which includes the temperature, wind and ride report elements of

a meteorological report.

PRC p30 ORCN / ETOPS Reference Card, Position Report (non-CPDLC)

"8 - Temp: "Minus [SAT]"

No published source located, ask the check airman. Confirm the Comply365 path with the check airman.

Workbook wording: Where in Comply365 can the AIREP form be found?

CRITICAL OBSERVABLE ITEM

5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE

Must you complete the AIREP form on every flight?

No published source located, ask the check airman

No FOM requirement for an AIREP form was located. The FOM does require meteorological reports with oceanic position reports and reports of adverse weather to Dispatch and ATC.

FOM 7.1.12 Reports to Dispatch

"Relay adverse weather reports to Dispatch and Air Traffic Control."

No published source located, ask the check airman

Workbook wording: Do you need to complete the AIREP form for each flight?

CRITICAL OBSERVABLE ITEM

5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE

Must you save the AIREP form after each flight?

No published source located, ask the check airman

No retention requirement for an AIREP form appears in the FOM.

FOM 7.1.12 Reports to Dispatch

No published source located, ask the check airman

Workbook wording: Do you need to save the AIREP form after each flight?

Company Reporting

CRITICAL OBSERVABLE ITEM

5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE

Are company position reports required?

Yes, and holding always requires one

FOM 7.1.12 Reports to Dispatch carries the requirement set. One that is explicit: through ACARS or radio the Captain will ensure Dispatch is contacted with the holding fix, Expected Further Clearance time, and fuel on board, and diversion plans should be discussed if applicable. Abnormalities require notification under 11.1.2.

FOM 7.1.12 Reports to Dispatch / 5.5.15.1 Dispatch Notification of Holding

"the Captain will ensure that Dispatch is contacted with the holding fix, Expected Further Clearance (EFC) time, and fuel on board."

Three items for the holding report: fix, EFC, fuel. read 7.1.12 in full for the complete routine position-reporting requirement.

Workbook wording: Are we required to make company position reports?

CPDLC Logon Procedures

CRITICAL OBSERVABLE ITEM

5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE

Departing the lower 48: how early do you log on for a CPDLC departure clearance?

During preflight: ETD minus 30 minutes with DCL, minus 5 without, logon KUSA

The PRC CPDLC card: in CONUS log on to KUSA during preflight, at ETD minus 30 minutes if the airport has Departure Clearance service, ETD minus 5 minutes if it does not. International logons are by FIR: climbing through 10 000 ft for most Pacific FIRs, 15 to 45 minutes prior to the oceanic FIR for KZAK and RJJJ.

PRC p38 Controller Pilot Data Link Communication (CPDLC)

"CONUS (lower 48) During Preflight KUSA i ETD—30' if DCL i ETD—5' if no DCL"

Manual entry of the SID with its runway and enroute transition is required when loading the DCL. Load and review any route change in the SEC F-PLN before you accept.

Workbook wording: When departing the 48 Contiguous states, to assist ATC, how many minutes prior to departure is an appropriate time to attain a departure clearance via CPDLC LOGON?

Cruise Altitude Considerations**CRITICAL OBSERVABLE ITEM****5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE****OPT versus REC MAX altitude?**

OPT: most economic level for cost index and weight; REC MAX: highest level the aircraft can hold with margins.

OPT FL is the most economic flight level for the cost index, weight and weather data, a compromise between fuel and time, shown on the PROG page and updated through cruise. REC MAX is the lowest of the altitudes at which the aircraft keeps a 0.3 g buffet margin, can fly level at MAX CRZ, can hold 300 ft/min at MAX CLB, stays between green dot and VMO/MMO, and is certified. Anti-ice is not included in REC MAX.

FCOM DSC-22_20-10-40-30 P 10/16 Recommended Maximum Altitude (REC MAX)

"The recommended maximum altitude is the lowest of the maximum altitude that: ■ The aircraft can reach with a 0.3 g buffet margin ■ The aircraft can fly in level flight at MAX CRZ rating ■ The aircraft can maintain a V/S of 300 ft/min at MAX CLB thrust"

The FCTM says PROG page MAX REC and OPT are what you use to answer CAN YOU CLIMB TO FL XXX. With an engine out the PROG page shows EO REC MAX instead.

Workbook wording: What is the difference between OPT/Max/Recommended altitudes?

FOM Electronic Plotting Chart Procedure**CRITICAL OBSERVABLE ITEM****5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE****When is an electronic plot required?****On all ORCN segments**

An electronic plotting chart should be used on all Oceanic and Remote Continental Navigation segments. The Flight Crew should load the entire cleared route into the approved charting services app, such as FD Pro X, on the EFB.

FOM 5.7.3.15 Electronic Plotting Chart

"An electronic plotting chart should be used on all ORCN segments."

Regulatory basis AC 91-70 and Ops Spec B036. Note the word is should, and the requirement is the ENTIRE cleared route, not just the oceanic portion.

Workbook wording: When and under what conditions are we required to do an electronic plot?

CRITICAL OBSERVABLE ITEM**5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE****How do you plot on the EFB?****Load the entire cleared route into the charting app**

The Flight Crew should load the entire cleared route into the approved charting services app on the EFB. FD Pro X is the app named in the FOM.

FOM 5.7.3.15 Electronic Plotting Chart

"load the entire cleared route into the approved charting services app (such as FD Pro X) on the EFB."

The FD Pro X screen-by-screen plotting workflow is vendor documentation, not FOM content. Confirm with the FD Pro Pubs Quick Reference guide in Comply365.

Workbook wording: How do you plot using the EFB?

CRITICAL OBSERVABLE ITEM**5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE****Must you save or submit plots after every flight?****No published requirement to save or submit**

FOM 5.7.3.15, written against AC 91-70 and Ops Spec B036, requires an electronic plotting chart on all ORCN segments and requires the full cleared route be loaded into the approved charting services app on the EFB. It stops there. Nothing in the section, or in the surrounding oceanic material, requires the crew to save, export or submit the plot afterward.

FOM 5.7.3.15 Electronic Plotting Chart

"An electronic plotting chart should be used on all ORCN segments."

5.7.3.15 requires the plot be used and the entire cleared route loaded into the approved charting app such as FD Pro X, but it sets no retention, save or submission requirement. Screenshotting a plot is personal practice, not policy.

Workbook wording: Are you required to save/submit your plots and route after every flight?

CRITICAL OBSERVABLE ITEM**5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE****When would you save a screenshot of your plotted route?****No published source located, ask the check airman**

FOM 5.7.3.15 requires the electronic plotting chart on ORCN segments and the full cleared route loaded into the charting app. It sets no save or screenshot requirement.

FOM 5.7.3.15 Electronic Plotting Chart

No published source located, ask the check airman. A screenshot after a deviation or a position doubtful event is common practice, not policy.

Workbook wording: When would it be prudent to save your plotted route (take screenshot postflight)?

SATCOM / HF Procedures**CRITICAL OBSERVABLE ITEM****5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE****Total VHF and HF failure. Any other way to reach ATC?****Yes. SATCOM voice, and CPDLC datalink**

Satellite voice is the fallback: it is not authorized for routine ATC use, but the FOM specifically permits it when HF or CPDLC are not available (FOM 7.1.7), and all areas of the route structure have satellite voice coverage. CPDLC over SATCOM datalink is the other path, and it is the primary means in oceanic airspace when paired with ADS-C.

FOM 7.1.7 SATCOM (if installed)

"Routine satellite voice communication with ATC is not authorized; however, it may be used when HF or CPDLC are not available."

If it is a datalink failure rather than voice, FOM 5.7.4.9 says revert to HF position reporting and tell ATC. For contingencies, FOM 5.7.4.2 also has you alert nearby aircraft on 121.5, backup 123.45, for relay.

Workbook wording: If you have a total VHF/HF radio failure, is there another way to contact ATC?

CRITICAL OBSERVABLE ITEM**5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE****Where are the current Pacific HF frequencies?****Jeppesen FD Pro (ARINC Services) and the enroute chart communication boxes; ARINC gives the day's primary and secondary**

FOM 20.4.3 sends you to Jeppesen FD Pro for oceanic communication procedures and frequencies (search ARINC, ARINC Services). FOM 20.4.3.3 adds that HF frequencies are in the communications boxes on the Enroute Navigation Charts and that ARINC provides the primary and secondary frequencies predicted for the day. Rev 125.3 deleted FOM 6.2.7 HF Frequency Assignments, which named the JetPack HF Frequencies tab and radio.arinc.net/pacific; neither is in the FOM any more.

FOM 20.4.3 Non-CPDLC Oceanic Communication / 20.4.3.3 Frequency Selection

"Communication procedures and frequencies can be found in Jeppesen FD Pro."

ARINC gives you the primary and secondary predicted for that day at check in. If neither works, try any listed frequency, and try several times on one before changing.

Workbook wording: Where can you find the most current (pacific) HF FREQUENCIES in use for your flight?

CRITICAL OBSERVABLE ITEM**5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE****Rule of thumb for HF frequencies and the sun?****Not published; reception varies with time of day, and ARINC gives the day's primary and secondary**

The FOM does not state the daytime-high, nighttime-low rule. It says HF reception varies greatly with time of day, atmospheric variations, sunspots and distance, that ARINC provides the primary and secondary frequencies predicted for the day, and that you may need to try several listed frequencies, trying several times on one before changing.

FOM 20.4.3.3 Frequency Selection

"HF reception varies greatly with time of day, atmospheric variations, sunspots, and distance. It is normal for HF radio to be unable to receive on all the listed frequencies at all times. The primary and secondary frequencies predicted to be in use for the day are provided by ARINC."

Frequencies print in the communications boxes on the enroute charts. Ionospheric disturbance from solar activity hits HF hardest near the poles (FOM 9.7.2.1).

Workbook wording: What is the rule of thumb with HF frequencies vs the Sun?

Assess Diversion and Alternate Airport Weather Conditions and Minima**CRITICAL OBSERVABLE ITEM****5.1.2.6 KNOW CONTINGENCY PLAN (I.E., ALTERNATES, WEATHER MONITORING, FUEL, ETC.)****How do you get updated enroute and descent winds?****WIND/TEMP REQUEST at 2R on the CRUISE or DESCENT WIND page, an ACARS uplink**

On the wind pages, pressing WIND/TEMP REQUEST at 2R sends a request for ACARS winds and temperature. The DES WIND page is reached from NEXT PHASE on the CRUISE WIND page or the WIND prompt on the VERT REV page and takes winds at five altitudes. The descent preparation SOP has the PF check the DES WIND page and enter winds from cruise level down.

FCOM DSC-22_20-20-10-25 P 13/154 Descent Wind Page

"[2R] WIND/TEMP Pressing this key sends a request for ACARS winds and temperature. REQUEST (Refer to DSC-22_20-30-80 Wind Data - Request for Wind Data)"

During cruise DIR TO is not available while uplinked wind data sits uninserted on the CRUISE WIND page. Insert or cancel it.

Workbook wording: How (aircraft specific) can you get updated enroute/descent winds?

Discuss Route Diversion scenario**CRITICAL OBSERVABLE ITEM****5.1.2.6 KNOW CONTINGENCY PLAN (I.E., ALTERNATES, WEATHER MONITORING, FUEL, ETC.) WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL****Talk through a route diversion****Complete the ECAM or QRH, decide whether the failure requires diversion, weigh continuing against diverting, coordinate with SOCC**

The PRC Diversion Guidance card: complete the non-normal procedure, then ask whether the failure requires diversion. Failures that do: anything requiring landing such as LAND ASAP, fire or an engine failure, insufficient fuel after a component failure raises consumption, or one generator remaining. Otherwise weigh whether the risks of continuing exceed the risks of diverting, and coordinate the divert with SOCC. Amber or green ECAM messages and any single failure other than an engine have no impact on ETOPS.

FCOM PRO-SPO-40A P 5/8 Diversion Decision Making (PRC p2 Diversion Guidance)

"Failure cases requiring a diversion to the nearest airport (cases leading to a LAND ASAP message on the ECAM and/or in the QRH) ■ Failure cases resulting in increased fuel consumption, exceeding the available fuel reserves ■ Electrical generation. Diversion is required in the case of: • Only one generator (either one IDG, APU GEN or EMER GEN) remaining available following a multiple failure"

Then the mechanics: NEW DEST in the FMS, weather and landing performance, fuel prediction, Dispatch and MedLink if medical, and the NTSB brief to the cabin.

Workbook wording: Discuss Route Diversion scenario

CRITICAL OBSERVABLE ITEM

13.1.1.1.1 KNOW ENGINE FAILURE IN CRUISE PROCEDURE

Engine failure over the western US or other terrain critical regions. What is the plan?**Fly the release's terrain clearance plan: zero net gradient or driftdown segment to the listed airport**

The Enroute Terrain Clearance Program limits takeoff weight so the aircraft clears enroute terrain with an engine failed, and the plan is described on the Flight Plan. A Zero Net Gradient Segment keeps a net climb gradient 1000 ft above terrain; a Driftdown Segment clears terrain by 2000 ft on the routing to the indicated airport from the point of failure. HA releases carry the remark TERRAIN MORA CHECK PERFORMED, ONE ENG OUT DRIFTDOWN, PASSED. The PRC Engine Failure During Cruise card gives the flying: MCT, A/THR off, green dot, FL200 initially, then REC MAX EO.

FOM 8.3.7 Enroute Terrain Clearance Program (ETCP)

"A Driftdown Segment (Method 2) is one along which the aircraft can, with one engine inoperative, clear terrain by 2000 ft as it proceeds to the indicated airport. The plan indicates the routing to the chosen airport from the point of engine failure that ensures terrain clearance."

For the obstacle strategy the PRC says hold green dot until clear of obstacles. Depressurization escape routes are a separate product, the decompression polygons in FOM 11.2.9.

Workbook wording: What are the Engine Failure and diversion procedures while flying over the western part of the 48 contiguous states? Other terrain-critical regions in the world?

Diversion Procedures/Select New Destination

8.3.4.26 DEMONSTRATE ABILITY TO CORRECTLY PROGRAM AIRCRAFT GUIDANCE SYSTEMS INCLUDING APPROACH

How do you change to a new destination in the FMS?**LAT REV at a waypoint, type the new destination into NEW DEST, then INSERT**

From the F-PLN page take a lateral revision at a waypoint other than the FROM and TO, type the new destination into the scratchpad and put it in the NEW DEST field, then insert the temporary flight plan. The approach SOP uses the same field when the F-PLN has lost its destination after a missed approach: perform a LAT REV at the last waypoint and redefine the destination in NEW DEST.

FCOM PRO-NOR-SOP-18-A P 5/6 Approach General

"Perform a LAT REV at the last waypoint and redefine the destination in the NEW DEST field."

The PRC FMS Guide has the keystrokes under NEW DESTINATION (DIVERT). Load the diversion runway and approach next, then the weather and landing performance request.

Workbook wording: How (aircraft specific) do you "Change" or "Select" a new destination?

Alternate Navigation System

8.3.4.26 DEMONSTRATE ABILITY TO CORRECTLY PROGRAM AIRCRAFT GUIDANCE SYSTEMS INCLUDING APPROACH

What is backup navigation?**MCDU NAV B/UP: the MCDU navigates from its stored flight plan and the onside IRS when both FMGECs fail**

The MCDU continuously memorizes the active flight plan downloaded from the FMGEC. If both FMGECs fail, the crew selects the back up navigation mode on the MCDU MENU page and recovers flight planning, aircraft position from the onside IRS or IRS 3, the F-PLN on the ND with automatic sequencing, limited lateral revision, but no AP or FD NAV

mode. MCDU 3 cannot do it.

FCOM DSC-22_10-30 P 6/6 Back Up Navigation Mode

"If both FMGECs fail, the back up navigation provides the following functions: ■ Flight Planning ■ Aircraft position using onside IRS or IRS 3 ■ F-PLN display on ND ■ No AP/FD NAV mode ■ Limited lateral revision ■ F-PLN automatic sequencing."

It is also the answer to a permanent MAP NOT AVAIL on both NDs after a failed FM reset. The PRC Oceanic References card counts one FMGEC plus one BACK UP NAV as meeting RNP 4 and RNP 10 entry equipment.

Workbook wording: What is Alternate Navigation?

8.3.4.26 DEMONSTRATE ABILITY TO CORRECTLY PROGRAM AIRCRAFT GUIDANCE SYSTEMS INCLUDING APPROACH

How do you navigate with backup navigation to a point or destination?

NAV B/UP on the MCDU MENU; fly stored F-PLN in selected modes using bearing/distance to the TO waypoint.

Activate NAV B/UP from the MCDU MENU page with the FM source selector in NORM. It gives you the memorized F-PLN, its display on the ND, automatic sequencing, AP and FD selected modes if one FG remains, limited lateral revisions, magnetic or true bearing and distance from the aircraft to the TO waypoint, true track between waypoints, time estimates from the onside IRS ground speed, and total time and distance to destination.

FCOM DSC-22_20-40-20 P 1/10 MCDU Back Up Navigation, General

"■ MAG (True) bearing depending on the pilot selection, from aircraft position to the TO WPT and associated distance ■ True track between waypoints ■ Time estimates computed with current GS from onside IRS ■ Total time and distance to destination"

Fly heading or track to the bearing it gives you; there is no managed NAV. With NAV B/UP on both sides at least one FG must be available to engage AP and A/THR.

Workbook wording: How do you perform Alternate Navigation to a point or destination?

16.1.3.2 ANALYZE AND ASSESS TERRAIN CRITICAL ROUTES

Western US depressurization strategy?

Fly the charted decompression polygon escape procedure

FOM 11.2.9 covers the western high-terrain depressurization strategy: polygons show where a direct descent to 10,000 ft is not possible before passenger oxygen depletes.

FOM 11.2.9.1 Decompression Polygon Procedures

"Decompression polygon procedures provide a calculated safe method for descending to 10,000 ft before passenger oxygen supplies are depleted following a rapid decompression over high terrain."

Polygons live on the Jeppesen FD Pro IFR High/Low charts; click the polygon label to open the detail drawer. The three escape types are Exit Any Dir, Exit Twd <fix>, and To Fix then Route. Rev 125.3 renumbered the polygon procedures to 11.2.9.1 under a new 11.2.9 Decompression section: plan to be at 10,000 ft before passenger oxygen is depleted.

Workbook wording: Western U.S. Depressurization Strategy

8.3.5.23 [PM] DEMONSTRATE ABILITY TO CORRECTLY PROGRAM AIRCRAFT GUIDANCE SYSTEMS INCLUDING APPROACH PROCEDURE SELECTION, WAYPOINT CODING, NAVIGATIONAL PERFORMANCE, AND TRACK OFFSETS. WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

PM: select a new destination in the FMS

LAT REV at a waypoint, NEW DEST, insert; then runway, approach, weather and landing data

As PM you build the diversion: lateral revision at a waypoint other than FROM and TO, new destination into NEW DEST, insert the temporary flight plan, then load the runway and approach, request the weather and the landing performance, and update the fuel prediction. The PRC Diversion Guidance card frames the decision: complete the ECAM or QRH procedure, then weigh the risk of continuing against diverting and coordinate with SOCC.

FCOM PRO-NOR-SOP-18-A P 5/6 Approach General

"Perform a LAT REV at the last waypoint and redefine the destination in the NEW DEST field."

Failures that require a diversion per the PRC: LAND ASAP, fire, engine failure, insufficient fuel after a component failure, or one generator remaining.

Workbook wording: Diversion Procedures/Select New Destination

8.3.5.23 [PM] DEMONSTRATE ABILITY TO CORRECTLY PROGRAM AIRCRAFT GUIDANCE SYSTEMS INCLUDING APPROACH PROCEDURE SELECTION, WAYPOINT CODING, NAVIGATIONAL PERFORMANCE, AND TRACK OFFSETS. WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

PM: demonstrate the backup navigation system

MCDU MENU, NAV B/UP, with FM source selector in NORM; fly selected modes to the stored F-PLN

Select NAV B/UP on the MCDU MENU page with the FM source selector at NORM. The MCDU links to its IRS and shows the memorized F-PLN on the ND with sequencing, bearing and distance to the TO waypoint, and time estimates from IRS ground speed. Use AP or FD selected modes if one FG remains. It can be selected temporarily on one side after a single FM failure to prove it is available on that side.

FCOM DSC-22_20-40-20 P 1/10 MCDU Back Up Navigation, General

"The MCDU NAV B/UP is to be used in case of FM 1 + 2 failure. It can be selected temporarily in case of FM 1 or 2 only failure, in order to ensure that the function is available on the failed side."

MCDU 3 cannot run back up navigation even when it replaces MCDU 1 or 2.

Workbook wording: Alternate Navigation System

ETOPS (23)

Dispatch Weather Minimums

1.1.2.9 KNOW WEATHER MINIMA & LIMITATIONS FOR DISPATCH, TAKEOFF, APPROACH, AND LANDING

ETOPS alternate weather before departure?

C055 minima, earliest to latest landing time

The Dispatcher will not list an airport as an ETOPS alternate unless weather is forecast at or above the ETOPS alternate minima in Ops Spec C055 (see 8.2.2). That must hold for the whole window it might be used, earliest to latest possible landing time. Field condition reports must indicate a safe landing can be made.

FOM 6.2.2 ETOPS Alternate Airport

"Field condition reports indicate that a safe landing can be made."

Each designated ETOPS alternate must have RFFS equivalent to ICAO Category 4 or higher, with a 30-minute response time, which means equipment available on arrival of the diverting airplane, not equipment parked there in advance. Rev 125.3: airports listed in 8.1.3 Authorized Airports meet the 30-minute requirement.

Workbook wording: Weather required for ETOPS alternate airport(s) prior to departure?

Enroute Weather Minimums

1.1.2.9 KNOW WEATHER MINIMA & LIMITATIONS FOR DISPATCH, TAKEOFF, APPROACH, AND LANDING

ETOPS alternate weather in flight?

Crew operating minima

From after takeoff until approaching the EEP, alternate weather must be at or above the crew's operating minima. That means what the operating crew needs to fly the approach, given their minima and any applicable MEL. The alternate must remain suitable to conduct an instrument approach.

FOM 6.3.5 In-Flight ETOPS Alternate Airport Weather Requirements

"Note that these weather requirements are less restrictive than the criteria used by Dispatch in planning."

This is a favourite oral question because the answer changes at the EEP. If a METAR for an ETOPS alternate cannot be obtained after the EEP, a new ETOPS alternate must be designated.

Workbook wording: Weather required for ETOPS alternate airport(s) in flight?

ETOPS Verification Flight**16.1.1.7.6 ETOPS VERIFICATION FLIGHT PROCEDURE****Where are the ETOPS verification flight procedures?****Flight Operations Manual, ETOPS chapter, Verification Flight section**

The Hawaiian procedure sits in the FOM ETOPS chapter under ETOPS Verification Flight, with the Alaska system verification flight section just ahead of it. Maintenance notifies you and makes a logbook entry. You report the result to Dispatch between 30 and 60 minutes after takeoff.

FOM 6.5.10 ETOPS Verification Flight (HA)

"An ETOPS Verification Flight (EVF) is a revenue or non-revenue flight during which the Flight Crew is asked to assist Maintenance to confirm that an ETOPS Significant System is operating normally or that an ETOPS significant problem has been resolved."

That report and the Dispatch acknowledgment must be complete before the ETOPS entry point. If the item is not operating normally you cannot go past it, and the Captain writes the result in the logbook.

Workbook wording: Where can the ETOPS verification flight procedures be found?

16.1.1.7.6 ETOPS VERIFICATION FLIGHT PROCEDURE**Any Dispatch call required after takeoff, before the EEP?****Yes. 30 to 60 minutes after takeoff**

Between 30 and 60 minutes after takeoff and prior to reaching the EEP, the Flight Crew will call or send an ACARS message to Dispatch to report the status of the affected component or system.

FOM 6.5.10 ETOPS Verification Flight (HA)

"This report and the return acknowledgment from Dispatch must be complete before reaching the EEP."

Two-way requirement. Your report alone is not enough; Dispatch's acknowledgment must also be received before the EEP.

Workbook wording: Is there a communication requirement to Dispatch between takeoff and prior to the EEP?

16.1.1.7.6 ETOPS VERIFICATION FLIGHT PROCEDURE**How do you send the pre-EEP component status report?****Phone call or ACARS**

The Flight Crew will call or send an ACARS message to Dispatch to report the status of the affected component or system, between 30 and 60 minutes after takeoff and prior to the EEP.

FOM 6.5.10 ETOPS Verification Flight (HA)

"the Flight Crew will call or send an ACARS message to Dispatch to report the status of the affected component or system."

Workbook wording: How can/should you make that communication of system performance?

16.1.1.7.6 ETOPS VERIFICATION FLIGHT PROCEDURE**Component fails before the EEP. Can you continue?****No**

If the item being verified is not operating normally, the aircraft cannot proceed beyond the EEP and must land for repairs.

FOM 6.5.10 ETOPS Verification Flight (HA)

"the aircraft cannot proceed beyond the EEP and must land for repairs."

Unambiguous. The verification failing before the EEP removes ETOPS entry as an option.

Workbook wording: If the component being verified fails prior to EEP can you continue?

16.1.1.7.6 ETOPS VERIFICATION FLIGHT PROCEDURE**Component fails after the EEP. Can you continue?****No published source located, ask the check airman**

FOM 6.5.10 says the aircraft cannot proceed beyond the EEP if the item being verified is not operating normally. It does not address a failure after the EEP. Treat a post-EEP failure as an in-flight system failure under FOM 6.4.4 and the

diversion logic in 6.4.1, with Dispatch.

FOM 6.5.10 ETOPS Verification Flight (HA)

"the aircraft cannot proceed beyond the EEP and must land for repairs."

No published source located, ask the check airman. Have the contingency plan agreed with Dispatch and Maintenance Control before departure, as the FOM verification flight section recommends.

Workbook wording: If the component being verified fails after EEP entry can you continue?

ETOPS APU Verification Flight

16.1.1.7.6 ETOPS VERIFICATION FLIGHT PROCEDURE

What is the APU verification procedure?

Cold soak 2 hours, start within 1 hour of top of descent

Either stay clear of ETOPS airspace, or complete the verification before the ETOPS entry point if you can hold cruise altitude two hours or more beforehand. Cold soak the APU at least two hours shut down at cruise, then start it within one hour of top of descent.

FOM 6.5.10 ETOPS APU Verification Flight

"Initiate the APU start within 1 hour of top of descent."

The cold soak requires the APU shut down at least 2 hours at a normal ETOPS cruising altitude, ideally between FL270 and FL350. That altitude band is the point of the test: it proves a cold start at altitude.

Workbook wording: What are the APU verification flight procedures?

16.1.1.7.6 ETOPS VERIFICATION FLIGHT PROCEDURE

How does APU verification differ from a single component EVF?

A scheduled cold soak and start, not a status report

A single-component EVF is a monitor-and-report task: confirm the system works, report to Dispatch between 30 and 60 minutes after takeoff and before the EEP, and land if it is not normal.

FOM 6.5.10 ETOPS Verification Flight (HA) and ETOPS APU Verification Flight

"Allow the APU to cold soak for at least two hours."

If the APU verification is unsuccessful, the Dispatcher and PIC must confer to determine the best course of action, as ETOPS entry may not be possible. That is a joint decision, unlike the single-component case where the EEP rule is automatic.

Workbook wording: How do those procedures differ from an ETOPS Verification (single component) flight?

16.1.1.7.6 ETOPS VERIFICATION FLIGHT PROCEDURE

How many APU start attempts?

Three attempts maximum

FOM 6.5.6 sets up the APU In-Flight Start Program. After engine start or takeoff the APU is shut down and cold soaked at least two hours at ETOPS cruise altitude. The crew then attempts a start during the 1-hour window before top of descent.

FOM 6.5.6 APU In-Flight Start Program (HA)

"• If the APU starts within the maximum allowed three attempts, the start is deemed to be successful."

Subsequent attempts after a failed first attempt may be made at lower altitudes, minimum FL270. Do not confuse this with the ground start rules.

Workbook wording: How many start attempts?

16.1.1.7.6 ETOPS VERIFICATION FLIGHT PROCEDURE

Must the APU be shut down before descent?

No. It may be shut down, not required

Under FOM 6.5.6 the test ends once the APU finishes its start sequence or after three failed attempts, and the APU may then be shut down; it is permissive, not required.

FOM 6.5.6 APU In-Flight Start Program (HA)

"Once the APU finishes its start sequence or after three failed start attempts, the test is complete, and the APU may be shut down."

FOM 6.5.10 ETOPS APU Verification Flight sets the same start window (within 1 hour of top of descent) and likewise imposes no shutdown requirement before descent.

Workbook wording: After APU start at altitude, is it required to shut the APU down before descent?

16.1.1.7.6 ETOPS VERIFICATION FLIGHT PROCEDURE

APU shut down before descent, then fails to start after landing. Does the verification stand?

No published source located, ask the check airman

FOM 6.5.10 judges the APU verification on the in-flight cold-soak start within one hour of top of descent and requires the Captain to detail the result in the logbook. The FOM does not say whether a later ground start failure changes that result.

FOM 6.5.10 ETOPS APU Verification Flight

"After an APU verification attempt, the Captain must detail the results in an Aircraft Maintenance Logbook entry."

No published source located, ask the check airman. Write up the ground start failure as its own discrepancy and let Maintenance Control decide.

Workbook wording: If APU is shutdown prior to top of descent, then subsequently the APU fails to start after landing, how/does that affect the verification portion of the flight? Successful, with a separate write up, or does the verification fail?

Assess Diversion and Alternate Airport Weather Conditions and Minima**CRITICAL OBSERVABLE ITEM****5.1.2.6 KNOW CONTINGENCY PLAN (I.E., ALTERNATES, WEATHER MONITORING, FUEL, ETC.)**

How do you get current METAR and TAF oceanic?

Ask Dispatch by datalink, satellite phone or high frequency radio

Dispatch is your weather shop once you are off very high frequency radio range. They send updates by datalink, satellite phone or high frequency radio, and they must relay anything that would keep you out of an ETOPS alternate. Ask for the airport and the time window you care about.

FOM 6.3.1 Dispatcher Duties

"Relay via ACARS, SATCOM, or HF any significant changes in airport status, forecasted weather conditions, or any other conditions that would preclude a safe approach and landing."

The alternates page in the flight computer also has a weather request prompt that sends a datalink request for alternate airport weather.

Workbook wording: How can you get the most current METAR/TAF when in oceanic airspace?

CRITICAL OBSERVABLE ITEM**5.1.2.6 KNOW CONTINGENCY PLAN (I.E., ALTERNATES, WEATHER MONITORING, FUEL, ETC.)**

Who monitors destination and alternate weather?

You and Dispatch, in parallel

The FOM frames this as joint PIC and Dispatcher responsibility rather than a fixed interval.

FOM 6.3.5 In-Flight ETOPS Alternate Airport Weather Requirements

"Dispatch will update ETOPS alternates if any conditions preclude safe approach and landing at existing alternates."

No fixed interval is published. The practical standard is that you should never be surprised by destination or alternate weather at top of descent. Tie to the descent gouge: ATIS, Build, Bug, Brakes, Brief, RNP/Recall.

Workbook wording: How often should you be checking the weather at your destination and alternates?

Fuel Management

6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL

What is the ETOPS critical fuel scenario?

Worst-case diversion fuel required from the critical point

The critical fuel scenario is the most fuel-critical of three diversion cases from the critical point to the ETOPS alternate. The cases: immediate descent to 10,000 ft with both engines (depressurization), descent to best single-engine altitude with one engine out, and immediate descent to 10,000 ft with one engine out (FOM 6.2.5).

FOM 6.2.5 Critical Fuel Planning

"ETOPS (AS) FUEL REQD OVER CP (HA) CRITF is the most critical fuel scenario from the critical point of the flight (CP/ETP to the diversion airport) chosen from the fuel scenarios outlined above."

It is a dispatch requirement, not an inflight limit (FOM 6.2.6 area, around FOM 6.4). If before the EEP it looks like actual FOB at the CP/ETP will be below the critical fuel requirement, coordinate with Dispatch.

Workbook wording: What is the ETOPS Critical Fuel Scenario?

APU In-Flight Start Program (APU component currency)

CRITICAL OBSERVABLE ITEM

16.1.1.7 PERFORM ETOPS PROCEDURES

EVF vs the routine APU in-flight start program?

EVF confirms a specific repair; the start program is routine currency

FOM 6.5.10 defines an EVF as a flight where the crew assists Maintenance in confirming an ETOPS Significant System works normally, or that an ETOPS significant problem has been fixed. Maintenance makes an AML release entry which the crew must verbally acknowledge with Dispatch.

FOM 6.5.10 ETOPS Verification Flight (HA) vs FOM 6.5.6 APU In-Flight Start Program (HA)

"An ETOPS Verification Flight (EVF) is a revenue or non-revenue flight during which the Flight Crew is asked to assist Maintenance to confirm that an ETOPS Significant System is operating normally or that an ETOPS significant problem has been resolved."

EVF adds a required report to Dispatch between 30 and 60 minutes after takeoff and before the EEP; if the item is not operating normally the flight may not proceed past the EEP. The APU In-Flight Start Program has no such Dispatch report requirement.

Workbook wording: How does this differ from the APU In-Flight Start Program?

CRITICAL OBSERVABLE ITEM

16.1.1.7 PERFORM ETOPS PROCEDURES

How does the release tell you which APU procedure applies?

ACI on the release, MIC sheet and placard: In-Flight Start Program; ETOPS release logbook entry: verification flight.

The routine APU In-Flight Start Program arrives as an Administrative Control Item from MEL section 05, reflected on the Dispatch Release, with an open item on the MIC sheet and the HAL-46-1333 placard card in the logbook. An ETOPS Verification Flight is a repair follow-up: MIC sheet item, verbal notification, the logbook entry AIRCRAFT RELEASED TO ETOPS SERVICE for verification flight, and your verbal acknowledgment with Dispatch.

FOM 6.5.6 APU In-Flight Start Program (HA)

"The Flight Crew will be notified of an APU start request through an open item on the MIC sheet and an aircraft placard card (form HAL-46-1333) in the Aircraft Maintenance Logbook."

Check for a conflicting MEL first. If an MEL requires the APU to run the whole flight, the MEL wins and there is no in-flight start test.

Workbook wording: Using the Dispatch Release, how can the Flight Crew determine which procedure to use (APU Verification Flight vs. APU In-Flight Start Program)?

CRITICAL OBSERVABLE ITEM

16.1.1.7 PERFORM ETOPS PROCEDURES

What do you log during the APU start?

Flight level, outside air temperature, attempts, result, seconds to available.

The Captain writes the result in the logbook in a set format: flight level and outside air temperature at the attempt, the number of attempts, the word successful or unsuccessful, and the seconds from start sequence to APU available. Successful means it started within three attempts. List any discrepancies if it failed.

FOM 6.5.6 APU In-Flight Start Program

“APU IN-FLIGHT START FL ____ (flight level during start attempt) OAT ____ °C (Outside Air Temperature)”

Shut the APU down after start or takeoff, let it cold soak at least two hours at cruise, then attempt the start in the hour before top of descent.

Workbook wording: What items must be logged/noted during the APU start?

CRITICAL OBSERVABLE ITEM

16.1.1.7 PERFORM ETOPS PROCEDURES

How many start attempts are allowed?

Three

FOM 6.5.6 allows a maximum of three start attempts; a start within three attempts is successful. The first attempt is made during the 1-hour period prior to top of descent after at least a two-hour cold soak at ETOPS cruise altitude, and follow-on attempts may be flown at lower altitude but no lower than FL270.

FOM 6.5.6 APU In-Flight Start Program (HA)

“• If the APU starts within the maximum allowed three attempts, the start is deemed to be successful.”

If the first attempt fails, subsequent attempts may be made at lower altitudes, minimum FL270. FOM 5.1.8.1 uses the same three-attempt rule to define an APU start failure.

Workbook wording: How many start attempts can be made?

Flight Crew actions prior to ETOPS entry point (EEP)?

CRITICAL OBSERVABLE ITEM

16.1.1.7 PERFORM ETOPS PROCEDURES

Crossing the EEP. What must be complete?

ETOPS alternate forecast at crew minima; complete ETOPS verification checks

FOM 6.3.2 lists what is required before crossing the EEP. For the A330 the applicable items are forecast weather at the ETOPS alternate at or above the operating crew's minima, accounting for crew minima and any MEL, and completion of any ETOPS verification flight checks. After the EEP monitor distance from the ETOPS alternate; at the EXP and before top of descent complete any APU verification or in-flight start steps.

FOM 6.3.2 Flight Crew Duties, Prior to Crossing ETOPS Entry Point (EEP)

“• (787/A321/A330) Complete any ETOPS verification flight checks.”

Applicable items are CAT II and CAT III, RNAV, GPS, RNP and RNP AR currency, plus autoland and APU in flight start when Maintenance asks. Check the release for the request before you plan the leg.

Workbook wording: What steps are required?

CRITICAL OBSERVABLE ITEM

16.1.1.7 PERFORM ETOPS PROCEDURES

How is the EEP determined and shown to the crew?

60 minutes from an adequate airport at .82M/290 KIAS, 408 nm; airports listed on the release

Per FOM 6.3.3 the EEP is the point 60 minutes from an adequate airport at the approved one-engine-inoperative cruise speed in still air under standard conditions. For the A330-200 that is 408 nm at .82M/290 KIAS from the FOM 6.2.3 table. The EXP is the same fixed distance leaving the area. The adequate airports used must be listed on the Dispatch Release and crew and Dispatch must use the same ones.

FOM 6.3.3 ETOPS Entry/Exit Point/Equal Time Point

“(HA) The adequate airports used to define EEP and EXP during dispatch planning must be listed on the Dispatch Release.”

Enter and monitor the ETOPS waypoints with fleet procedures and crosscheck them against the route or the range rings. Identify arrival at EEP, ETP and EXP aloud.

Workbook wording: How is the EEP determined and depicted to the flight crew?

CRITICAL OBSERVABLE ITEM**16.1.1.7 PERFORM ETOPS PROCEDURES****Max diversion distance for ETOPS 120 and 180?****A330-200: 803 nm at 120 minutes, 1200 nm at 180 minutes**

Table 6.2.3(1) in FOM 6.2.3 lists the A330-200 at .82M/290 KIAS single engine speed: 408 nm for 60 minutes, 803 nm for 120 minutes and 1200 nm for 180 minutes. These are Maximum Diversion Distances from the approved one-engine-inoperative cruise speed.

FOM 6.2.3 ETOPS Area of Operation, Table 6.2.3(1)

"A330-200 .82M/290 KIAS 408 nm 803 nm 1200 nm"

A reroute in an ETOPS segment must be coordinated with Dispatch so you do not fly beyond the MDD. The MDT is a planning number, not an operational time limit for the diversion itself.

Workbook wording: What is the maximum diversion distance, ETOPS 120? ETOPS 180?

CRITICAL OBSERVABLE ITEM**16.1.1.7 PERFORM ETOPS PROCEDURES****ETOPS diversion speed?****.82M/290 KIAS, the approved one-engine-inoperative cruise speed**

FOM 6.4.5 ties the diversion speed to the approved One-Engine-Inoperative Cruise Speed in FOM 6.2.8 (renumbered from 6.2.9 in Rev 125.3), which for the A330 is .82M/290 KIAS at a 470k lb reference weight and FL350. The critical fuel scenario, the EEP and EXP distances and the MDD are all built on it. The PRC engine failure card (8/31/26) no longer repeats it: for ETOPS and standard strategy it now says set 300 kt or M 0.78; the FOM 125.3 table still reads .82M/290 KIAS.

FOM 6.2.8 One-Engine-Inoperative Cruise Speed

"A330 .82M/290 KIAS 470,000 350"

The Captain may deviate from the planned speed profile after assessing the emergency and fuel remaining. ETOPS and standard strategy on the PRC card (8/31/26) are 300 kt or M 0.78, obstacle strategy green dot; confirm which speed Dispatch planned with.

Workbook wording: ETOPS Diversion speed?

Oceanic (26)**Dispatch Release: Review all sections****CRITICAL OBSERVABLE ITEM****1.1.1 PERFORM FLIGHT PLANNING****Named vs unnamed waypoint?****Named is in the database. Unnamed is lat/long or hand entered**

Non-defined or user-named waypoints are waypoints that are either not in the aircraft database, unnamed such as a latitude/longitude like 26N170W, or manually entered by the crew. These may require specific verification procedures, and the FOM sends you to fleet-specific manuals for the route verification procedure.

FOM 5.2.11.5 Non-Defined or Unnamed Waypoints

"waypoints that are either not in the aircraft database, unnamed (a latitude/longitude, such as "26N170W" or "26N70"), or a waypoint that the Flight Crew manually enters."

Oceanic tie-in: FOM 22.3.4 Unnamed Waypoint Verification carries the NAT-specific check.

Workbook wording: What is the difference between a named vs non-defined/unnamed waypoint?

RVSM Requirements

CRITICAL OBSERVABLE ITEM

5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE

What altitudes are RVSM airspace?

FL290 through FL410

RVSM airspace is any airspace or route where aircraft are separated by 1000 ft vertically, between FL290 and FL410 inclusive.

FOM 5.5.6.5 Enroute RVSM Altitude Monitoring Requirements

"between FL290 and FL410, inclusive), pilots shall verify that all altimeters are set to 29."

At initial level off, pilots shall verify all altimeters are set to 29.92 and crosscheck them. Use of the standby altimeter as a primary is PROHIBITED in RVSM airspace.

Workbook wording: What altitudes define RVSM airspace?

CRITICAL OBSERVABLE ITEM

5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE

RVSM equipment requirements?

Both primary altimeters, one autopilot, altitude alerting

The following shall be operating normally prior to entry into RVSM airspace: both primary altimeters, one autopilot altitude-control system, and the Altitude Alerting System. The autopilot should be engaged during altitude capture and level cruise, except where circumstances such as the need to retrim or turbulence require disengagement. Report reaching assigned altitudes if not in radar contact.

FOM 5.5.6.3 Entry Into RVSM Airspace

"Both primary altimeters • One autopilot altitude-control system"

Three items only. A common wrong answer adds transponder or TCAS to the list.

Workbook wording: Are there aircraft equipment requirements to operate in RVSM airspace?

CRITICAL OBSERVABLE ITEM

5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE

RVSM equipment fails in flight. What do you do?

Notify ATC and Dispatch

In the event of required RVSM equipment failure prior to or within RVSM airspace, notify ATC and Dispatch to coordinate a prudent course of action.

FOM 5.5.6.4 System Failure Enroute

"ATC may not require any action; however, a reroute or diversion may be required."

Both ATC and Dispatch, not one or the other. The FOM deliberately leaves the outcome open rather than mandating an immediate descent out of RVSM.

Workbook wording: What are you expected to do if you have a failure of required equipment while in RVSM airspace?

CRITICAL OBSERVABLE ITEM

5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE

Diverting without a clearance. What altitude do you plan?

Below FL290

Under the oceanic in-flight contingency procedures (FOM 5.7.4.2), if a revised clearance cannot be obtained, turn at least 30 degrees left or right. Establish a same-direction parallel track offset 5 nm. Be below FL290 before diverting from the assigned clearance.

FOM 5.7.4.2 General Procedures

"The aircraft should be below FL290 prior to diverting from the assigned clearance unless a revised ATC clearance has been obtained."

Same guidance is repeated in FOM 5.7.4.3. First turn at least 30 degrees off the cleared track to establish a parallel same-direction offset of 5 nm, then take a 500 ft vertical offset once established.

Workbook wording: When diverting to an alternate without a clearance, what altitude should you plan to be below when diverting re RVSM?

CPDLC Logon Procedures**CRITICAL OBSERVABLE ITEM****5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE****Airborne CPDLC logon window before the FIR?****Not the same for every FIR; KZAK and RJJJ want 15 to 45 minutes before the oceanic FIR**

The FOM sets the policy: log on during preflight or after takeoff above 10 000 ft, or as the FIR or OCA Pilot Reference Card requires. The PRC CPDLC card then gives the window by FIR: Hawaii, climbing through 10 000 ft and 15 to 45 minutes prior to the oceanic FIR, KZAK; Japan the same for RJJJ; Alaska, Australia, New Zealand and Tahiti climbing through 10 000 ft.

PRC p38 Controller Pilot Data Link Communication (CPDLC)

"Hawaii Climbing through 10k and 15' KZAK to 45' prior to Oceanic FIR"

Use FD Pro ROUTE INFO for the specific FIR logon window and code. For the NAT HLA FIRs use the Europe Airway Manual ATC data.

Workbook wording: Once airborne, what is the CPDLC logon time window prior to the FIR ("Flight Information Region")? Is it the same for all FIRs?

CRITICAL OBSERVABLE ITEM**5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE****How do you log on for a departure clearance, and enroute?****MCDU ATC COMM, ATC MENU, LSK 5L NOTIFICATION, verify flight number and ATC, then NOTIFY**

On the MCDU select ATC COMM, then ATC MENU, then LSK 5L to bring up the NOTIFICATION page. Verify the flight number and the ATC center code, KUSA for a domestic departure clearance, then select NOTIFY. Enroute the session is transferred automatically between US domestic airspace and the international FIR when CPDLC is in use in both; otherwise log on to the next FIR in its window.

PRC p38 Controller Pilot Data Link Communication (CPDLC), Logon Procedures

*"On the MCDU: select ATC COMM / ATC MENU / LSK 5L to bring up the NOTIFICATION page. i VERIFY correct Flight Number and ATC; select NOTIFY *"*

If ATC CTR shows KUSA and you get NO COMM, do not re-log on; wait for reconnection and use voice. If it does not show KUSA, attempt a re-logon.

Workbook wording: How, specifically in your aircraft, do you logon for a departure clearance? Enroute?

CPDLC Menus**CRITICAL OBSERVABLE ITEM****5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE****What can you request through CPDLC?****Direct to a fix, altitudes, climb or descent, block altitude, voice contact, emergency messages**

Pilot-initiated CPDLC requests on the PRC card: REQUEST DIRECT TO a position that is on the current ATC assigned route, REQUEST an altitude, CLIMB TO, DESCENT TO, BLOCK altitude to altitude, VOICE CONTACT, and emergency messages. Do not send multiple requests in one message, do not repeat a request of the same type before the first is answered, and avoid free text except for MAYDAY or PAN PAN.

PRC p39 Pilot-Initiated REQUEST Messages Using CPDLC

"Voice is always the backup to CPDLC and PRIMARY for emergencies."

Lateral offsets and weather deviations are also datalink requests in the ATC LAT REQ page. Voice is always the backup and primary for emergencies.

Workbook wording: What types of requests can be made through CPDLC?

FOM Waypoint Transition Procedure**CRITICAL OBSERVABLE ITEM****5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE****Two minutes before a waypoint, what do you do?****Confirm the next waypoint and the one after it**

Approximately 2 minutes prior to reaching a waypoint, confirm that the upcoming waypoint and the subsequent waypoint (next plus one) agree with the Flight Plan and ATC clearance.

FOM 5.7.3.16 Waypoint Transition

"confirm that the upcoming waypoint and the subsequent waypoint (next + 1) agrees with the Flight Plan/ATC clearance."

The next-plus-one check is the whole point. Verifying only the active waypoint will not catch a gross navigation error that begins one leg downstream.

Workbook wording: Approximately 2 minutes prior to reaching a waypoint, what crew actions are required?

CRITICAL OBSERVABLE ITEM**5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE****What gets checked after waypoint transition?****Time and fuel recorded, difference computed, navigation mode verified, waypoint slashed**

After crossing, record time and fuel from the flight management computer on the flight plan or release and compare with the estimate. Note the difference: plus is underburn, minus is overburn. Confirm the navigation mode on the flight mode annunciator, slash the waypoint, and send the position report if required.

FOM 5.7.3.16 Waypoint Transition

"Record the time and fuel remaining from the FMC/FMS on the Flight Plan/Dispatch Release or the ACARS HOWGOZIT/Flight Plan Review and compare them with the estimates. Record the difference (+ is ahead/underburn and - is behind/overburn)."

Two minutes before the fix you confirm the next waypoint and the one after it against the clearance. Ten minutes after passage you run the navigation crosscheck.

Workbook wording: What items are required to be checked on the flight plan after waypoint transition?

CRITICAL OBSERVABLE ITEM**5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE****Where is fuel remaining recorded at waypoint transition?****The flight plan, the release, or the ACARS HOWGOZIT**

Time and fuel remaining from the FMS are recorded on the Flight Plan/Dispatch Release, or alternatively on the ACARS HOWGOZIT/Flight Plan Review, and compared against the estimates. On the A330 read fuel remaining from the FUEL PRED page, not the ECAM tank total, and keep the same source at every waypoint.

FOM 5.7.3.16 Waypoint Transition

"Record the time and fuel remaining from the FMC/FMS on the Flight Plan/Dispatch Release or the ACARS HOWGOZIT/Flight Plan Review"

Workbook wording: On which document is the fuel remaining (at waypoint transition) recorded?

CRITICAL OBSERVABLE ITEM**5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE****Ten minutes after waypoint transition, what do you do?****Crosscheck navigation performance and course**

Approximately 10 minutes after waypoint passage, crosscheck navigational performance and course compliance.

FOM 5.7.3.17 10 Minutes After Waypoint Passage

"Take immediate corrective action for any anomalies or deviations."

Cross-reference 5.7.4.11 Position Doubtful if the crosscheck does not resolve. If you are on a weather deviation, the FOM requires this process to be completed immediately rather than waiting for the 10-minute point.

Workbook wording: Approximately 10 minutes after waypoint transition, what crew actions are required?

CRITICAL OBSERVABLE ITEM**5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE****Why track fuel score?****It reveals fuel leaks and burn trends early**

Scoring fuel at every waypoint is the primary in-flight leak detector: a growing negative trend that fuel flows do not explain is the signature of a leak or fuel system malfunction (FOM 5.5.3).

FOM 5.5.3 Flight Plan Time and Fuel Monitoring - Enroute

"The difference between the actual and the estimated fuel remaining is an important factor in identifying and troubleshooting fuel leaks or other fuel system malfunctions."

FOM 7.1.12 requires a Dispatch call when actual fuel differs from planned by 3k lb. Call early rather than at the gate, so Dispatch can rerun the numbers while you still have options.

Workbook wording: Why is keeping track of the actual fuel remaining vs planned fuel remaining ("FUEL SCORE") important?

SATCOM / HF Procedures**CRITICAL OBSERVABLE ITEM****5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE****Where is the HF position report template?****Jeppesen FD Pro X, Pubs, Pacific, Oceanic Position Reporting Procedures**

For the Oakland FIR the full format sits in Jeppesen FD Pro X under Pubs, Pacific, Oceanic Position Reporting Procedures. The FOM carries the same skeleton: present position, estimated next position, ensuing fix, then preplanning flight levels. Oakland also wants a meteorological report with every position report.

FOM 5.7.3.19 Position Reporting, Oceanic (non-CPDLC)

"For operations in the Oakland FIR, oceanic position reporting procedures are clearly delineated in the Jeppesen FD Pro X app > Pubs > Pacific > Oceanic Position Reporting Procedures Oakland Oceanic FIR."

Present position is callsign, the word position, the fix, time in four digits and altitude. Ensuing fix is name only. The meteorological part is temperature, wind and any significant weather.

Workbook wording: Where can you find a 'template' to make an HF Radio Position report?

FOM and FCTM guidance on Weather avoidance**15.1.2.3.4 ANALYZE WEATHER CONDITIONS USING WEATHER RADAR SYSTEM****Where are the ICAO oceanic weather deviation procedures?****Our Flight Operations Manual oceanic chapter, reprinted from ICAO Doc 4444**

The oceanic weather deviation procedure sits in the oceanic chapter of our Flight Operations Manual, taken from ICAO Doc 4444. It covers the request, what to do when a clearance cannot be obtained, and the offset table. Under 5 nm hold your level. At 5 nm or more, change level 300 ft.

FOM 5.7.4.4 Weather Deviation Procedures for Oceanic Airspace

"When weather deviation is required, the pilot should initiate communications with ATC via voice or CPDLC. A rapid response may be obtained by either:"

Eastbound, deviating left descend 300 ft, right climb 300 ft. Westbound is the opposite. Start the offset about 5 nm off track and be back at level within 5 nm of centerline.

Workbook wording: Where can you find the ICAO Weather Deviation procedures for Oceanic Controlled Airspace?

15.1.2.3.4 ANALYZE WEATHER CONDITIONS USING WEATHER RADAR SYSTEM

How do you request a weather deviation?

State WEATHER DEVIATION REQUIRED by voice, or send a datalink request

Two ways to get a quick answer. Say WEATHER DEVIATION REQUIRED on the frequency, which tells the controller you want priority, or send a lateral deviation request over datalink. Tell ATC how far you need. If no clearance is coming and the weather is dangerous, use PAN PAN or the urgency downlink.

FOM 5.7.4.4 Weather Deviation Procedures for Oceanic Airspace

"Stating "WEATHER DEVIATION REQUIRED" to indicate that priority is desired on the frequency and for ATC response; or • Requesting a weather deviation using a CPDLC lateral downlink message."

Tell ATC when the deviation is over and you are back on the cleared route. On the A330 the datalink route is the ATC LAT REQ page on the MCDU, and the PRC oceanic card says broadcast on 121.5 and 123.45 if no clearance can be had.

Workbook wording: How, specific to your aircraft, do can you make a weather deviation request?

15.1.2.3.4 ANALYZE WEATHER CONDITIONS USING WEATHER RADAR SYSTEM

How do you insert the offset in the FMS?

LAT REV at the FROM waypoint, OFFSET, type the value and side such as 5L or L5, INSERT

The OFFSET page is reached from the Lateral Revision at the FROM waypoint or any downpath waypoint except the destination. An offset of 1 to 50 NM in one mile steps, left or right, immediate or deferred, ending by default or at a planned waypoint. The PRC FMS Guide: F-PLN key, 1L, type the offset and direction, for example L5 or 5L, insert at 2L, then 6R INSERT. Clear it with CLR or 0.

FCOM DSC-22_20-10-30-12 P 32/60 Offset

"The OFFSET page is accessed from the Lateral Revision (LAT REV) page at the FROM waypoint or at any waypoint downpath the flight plan, except the destination airport. An offset may be defined between 1 and 50 NM in one-nautical-mile steps."

The PRC engine failure card uses the same page for the 5 NM contingency offset: pull heading or insert a 5 NM left or right offset on the F-PLN page.

Workbook wording: Once received, how (aircraft CDU/FMS specific) can you insert the offset distance?

Oceanic Contingency Procedures**CRITICAL OBSERVABLE ITEM**

13.1.1.1.1 KNOW ENGINE FAILURE IN CRUISE PROCEDURE

Engine failure in oceanic airspace. What do you do?

Nearest suitable airport, after getting a revised oceanic clearance

Land at the nearest suitable airport where a safe landing can be made. Get a revised clearance before leaving the track if you can. If you cannot, turn at least 30 degrees left or right, establish a parallel track offset 5 nm, and get below FL290. Tell air traffic control what you have and ask for expeditious handling.

FOM 6.4.2 Engine Failure

"14 CFR 121.565 requires the Captain of a two-engine aircraft with one engine inoperative to land at the nearest suitable airport where, in the Captain's judgment after considering all relevant factors, a safe landing can be made."

Squawk 7700, exterior lights on, and broadcast identification, condition, position and intentions on 121.5. The PRC engine failure card (8/31/26) adds the A330 speeds: 300 kt or M 0.78 for ETOPS and standard strategy, green dot for obstacles.

Workbook wording: What are the engine failure and diversion procedures while in oceanic airspace?

CRITICAL OBSERVABLE ITEM

13.1.1.1.1 KNOW ENGINE FAILURE IN CRUISE PROCEDURE

Driftdown off the route. Which way do you turn?

At least 30 degrees, left or right per listed factors

FOM 5.7.4.2 has you leave the cleared track by initially turning at least 30 degrees left or right to establish a parallel, same-direction track offset 5 nm. Direction is a judgment call driven by where you sit in the track system, what is allocated on adjacent tracks, which way the alternate lies, the SLOP you are already flying, and terrain.

FOM 5.7.4.2 General Procedures (Oceanic Emergency/Abnormal)

"The direction of the turn should be based on one or more of the following factors:"

The PRC engine failure card: leave the cleared track by turning at least 30 degrees left or right to a parallel same-direction track offset 5 NM, and once below FL290 or cleared by ATC maneuver as required. Direction weighs adjacent tracks, the alternate, any SLOP and terrain.

Workbook wording: When turning off the route during a drift-down, which direction will you turn? Why?

SLOP

8.3.4.26 DEMONSTRATE ABILITY TO CORRECTLY PROGRAM AIRCRAFT GUIDANCE SYSTEMS INCLUDING APPROACH

What is SLOP for?**Collision risk and wake turbulence**

The Flight Crew should use SLOP as standard operating practice in normal oceanic operations to mitigate collision risk and wake turbulence. It addresses both wake vortex encounters and the heightened collision risk when non-normal events such as operational altitude deviation errors and turbulence-induced altitude deviations occur.

FOM 5.7.4.7 Strategic Lateral Offsets in Oceanic Airspace

"to mitigate collision risk and wake turbulence."

SLOP is generally accepted but disallowed in specific regions; check the AIP. Any aircraft overtaking another should offset within the procedure to create the least wake turbulence for the aircraft being overtaken.

Workbook wording: What is the purpose of SLOP?

8.3.4.26 DEMONSTRATE ABILITY TO CORRECTLY PROGRAM AIRCRAFT GUIDANCE SYSTEMS INCLUDING APPROACH

How far can you offset with SLOP?**Up to 2 nm right of centerline. Never left**

SLOP is applied in relation to a route or track. The aircraft may fly zero, or up to a maximum of 2 nm right of course only, from 0.0 to 2.0 nm right of centerline in 0.1 nm increments.

FOM 5.7.4.7 Strategic Lateral Offsets in Oceanic Airspace

"SLOP left of course is not authorized."

Three legal positions in practice: centerline, 1 nm right, 2 nm right, though the FOM permits any 0.1 increment. No ATC clearance is needed. Must be back on centerline by the oceanic exit point.

Workbook wording: Up to what distance/direction from the route is SLOP used/acceptable?

In Trail Procedures (OpSpec A354)

8.3.4.26 DEMONSTRATE ABILITY TO CORRECTLY PROGRAM AIRCRAFT GUIDANCE SYSTEMS INCLUDING APPROACH

What is ITP?**In Trail Procedure: a CPDLC climb or descent through an occupied level using ADS-B reference aircraft**

An ITP is a flight level change through the level of one or two reference aircraft in trail, using qualified ADS-B data, TCAS and ADS-B combined. The PRC card lists the criteria: no more than two reference aircraft, altitude difference at or below 2000 ft, ITP distance at least 15 NM with a closing ground speed differential at or below 20 kt or 20 NM with 30 kt, climb or descent at 300 ft/min minimum, same direction, and the ITP aircraft able to hold its Mach.

FCOM DSC-34-20-20-20 P 2/2 ATSA In Trail Procedure (ATSA ITP)

"The ITP enables aircraft in oceanic areas or areas that do not have radar coverage to change flight levels on a more frequent basis with a longitudinal separation that is temporarily reduced during the climb"

ATC remains responsible for separation. If anything goes wrong during the maneuver apply the regional contingency procedures from the AIP. The criteria list is on PRC p40.

Workbook wording: What is "ITP"?

8.3.4.26 DEMONSTRATE ABILITY TO CORRECTLY PROGRAM AIRCRAFT GUIDANCE SYSTEMS INCLUDING APPROACH

When would you use an ITP?

When a standard climb or descent is blocked by traffic in trail and the MCDU says ITP POSSIBLE

Use it when you want a level change in procedural oceanic airspace and the only conflict is one or two same-direction aircraft ahead or behind at the intervening levels. First check REC MAX FL on the PROG page, then MCDU MENU, TRAF at 5R, IN TRAIL PROCEDURE, enter the desired flight level and check ITP POSSIBLE with its time limit and the reference aircraft in green.

FCOM PRO-SPO-90-10 P 2/4 In-Trail Procedure, Procedure (PRC p40 CPDLC In-Trail Procedures)

"Check if the ITP procedure is possible: ■ Check that "ITP POSSIBLE" is displayed in green next to the LSK 1R, and the associated time limit for this ITP procedure ("ITP POSSIBLE UNTIL NN:NN Z"), ■ Check that the reference aircraft are displayed in green in the ITP TRAFFIC list."

If ITP is no longer possible when the clearance arrives, send UNABLE. If still possible send WILCO and perform it without delay, holding current Mach and at least 300 ft/min.

Workbook wording: When might you consider using the ITP?

8.3.4.26 DEMONSTRATE ABILITY TO CORRECTLY PROGRAM AIRCRAFT GUIDANCE SYSTEMS INCLUDING APPROACH

How do you request an ITP climb or descent?

ATC VERT REQ page: CLB TO or DES TO the level, then FREE TEXT with ITP and reference aircraft.

From the ATC VERT REQ page enter CLB TO or DES TO the desired level at 1L or 2L, then add the ITP information in the free text: line 1 ITP, line 2 the ITP distance NM AHEAD or BEHIND and the flight ID of the first reference aircraft, line 3 AND plus the second reference aircraft if there is one. The flight IDs must exactly match the ITP TRAFFIC LIST page.

FCOM PRO-SPO-90-10 P 3/4 In-Trail Procedure, Procedure (PRC p41 CPDLC In-Trail Procedures cont'd)

"Request an ITP clearance to ATC by CPDLC in the ATC VERT REQ page of the MCDU. Apply the ITP phraseology (Vertical request and FREE TEXT), as follows:"

Write ITP, not In Trail, and do not add slashes or extra spaces; ATC's system parses the text.

Workbook wording: How do you (specifically) request an ITP CLB/DES clearance?

16.1.2 PERFORM LONG RANGE COMMUNICATION PROCEDURES

When is an HF position report required?

Only when you are not on CPDLC

When using CPDLC you must contact ARINC after a successful CPDLC logon but prior to the initial Gateway Fix to obtain a SELCAL check on HF, establishing a backup if CPDLC fails. No other HF reports need be made while using CPDLC. Non-CPDLC position reporting varies by area and is specified on the enroute chart.

FOM 5.7.3.18 Position Reporting Oceanic (CPDLC)

"Flight Crews should not make HF position reports when using CPDLC procedures."

REV 125.1 ADDED: if SELCAL is not functioning, one member of the crew must listen to HF continuously. That is the new sentence in this revision and a likely oral question.

16.1.2 PERFORM LONG RANGE COMMUNICATION PROCEDURES

Can you enter oceanic airspace with both HF radios out?

No. At least one HF radio is required

FOM 20.4.1 sets dispatch minimums of two HF radios, or a single HF plus SATVOICE. After pushback and prior to entering oceanic airspace the requirement is dual LRNS plus two HF radios, or a single HF with SATCOM data link, or a single HF with SATVOICE.

FOM 20.4.1 Required Equipment for OAK, ANC, and FUK FIRs

"After pushback, prior to entering oceanic airspace: • Dual Long Range Navigation Systems (to include 2 MCDUs) • Two HF radios, or – A single HF radio and SATCOM data link, or – A single HF radio and SATVOICE"

Every authorized combination for OAK/ANC/FUK includes at least one working HF. Same story in the NAT: FOM 22.4.3 requires two independent LRCS, one of which must be HF. FOM 20.4.4.1 covers what to do if both HF fail after you are already established oceanic.

Workbook wording: Can you enter Oceanic airspace with both HF radios not working?

International Theaters (4)

1.2.3 PERFORM INTERIOR/EXTERIOR INSPECTION

Where is the aircraft Sanitation Certificate?

Not named in our manuals; the disinsection certificate is in Comply365

The FOM does not name a sanitation certificate. The closest document is the Certificate of Residual Disinsection, which is placed in the Comply365 app under Flying, Regulatory and Operating Docs, International Documents, Europe, and is also carried on the General Declaration.

FOM 18.1.10.1 Residual Spraying

"a Certificate of Residual Disinsection will be placed into the Comply365 app under Flying > Regulatory & Operating Docs > International Documents > Europe."

Confirm what the Japan ramp inspector calls a sanitation certificate with the check airman before your first Japan turn.

Workbook wording: Where is the aircraft Sanitation Certificate located? (inspected during Japanese JCAB Ramp Checks)

1.2.3 PERFORM INTERIOR/EXTERIOR INSPECTION

Is HA approved for the NADP 1 departure?

Yes. Normal takeoff procedures satisfy it

FOM 5.4.7 carries the noise abatement departure procedure under Ops Spec C068. Compliance with published noise abatement procedures is required at noise-sensitive airports. Because normal takeoff procedures already satisfy both the close-in (NADP 1) and distant (NADP 2) profiles, No separate NADP profile to fly.

FOM 5.4.7 Noise Abatement Departure Procedure (NADP), Ops Spec C068

"(AS) Normal takeoff procedures satisfy either close-in or distant NADP profile requirements."

The FOM tags this (AS), and post-merger it is the governing procedure. Specific gradients and configuration-change altitudes come from the fleet manuals, the 10-7, or the Jeppesen Airway Manual noise abatement pages.

Workbook wording: Is HA allowed to perform the NADP 1 Departure Procedure? (Japanese JCAB Ramp Check question; they'll want to see the OpSpec)

Dispatch Release: Review all sections

CRITICAL OBSERVABLE ITEM

1.1.1 PERFORM FLIGHT PLANNING

Where is country-specific guidance?

FOM theater chapters plus Jeppesen State Rules and Procedures

FOM chapters 19 through 24 are organized by theater and carry country-specific airspace, navigation, communication and procedural rules.

FOM chapters 19-24 Theater chapters; example FOM 19.3.1.2 Holding Airspeeds

"For holding procedures and airspeeds, see Jeppesen FD Pro app at the following location: Pubs > North America > North America Airway Manual > State Rules and Procedures – Canada > Instrument Flight Rules – Holding Procedures."

The six theater chapters are 19 North America, 20 Pacific, 21 Caribbean, 22 Atlantic, 23 Europe, 24 Asia. They repeatedly hand off to Jeppesen FD Pro State Rules and Procedures, and airport-specific items live in the Jeppesen 10-7 pages.

Workbook wording: International Planning: Where can you find country-specific guidance?

Holding (Procedures, speeds)

MISC MISCELLANEOUS, NOT ASSOCIATED WITH GRADE SHEET TASKS

Where are international holding speeds?**ICAO Doc 8168 and the theater chapters**

FOM 5.5.15 cites AIM 5-3-8 and ICAO Doc 8168. Country-specific holding speeds and procedures appear in the theater chapters 19 through 24 and in the Jeppesen country pages, for example the Heathrow holding restriction in the Europe chapter.

FOM 5.5.15 Holding, ICAO DOC 8168

"ICAO DOC 8168 Flights anticipating holding at an enroute fix or flights that have not received clearance beyond a fix should begin a speed reduction no earlier than three minutes prior to reaching such fix."

Rev 125.1 renumbering: the United Kingdom chapter is 23.17 with LHR subsections under 23.17.1, and Italy is 23.18.

Workbook wording: Where can you find international country-specific holding speeds?

TCAS and Traffic (5)**TCAS**

MISC MISCELLANEOUS, NOT ASSOCIATED WITH GRADE SHEET TASKS

Where are TCAS procedures in the FOM?**TCAS Policy, in the Operating - General chapter**

FOM 2.3.17 TCAS Policy is the FOM home for TCAS. It requires compliance with an RA unless the pilot considers it unsafe, prohibits maneuvering opposite an RA unless visually determined to be the only means of safe separation, and reminds crews the aircraft in sight may not be the intruder.

FOM 2.3.17 TCAS Policy

"Compliance with a TCAS Resolution Advisory (RA) is required, unless the pilot considers it unsafe to do so."

The FOM carries policy only. The system, symbols and TA or RA logic are in FCOM DSC-34-20-60, and the traffic avoidance maneuver is the ECAM and FCOM abnormal procedure. Oceanic TCAS guidance is at FOM 5.7, TA/RA whenever possible.

Workbook wording: Where can you find TCAS procedures in the FOM?

MISC MISCELLANEOUS, NOT ASSOCIATED WITH GRADE SHEET TASKS

What is a TA?**Traffic Advisory: a potential collision threat, TAU about 40 seconds, amber circle and TRAFFIC TRAFFIC**

TCAS sorts intruders into other, proximate, TA and RA. A Traffic Advisory is a potential collision threat with a TAU, time to closest point of approach, of about 40 seconds. It shows as an amber circle on the ND with the TRAFFIC, TRAFFIC aural, and no vertical order on the PFD.

FCOM DSC-34-20-60-10 P 5/10 TCAS Description

"Traffic Advisory (TA) ■ Potential collision threat ■ ND: Intruder position ■ TAU is about 40 s ■ Aural messages"

A proximate intruder is within 6 NM and 1200 ft with no threat, shown as a filled white diamond. TCAS sees 30 NM laterally and 9900 ft above and below.

Workbook wording: What is an "TA"?

MISC MISCELLANEOUS, NOT ASSOCIATED WITH GRADE SHEET TASKS

What is an RA?**Resolution Advisory: a real collision threat, TAU about 25 seconds, red square with vertical orders on the PFD**

A Resolution Advisory is a real collision threat with a TAU of about 25 seconds. The intruder is a red square on the ND, aural messages sound, and the PFD shows vertical speed orders: maintain the current vertical speed for a preventive advisory, or change it for a corrective advisory. FOM 2.3.17 requires compliance with an RA unless the pilot considers it unsafe.

FCOM DSC-34-20-60-10 P 5/10 TCAS Description

“Resolution Advisory ■ Real collision threat ■ ND: Intruder position (RA) ■ TAU is about 25 s ■ Aural messages ■ PFD: Vertical orders”

TA mode converts every RA into a TA; the panel says use it for degraded performance such as an engine failure, gear down, or parallel runway approaches where the 10-7 calls for it.

Workbook wording: What is an “RA”?

MISC MISCELLANEOUS, NOT ASSOCIATED WITH GRADE SHEET TASKS

What is a phantom TA?

No published source located, ask the check airman

Neither the FOM nor the FCOM uses the term. The FCOM TCAS description does note the display limits behind spurious-looking traffic: only the eight most threatening intruders are shown, an intruder without range is not displayed, and one without bearing appears as digital data at the bottom of the ND.

FCOM DSC-34-20-60-20 P 2/8 ND Indications

“The traffic is displayed in all ROSE modes and ARC mode whatever NM range is selected. Only the eight most threatening intruders are displayed.”

No published source located, ask the check airman

Workbook wording: What is phantom TA?

MISC MISCELLANEOUS, NOT ASSOCIATED WITH GRADE SHEET TASKS

Where are the TCAS symbols defined?

FCOM DSC-34-20-60-20 ND Indications: white diamond other, filled diamond proximate, amber circle TA, red square RA

The TCAS Controls and Indicators section defines the ND symbols: an empty white diamond for other intruders, a filled white diamond for proximate, an amber circle for a TA with the TRAFFIC TRAFFIC aural, a red square for an RA with PFD vertical orders. Relative altitude in hundreds of feet and a vertical speed arrow above 500 ft/min appear in the intruder's color, and a 2.5 NM white range ring shows on the 10 and 20 NM ranges.

FCOM DSC-34-20-60-20 P 3/8 ND Indications

“RA intruder Indicated by a red square. Associated with vertical orders displayed on the PFD and aural messages.”

TRAFFIC selector: THRT shows others only during a TA or RA within 2700 ft; ALL shows them always; ABV and BLW extend the band to 9900 ft above or below.

Workbook wording: Where can you find TCAS symbology definitions?

Medical (7)

MedLink

MISC MISCELLANEOUS, NOT ASSOCIATED WITH GRADE SHEET TASKS

What is MedLink?

Contract ER physicians giving 24-hour inflight medical advice

FOM 11.1.9.1 describes MedLink as emergency room physicians available 24 hours a day with aviation-specific experience, providing recommendations on onboard medical issues, and notes contact with MedLink removes Company liability for passenger health.

FOM 11.1.9.1 MedLink Overview

"MedLink utilizes emergency room physicians 24 hours a day to provide crewmembers with informed recommendations regarding on-board medical issues."

Company policy is to follow MedLink's recommendation, but the Captain keeps final authority on operational requests. The one-way exception: if MedLink says a passenger is not fit to fly, the Captain will not override MedLink.

MISC MISCELLANEOUS, NOT ASSOCIATED WITH GRADE SHEET TASKS

When do you contact MedLink?**Any medical support need, aircraft oxygen use, fitness-to-fly doubt**

FOM 11.1.9.2 makes MedLink contact mandatory, except during critical phases of flight, for any medical situation where the crew requires support or advice. It is also mandatory for any passenger or crew use of aircraft oxygen, and any uncertainty about whether a passenger is contagious or fit to fly (11.1.8.2).

FOM 11.1.9.2 When to Contact MedLink

"MedLink shall be contacted for the following, except during critical phases of flight:"

Note the carve-out: not during critical phases of flight (sterile flight deck per 2.3.12). Passenger or crew use of aircraft oxygen is a mandatory trigger even if the person seems fine.

Workbook wording: When should you contact MedLink?

MISC MISCELLANEOUS, NOT ASSOCIATED WITH GRADE SHEET TASKS

Who contacts MedLink on the ground?**CSA at the gate; after door close, F/As or crew**

FOM 11.1.9.3 On Ground has the Flight Crew, F/As, station personnel, and MedLink work together to decide fitness to fly, and requires MedLink contact to clear a passenger even if paramedics already responded. At the gate or prior to door closure the CSA should initiate contact, and may deplane the passenger for privacy while assessing fitness to fly.

FOM 11.1.9.3 Responsibility for Contacting MedLink - On Ground

"The CSA should initiate contact with MedLink."

After door closure and after landing the split is by equipment: with operational broadband or a SATCOM cabin device the F/As initiate; without it the Flight Crew initiates. Advise Operations if medical assistance is expected at the gate.

Workbook wording: Who is responsible for contacting MedLink while on the ground?

MISC MISCELLANEOUS, NOT ASSOCIATED WITH GRADE SHEET TASKS

Who contacts MedLink in flight?**F/As if cabin SATCOM works; otherwise the Flight Crew**

Per FOM 11.1.9.3 In Flight, on HA aircraft with available and operational cabin SATCOM handsets the F/As advise the Flight Crew and then call MedLink directly; if the cabin device is unavailable, the Flight Crew initiates.

FOM 11.1.9.3 Responsibility for Contacting MedLink - In Flight

"On aircraft with operational SATCOM handsets in the cabin, the F/As call MedLink directly."

MedLink is reached directly, with Dispatch conferenced in when a diversion looks possible. Get the flight attendants gathering vitals while you set up the call, so the doctor has data on first contact.

Workbook wording: Who is responsible for contacting MedLink while in flight?

MISC MISCELLANEOUS, NOT ASSOCIATED WITH GRADE SHEET TASKS

Who does the flight crew call first for a medical event, MedLink or Dispatch?**MedLink directly; Dispatch is conferenced in if a diversion is possible.**

FOM 11.1.9.3 has you go to MedLink first, not Dispatch. With operational cabin SATCOM handsets the F/As call MedLink directly. With SATCOM but no cabin handset the A330 Flight Crew calls MedLink at (602) 282-4910 and relays information between the cabin and MedLink. Only with no operational SATCOM does the call route through ARINC for a patch, and Dispatch is conferenced in or called afterward when a diversion is possible.

FOM 11.1.9.3 Responsibility for Contacting MedLink - In Flight

“(A321/A330) The Flight Crew will relay info between cabin and MedLink.”

Only with no operational SATCOM does the call route through a relay: ARINC patch, or Dispatch if ARINC fails. Dispatch always gets notified for a potential diversion, either by MedLink conferencing them in or by you calling after.

Workbook wording: Do we as flight crew contact MedLink directly, or do we call dispatch first?

MISC MISCELLANEOUS, NOT ASSOCIATED WITH GRADE SHEET TASKS

How do you make a MedLink call from the aircraft?**SATCOM to (602) 282-4910, or ARINC phone patch**

Per FOM 11.1.9.3, with operational SATCOM the Flight Crew calls MedLink Inflight Medical Assistance at (602) 282-4910 and, on the A330, relays information between the cabin and MedLink. Without SATCOM, contact ARINC for a phone patch to the same number. On the A330 the number sits in the SATCOM DIRECTORY SAFETY list; dial from the MCDU and talk on the ACP SATCOM key.

FOM 11.1.9.3 Responsibility for Contacting MedLink - In Flight

“The Flight Crew will contact ARINC for a patch to MedLink Inflight Medical Assistance at (602)”

The A330 has no cabin headset conference: you are the relay, so have the F/A pass vitals to you before you dial. Ground Medical Assistance is (602) 282-6613 or (800) 961-4847.

MISC MISCELLANEOUS, NOT ASSOCIATED WITH GRADE SHEET TASKS

Medical event on an international flight. Any special notifications?**Yes. General Declaration entry; Australia notified 30 minutes before landing**

FOM 18.1.15 uses the US Public Health Service ill person definition. Temperature of 100F (38C) or greater persisting more than 48 hours counts. So does fever with rash, swollen glands, jaundice, persistent cough, vomiting, difficulty breathing, muscle pain, headache with stiff neck, decreased consciousness, or unexplained bleeding, and persistent diarrhea.

FOM 18.1.15 Death or Illness Reporting - International

“For Flights to Australia: The Captain must report to the Australian authorities the presence on board the aircraft of any apparent death or any ill person among passengers or crew at least 30 minutes before landing.”

The General Declaration requirement applies to countries other than the US and Canada. For a suspected communicable disease inbound to the US, 11.1.8.2 adds the CDC/public health authority path, usually conferenced in by MedLink or set up through Dispatch.

Workbook wording: If on an international flight, are there special procedures/notifications that must be made by the flight crew to either ATC or country officials?

HAZMAT ⁽⁶⁾**Load Closeout/TPR Update and review procedure**

1.2.2 PERFORM ACARS AND FMS INITIALIZATION AND VERIFICATION

Dangerous goods on board. What governs?**Review and acknowledge the eNOTOC, then retain it**

Dangerous goods on board means the PIC must receive a NOTOC (FOM 14.1.1).

FOM 14.1.6 Captain Responsibilities (HA)

“• Acknowledging the eNOTOC by selecting either “ACCEPT” on the MCDU or “ACKNOWLEDGE NOTOC” on the Comm page (depending on fleet type).”*

HA rules differ from AS: hand edits to the eNOTOC are not permissible (AS allows Ramp Lead edits to tail number and DG location on the paper NOTOC, FOM 14.1.5). eNOTOC arrives at departure minus 30 min international, minus 15 min interisland (FOM 14.1.4).

Workbook wording: Describe the Dangerous goods procedures if you have any on your flight

15.2.1 DEMONSTRATE DANGEROUS GOODS POLICIES AND PROCEDURES

Where is the dangerous goods policy in the FOM?**The HAZMAT chapter, under Dangerous Goods**

Dangerous goods policy lives in FOM Chapter 14, HAZMAT, section 14.1 Dangerous Goods, starting with 14.1.1 Regulatory Compliance under Ops Spec A055, 49 CFR Parts 171 to 175 and the IATA DGR. Key HA subsections: 14.1.4 eNOTOC, 14.1.6 Captain Responsibilities, 14.1.8 acceptable NOTOC versions, 14.1.17 rejected shipments, 14.1.20 ICAO ERG drill codes and 14.1.26 damaged or leaking DG.

FOM 14.1 Dangerous Goods (Chapter 14 HAZMAT)

“The Company accepts Dangerous Goods (DG) in accordance with International Air Transport Association (IATA) Dangerous Goods Regulations and the domestic Code of Federal Regulations (CFR) Title 49.”

FOM Chapter 14 is titled HAZMAT and section 14.1 is Dangerous Goods, running 14.1.1 through 14.1.28. eNOTOC and HA Captain responsibilities are 14.1.4 and 14.1.6.

Workbook wording: Where can the dangerous goods policy be found in the FOM?

15.2.1 DEMONSTRATE DANGEROUS GOODS POLICIES AND PROCEDURES

What is an eNOTOC?**The electronic Notice to Captain of dangerous goods, sent by ACARS**

Per FOM 14.1.4 the eNOTOC is the Hawaiian electronic Notice to Captain for dangerous goods, uplinked at departure minus 30 minutes for trans-Pacific and international flights and departure minus 15 minutes for inter-island, and it prints automatically.

FOM 14.1.4 eNOTOC (HA)

“The eNOTOC will be sent to the Flight Crew at departure time minus 30 minutes for trans-Pacific and international flights and at departure time minus 15 minutes for inter-island flights.”

It prints automatically on all aircraft. Versions per FOM 14.1.8: eNOTOC (primary), hard-copy eNOTOC from the Load Master if ACARS/printer is inop but FM Amadeus works, and HAL Form 902 (freighter DG COMAT only).

15.2.1 DEMONSTRATE DANGEROUS GOODS POLICIES AND PROCEDURES

How do you acknowledge an eNOTOC?**Select ACCEPT on the MCDU eNOTOC uplink**

FOM 14.1.6 requires the HA Captain to acknowledge the eNOTOC by selecting ACCEPT* on the MCDU. The eNOTOC uplinks at departure minus 30 minutes for trans-Pacific and international flights and minus 15 for inter-island, and prints automatically.

FOM 14.1.6 Captain Responsibilities (HA)

“Acknowledge the eNOTOC by selecting either “ACCEPT” on the MCDU or “ACKNOWLEDGE NOTOC” on the Comm page (depending on fleet type).”*

If digital acknowledgment is impossible, acknowledge by voice to Dispatch and note it; the acknowledgment is the record that you saw the load. Hand edits to the eNOTOC are not permitted at HA.

15.2.1 DEMONSTRATE DANGEROUS GOODS POLICIES AND PROCEDURES

How do you fight a lithium battery fire?**HALON, then water. Never ice**

FIGHT THE FIRE with a HALON extinguisher. Water may be used to fight a lithium ION battery fire but should NOT be used on a lithium METAL battery fire. It is very important to prevent the fire spreading to adjacent flammable materials.

FOM 11.1.10.2 PED Fire Containment Bag (Yellow Bag)

“AVOID USING ICE to cool the device. Ice will act as an insulator.”

It may take 15 or more minutes to completely cool the device. REV 125.1: this is the YELLOW bag section, renumbered from 11.1.10.

15.2.1 DEMONSTRATE DANGEROUS GOODS POLICIES AND PROCEDURES

Where are the cabin lithium battery limits?

The Alaska Airlines onboard PED and prohibited-items web pages

FOM 13.4 sends you to the [alaskaair.com](https://www.alaskaair.com/content/travel-info/policies/onboard-policies-portable-electronic-devices) onboard-policies portable electronic devices page and the prohibited-items page for the list of permitted and restricted devices and battery types/sizes.

FOM 13.4 Portable Electronic Devices / FOM 13.4.1 Lithium Batteries in PEDs

"See the list of portable electronic devices that are permitted and restricted on board at: <https://www.alaskaair.com/content/travel-info/policies/onboard-policies-portable-electronic-devices> and <https://www.alaskaair.com/content/travel-info/policies/prohibited-items>."

The FOM itself does not publish watt-hour numbers; it points to the [alaskaair.com](https://www.alaskaair.com/content/travel-info/policies/onboard-policies-portable-electronic-devices) PED policy and prohibited-items pages. FOM 13.4.1 gives the carry-on vs checked rules, and 11.1.10 covers the containment bags.

Workbook wording: Where can you find lithium battery type/size limitations to be carried in the cabin?

Descent and Approach (36)

Descent Preparation

6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL

What goes into descent preparation?

Weather and landing info, charts, QNH preset on ISIS, ECAM status, landing performance, then the FMS pages

The Descent Preparation SOP: obtain weather and landing information for destination and alternate, prepare the Jeppesen charts, preset QNH on the ISIS, check the ECAM STATUS page for any landing capability degradation, request landing performance via ACARS, then complete the FMS: ARRIVAL page, F-PLN A check, RADIO NAV, DES WIND, PERF CRZ and DES, PERF APPR with QNH, temperature, wind, minimum and landing configuration, PERF GO AROUND, FUEL PRED, followed by the approach briefing.

FCOM PRO-NOR-SOP-16 P 1/10 Descent Preparation

"Check the weather reports and runway conditions at destination and alternate airports. Airfield data, if any, should include the runway to be used for arrival. When the aircraft operates in low OAT conditions, consider altitude corrections for low temperature."

Do not modify the final approach from the FAF to the runway in the F-PLN. Identify the FDP and check the FPA after it; a TOO STEEP PATH message means no FINAL APP guidance.

Workbook wording: What items are included in preparing for descent?

Descent Plan Execution

6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL

How do you start the descent?

Engage DES mode at the FMS top of descent; DES for an early descent, green dot if delayed

The standard method is to engage DES at the T/D computed by the FMS. For an early descent from ATC use DES, which guides down at a lower vertical speed to converge with the profile, or about 1000 ft/min in V/S. If ATC delays the descent past T/D, DECELERATE or T/D REACHED appears and you slow toward green dot with ATC permission, then engage DES with managed speed when cleared.

FCOM PRO-NOR-SOP-17 P 1/6 Descent Initiation

"The standard method to initiate the descent is to engage the DES mode at the Top of Descent (T/D) computed by the FMS."

Monitor with the PF MCDU on PROG for VDEV. In DES mode on or below the path do not use speed brakes, the A/THR will add thrust against them; use OP DES with speed brakes instead.

Workbook wording: How do you get the aircraft to begin the descent?

6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL

Cleared the arrival vs cleared to descend via?

No published source located, ask the check airman

The only place our manuals use the phrase is the Mexico section, where a descend via clearance means fly the published lateral routing and the published altitudes. Neither the FOM nor the FCOM defines it against a plain arrival clearance. That distinction lives in the Aeronautical Information Manual, not in our books.

FOM 21.3.3.5 Mexico Phraseology for Climbs and Descents

"ATC expects the pilot to navigate on the lateral routing and altitudes as published on a procedure. Typically, the controller will issue a "Descend Via" or "Climb Via" clearance to keep the radio transmission short and concise."

No published source located, ask the check airman. The FOM uses descend via only in the Mexico theater and says Central America does not use it at all; the US definition lives in the AIM.

Workbook wording: What is the difference between "Cleared XX arrival" vs "Cleared to descend via XX arrival"?

Analysis of Airport Conditions

6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL

Does the FOM table CAT II and III limitations?

No. It points to the fleet manuals, PRC and QRH

FOM 5.6.17 directs Flight Crews to use the approach procedures in the fleet-specific manuals for all CAT II/III approaches. Missed approach criteria for CAT II/III are also in the fleet-specific manuals.

FOM 5.6.17 Category II/III Operational Requirements

"The approach briefing references and required equipment tables in the PRC/QRH contain complete listings of the Flight Crew procedures and required equipment."

Ops Spec C060. Also: Alaska may be restricted by variant from CAT II/III at certain runway/airport combinations, see the tailored Jeppesen chart notes (5.6.17.1).

Workbook wording: Is there a table in the FOM categorizing CAT II/III Approach and Landing Limitations?

6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL

When can you autoland?

CAT II or III runways, autoland qualified PIC, within the FCOM autoland envelope

FOM 5.6.18 permits autoland only on CAT II or III runways, tells you to advise the tower so the ILS critical area is protected, and requires the PIC to be autoland qualified. The FCOM adds the envelope: CONF 3 or FULL, glide slope between 2.5 and 3.25 degrees, field elevation below 9200 ft, weight above 256k lb and below maximum landing weight, wind at most 35 headwind, 10 tailwind, 15 crosswind.

FOM 5.6.18 Autoland Operations

"Autoland approaches and landings are permitted on CAT II/III runways only. Pilots should advise the tower anytime Autoland operations are conducted to ensure the ILS Critical Zone is protected."

The critical area is protected automatically only below 800 ft or 2 miles. The FCOM allows autoland on a CAT I installation in better weather only with the airline's beam check and CAT 2 capability on the FMA.

Workbook wording: Under what conditions can you perform an autoland?

6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL

When must you autoland?

CAT II and CAT III approaches, and SA CAT II without a HUD

FOM 5.6.17.2 and 5.6.17.3 require CAT II and CAT III approaches to be flown with Autoland, so on the A330 every CAT II or III is an autoland. The FCOM SA CAT II limitation says without HUD the flight crew must perform an automatic landing. To be qualified the HA Captain must first complete one automatic landing in the aircraft.

FOM 5.6.17.2 CAT II/SA CAT II Operating Limitations

"CAT II approaches must be flown with either Autoland or (737) HGS AIII procedures. Rollout guidance is not required."

Currency bites new Captains: without that one automatic landing logged in the aircraft you are not CAT II/III qualified regardless of sim performance (FOM 5.6.17).

Workbook wording: Under what conditions must you perform an autoland?

6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL**How do you log an autoland?****Fill in every field on the datalink Autoland Report page**

Every autoland attempt gets reported on the ACARS Autoland Report page, logbook page information included. If the datalink is inoperative, write a logbook entry giving airport, landing runway, category, and whether it was successful. An unsuccessful autoland also needs an Info to Maintenance entry explaining the failure.

FOM 5.6.21.2 Autoland Reporting, ACARS

"For all Autoland attempts, the Captain is responsible for ensuring that all applicable fields on the ACARS Autoland Report page are filled in, including the logbook page information."

Autoland landings are also recorded on the flight summary page. A successful autoland that maintenance did not request needs no logbook entry at all.

Workbook wording: What are the requirements to log/report an autoland?

Diversion Considerations**6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL****When do you start the divert?****Early enough to reach the alternate with 30 minutes fuel**

Decide with Dispatch, and decide before you are into minimum fuel. Ask air traffic control for delay information early so you do not hold, then divert, and land at the alternate under 30 minutes. Minimum fuel is only declared once you are committed to a field, so it is not a decision tool.

FOM 11.2.5 Minimum Fuel

"The decision whether or not to divert to an alternate or other diversion airfield should be made in coordination with Dispatch prior to the declaration of minimum fuel."

If it becomes clear you will land with less than 30 minutes of fuel, declare MAYDAY MAYDAY MAYDAY FUEL. Thirty minutes is a floor, not a target.

Workbook wording: What factors should you consider when deciding when to start diverting to your alternate?

6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL**Not using your listed alternate. What do you weigh?****Aircraft state and fuel, enroute weather, runways and approaches, rescue and firefighting**

Weigh configuration, weight, systems status and fuel remaining, then wind and weather at the diversion altitude, minimum enroute altitudes, and fuel burn to get there. At the field, weigh terrain, weather, wind, runways and surface condition, approach aids and lighting, and rescue and firefighting. Your familiarity with the airport counts too.

FOM 11.2.7.1 Nearest Suitable Airport

"Aircraft configuration, weight, systems status, and fuel remaining • Wind and weather conditions enroute at the diversion altitude • Minimum altitudes enroute to the diversion airport • Fuel burn to the diversion airport"

With one engine out, having the fuel to go farther, passenger accommodation, and where maintenance sits are never justification for flying past the nearest suitable airport.

Workbook wording: If you are not diverting to your listed alternate, what factors should you consider before selecting the diversion airport?

LAHSO Policy

6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL

What is LAHSO?

Land and Hold Short Operations

LAHSO involves landing and holding short of an intersecting runway, an intersecting taxiway, or some other designated point on a runway. The Available Landing Distance (ALD) is measured from the landing threshold to the hold short point, and the hold short point is marked by a painted holding position marking and red and white hold short position signs.

FOM 5.6.32.1 Definitions LAHSO

"A point on the runway beyond which a landing aircraft with a LAHSO clearance is not authorized to proceed."

Revisions to this section require FAA approval, which signals it is regulatory text, not company preference.

6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL

What must you check before accepting LAHSO?

Stopping within the ALD, using AFM distance plus 1000 ft

The Captain may accept a LAHSO clearance only if every condition in 5.6.32.2 is satisfied. The HA determination is that existing conditions allow the aircraft to safely land and stop within the Available Landing Distance, where calculated landing distance equals the FAA-approved AFM distance plus 1000 ft for the actual configuration, environment and weight.

FOM 5.6.32.2 Operational Information LAHSO

"The calculated landing distance will be the FAA-approved Aircraft Flight Manual (AFM) distance plus 1000 ft"

Rev 125 revised LAHSO guidance to prohibit LAHSO on wet runways and require the tailwind to be calm, less than 3 kt. Read the current four factors in 5.6.32.2 before you quote them.

Workbook wording: What are the 4 different factors that must be considered prior to accepting a LAHSO clearance?

Complete Landing Performance Assessment

6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL

What is a TLR?

Takeoff and Landing Report

FOM 5.6.22.1 sets the surrounding policy. Captains shall ensure landing is allowed and executed within the appropriate performance data. The landing data used for in-flight analysis is based on input conditions, which is what makes an accurate runway assessment critical.

FOM 5.6.22.1 Landing Policy

"The landing data used for in-flight analysis is based on input conditions, this highlights the importance of an accurate runway assessment."

Acceptance is not weight-only: environmental conditions matter equally. Do not interchange the takeoff and landing crosswind tables, they serve different functions and the takeoff table is CG and weight sensitive while the TALPA landing table is not.

6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL

How do you request landing performance?

ACARS PERFORMANCE page, LDG CONDITIONS, enter wind and runway, then SEND

From the ACARS MENU select PERFORMANCE, then LDG CONDITIONS. The airport populates with the destination, ZFW and EFOA uplink from the weight and balance data, the only required entry is the steady wind, and gust, OAT and QNH default to the METAR. Set the RCAM code at 3L, 6 is dry, per runway third if needed, flaps FULL or 3, then SEND at 6R. View the reply under VIEW DATA. Page 3/3 carries anti-ice, fail factor, autoland, reverse credit, A/THR and VAPP additive options.

FCOM PER-ARD-ALP P 5/18 Landing Conditions

"To request landing performance access the LDG CONDITIONS page: PERFORMANCE → LDG CONDITIONS"

HA policy is no credit for reverse thrust. For a failure not on the release that affects landing performance, compute the distance with Flysmart (PRO-NOR-SOP-16).

Workbook wording: How can you request landing data performance?

6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL**What do HXX/TXX and LXX/RXX mean on the landing data printout?****No published source located, ask the check airman**

The landing data report layout is shown as a printed example figure in the FCOM PER-ARD-ALP section, which the text extract does not reproduce. The same section explains the two distances: SAFO Factored with a 15 percent margin, and Unfactored, the actual distance without margin.

FCOM PER-ARD-ALP P 15/18 Printed Example

"The SAFO Factored Distances include a safety margin of 15% as recommended by SAFO 19001."

No published source located, ask the check airman. Read the designators off a real printout with the check airman.

Workbook wording: On the landing data analysis printout, what do the designator/s "HXX/TXX" and "LXX/RXX" represent?

Altimeter procedures**6.1.1 PERFORM DESCENT AND TERMINAL ARRIVAL****When in the descent do you set QNH?****Approaching the transition level, when cleared to an altitude; set it on the EFIS panels and the standby**

When the aircraft approaches the transition level and is cleared for an altitude, both pilots set the QNH on the EFIS control panel and the standby altimeter and crosscheck the settings and altitudes. The QNH was preset on the ISIS during descent preparation. Any QNH change must also go on the FMS PERF APPR page.

FCOM PRO-NOR-SOP-17 P 3/6 Barometric Reference

"When the aircraft approaches the transition level, and when cleared for an altitude: BAROMETRIC REFERENCE..... SET/CROSSCHECK BOTH L2 ■ Set QNH on the EFIS control panel and on the standby altimeter ■ Crosscheck the barometric reference settings and the altitude indications."

A QNH that differs a lot from the one used in the approach preparation is a symptom of a barometric reference error; confirm it from all sources.

Workbook wording: When, in the descent, do you change your altimeter setting to QNH (local Altimeter)?

6.1.1 PERFORM DESCENT AND TERMINAL ARRIVAL**Transition level vs transition altitude?****Altitude is charted, flown on QNH; level is assigned, flown on standard**

Transition altitude is the climb boundary. At and below it you use the local altimeter setting and call the number an altitude. Transition level is the descent boundary. At and above it you use standard and call the number a flight level. The gap between them is the transition layer.

FOM 18.1.6 International Altimeter Settings

"The aircraft shall be operated with the appropriate QNH value set below the Transition Altitude and STD (1013.25 hPa/29.92 in Hg) above the Transition Level."

Outside North America both are usually lower than the 18,000 ft you are used to, and they vary by country. Central America uses FL200 and 19,000 ft unless charted otherwise.

Workbook wording: What is the difference between a transition level vs transition altitude?

6.1.1 PERFORM DESCENT AND TERMINAL ARRIVAL**What altimeter setting defines a level?****Standard, 29.92 in Hg or 1013.25 hectopascals**

A flight level is a height flown on the standard setting. Set 29.92 in Hg or 1013.25 hectopascals and the number you read is a flight level, spoken as flight level three five zero. It is a pressure surface, not a true height above the ground.

FOM 18.1.6 International Altimeter Settings

"The aircraft shall be operated with the appropriate QNH value set below the Transition Altitude and STD (1013.25 hPa/29.92 in Hg) above the Transition Level."

Two airplanes on standard stay separated from each other even when the actual pressure is far from standard, which is the whole point of using it in the upper airspace.

Workbook wording: What (altimeter setting) defines a "level"?

6.1.1 PERFORM DESCENT AND TERMINAL ARRIVAL

What altimeter setting defines an altitude?

QNH, the local altimeter setting

An altitude is a height above mean sea level flown on the local setting. Set the reported QNH, in inches of mercury or hectopascals, and the number you read is an altitude. You use it at and below the transition altitude, and it is what puts your indicator near field elevation on the ground.

FOM 18.1.6 International Altimeter Settings

"The aircraft shall be operated with the appropriate QNH value set below the Transition Altitude and STD (1013.25 hPa/29.92 in Hg) above the Transition Level."

Outside North America most settings come in hectopascals. Confirm the unit before you set it, because 1002 hectopascals and 30.02 inches are very different worlds.

Workbook wording: What (altimeter setting) defines an "altitude"?

6.1.1 PERFORM DESCENT AND TERMINAL ARRIVAL

What is QFE?

Setting referenced to field elevation; altimeter reads height above the field

QFE is the pressure at the airport itself. Set it and the altimeter reads zero on the runway and height above the field in the air, not height above sea level. We do not fly it. If a controller gives you QFE, ask for QNH and set that instead.

FOM 18.1.6 International Altimeter Settings

"Where QFE is given by ATC, the Flight Crews shall request QNH and set in the normal manner."

On the A330 the FCOM covers QFE under LIM-NAV Operations with QFE Barometric Reference and the MCDU approach minima insertion table. Where charted altitudes are built on QFE, use the correction tables on the chart.

Manually Perform Descent and Arrival (optional)

6.1.1 PERFORM DESCENT AND TERMINAL ARRIVAL

Is the autopilot required on an RNAV1 STAR?

No, but recommended in high workload environments

FOM 2.3.9.2 makes autopilot use a recommendation, not a requirement, and FOM 2.3.9.3 encourages hand flying in lower workload terminal environments to keep proficiency. The one place the FOM mandates automation for PBN is 5.1.4.1: RNP AR procedures with RNP values less than 0.3 or with RF legs shall use autopilot or flight director driven by the RNAV system.

FOM 2.3.9.2 Autopilot Basic Operating Procedures

"In high workload, high traffic environments, autopilot use is strongly recommended."

The only PBN autopilot mandate in the FOM is 5.1.4.1, RNP AR with RNP below 0.3 or RF legs, and the FCOM takeoff SOP repeats it: the AP must be engaged for RNP AR below 0.3 NM. Nothing requires the autopilot on an RNAV 1 STAR.

Workbook wording: Is the autopilot required to be engaged on an RNAV1 STAR?

8.2.1.5 CONDUCT APPROACH BRIEFING

What is in the arrival and approach TPC briefing?

Threats, Plan, Considerations. The PF starts it before descent

Prior to beginning the descent the PF initiates the approach briefing, which should be interactive, concise and focused on the big picture. T, Threats, shared: the PM identifies relevant threats using the threat list on the Crew Briefing Reference Card and any airport-specific threats in the 10-7, and the PF briefs additional threats as needed.

FOM 5.5.16 Threat-Based Approach Briefing/TPC

"Pilots jointly identify any relevant threats, followed by the PF presenting the arrival plan"

Threats is the only shared element; Plan and Considerations are PF. Management strategies are discussed as threats are identified rather than saved to the end. The departure equivalent is briefed the same way, see FOM 1.3.1 in the workbook.

Workbook wording: What items compose the Arrival and Approach TPC Briefing?

Flap and Gear Management

CRITICAL OBSERVABLE ITEM

8.1.1.5 CONFIGURE AIRPLANE FOR APPROACH/LANDING

Normal flap and gear schedule on approach?

Flaps 1 at green dot, flaps 2 by 2500 ft AGL, gear, flaps 3, flaps FULL, stable by 1000 ft.

At green dot select FLAPS 1, more than 3 NM before the final descent point and on the final descent with flaps 1 and S speed by 2500 ft AGL on a decelerated approach. At 2500 ft AGL minimum FLAPS 2, and when the flaps are at 2 gear down, autobrake confirmed and ground spoilers armed. When the gear is down FLAPS 3, then FLAPS FULL, check VAPP, ATHR in SPEED or off, and the landing checklist.

FCOM PRO-NOR-SOP-18-B P 3/6 Intermediate/Final Approach

"■ For decelerated approaches, the aircraft must reach or be established on the final descent with FLAPS 1 and S speed at or above 2 500 ft AGL"

If you intercept the glide below 2500 ft AGL, 2000 minimum, select FLAPS 2 at one dot below. Gear down early is the way to slow down; speed brakes raise VLS.

Workbook wording: What is the normal flap/gear approach configuration schedule?

Stabilized Approach Criteria

CRITICAL OBSERVABLE ITEM

8.1.1.5 CONFIGURE AIRPLANE FOR APPROACH/LANDING

Where are the stabilized approach criteria?

FCOM PRO-NOR-SOP-18-A stabilization criteria and the SCO gates; the FOM only points there

FOM 5.6.5 gives the policy and the warning and sends you to the fleet manual. The A330 criteria live in FCOM PRO-NOR-SOP-18-A, Stabilization Criteria: correct lateral and vertical path, landing configuration, thrust stabilized usually above idle at target speed, no excessive deviation. The gates and callouts are in PRO-NOR-SCO, Flight Parameters, Approach: 1000 ft AFE stabilized gate, 500 ft AFE go-around gate.

FCOM PRO-NOR-SOP-18-A P 3/6 Stabilization Criteria

"In order for the approach to be stabilized, all of the following conditions must be satisfied before, or at the stabilization height: ■ The aircraft is on the correct lateral and vertical flight path ■ The aircraft is in the desired landing configuration ■ The thrust is stabilized, usually above idle, and the aircraft is at target speed for approach ■ The flight crew does not detect any excessive flight parameter deviation."

The FOM's one hard number is runway alignment by 500 ft AFE, with 300 ft exceptions on the airport 10-7 page.

Workbook wording: Where is the Stabilized Approach Criteria parameters located?

CRITICAL OBSERVABLE ITEM

8.1.1.5 CONFIGURE AIRPLANE FOR APPROACH/LANDING

What is required at each stabilized approach gate?

1000 ft AFE: gear, final flaps, speedbrakes in, thrust set, sink 1000 max, speed +20/-5, STABLE; 500 ft: go-around gate.

The SCO approach table sets two gates. At 1000 ft AFE the stabilized gate: gear down, final flaps selected, speed brakes retracted, thrust appropriate for the configuration, vertical speed at or below 1000 ft/min unless briefed, airspeed target +20 and -5 kt. The PF calls STABLE, or CORRECTING SPEED or VERTICAL SPEED, and the PM answers CHECKED;

if the configuration criteria are not met the call is UNSTABLE, GO AROUND, FLAPS. At 500 ft AFE the go-around gate applies the same criteria with the PM calling STABLE.

FCOM PRO-NOR-SCO P 6/16 Flight Parameters, Approach

"Stabilized Gate 1 000 ft/min, must be briefed prior callouts, if Go (If 1 000 ft configuration to approach Around called by PF) criteria are not met) ■ Airspeed: Target +20 kt/-5 kt"

Only speed and vertical speed may be corrected through the 1000 ft gate; every other criterion is mandatory there. Runway alignment exceptions to 300 ft AFE are on the 10-7.

Workbook wording: What are the Stabilized Approach requirements at each "gate"?

CRITICAL OBSERVABLE ITEM

8.1.1.5 CONFIGURE AIRPLANE FOR APPROACH/LANDING

Why go around if not stabilized?

Unstable approaches must not be landed from

FOM 5.6.5 pairs the WARNING with the no-fault go-around policy: either crewmember may call for a go-around and the professional judgment of the flight crew is used to assess the probability of meeting the criteria.

FOM 5.6.5 Stabilized Approach

"Do not land from an unstable approach."

This is a WARNING in FOM 5.6.5, not a recommendation. Either crewmember may call the go-around under the Alaska no-fault policy, so an FO go-around call is not a challenge to your authority as PIC.

Workbook wording: Emphasize the importance of going around if the stabilized approach criteria are not met and the no fault go-around policy.

8.2.1 PERFORM CAT 1 ILS APPROACH WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

What are the ILS callouts and configuration steps?

ARM APPROACH, LOC/G/S blue then star, both APs, LAND at 350 ft, HUNDRED ABOVE, MINIMUM, CONTINUE or GO-AROUND.

Cleared, on the intercept and with LOC deviation shown, the PF presses APPR and calls the armed modes with their color, G/S blue and LOC blue, then engages the second AP and sets the go-around altitude. Captures are called without color, LOC, G/S. Configure per the schedule. At 350 ft RA check LAND engaged if autoland. At minimum plus 100 the PM calls ONE HUNDRED ABOVE, at minimum MINIMUM, and the PF answers CONTINUE or GO-AROUND. The PM calls LOC or GLIDESLOPE at half a dot.

FCOM PRO-NOR-SCO P 3/16 FMA

"■ All armed modes with their associated color (e.g. blue, magenta): "G/S blue", "LOC blue" ■ All active modes without their associated color (e.g. green, white): "NAV", "ALT". The PM should check and respond, "CHECKED" to all FMA changes called out by the PF."

Above 5000 ft AGL the FMA shows CAT 1 regardless; the real capability appears below 5000 ft. Captures engage no sooner than 3 seconds after arming.

Workbook wording: Discuss/demonstrate the callouts and configuration steps

8.2.1 PERFORM CAT 1 ILS APPROACH WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

What does the FMA show when the approach is armed?

LOC and G/S blue in the armed line; capability CAT 1 above 5000 ft, then CAT 2 or CAT 3.

Pressing APPR arms LOC and G/S, shown in blue in the armed lines of the FMA, and the PF calls G/S blue, LOC blue. Above 5000 ft AGL the FMA displays CAT 1; below 5000 ft it displays the actual approach capability, CAT 2, CAT 3 SINGLE or CAT 3 DUAL, once both APs are engaged.

FCOM PRO-NOR-SOP-18-C P 2/32 Initial/Intermediate Approach

"■ When APPR mode is selected, AP1 and AP2 should be engaged ■ Above 5 000 ft AGL, the FMA displays CAT 1 ■ Below 5 000 ft AGL, the FMA displays the correct approach capability for the intended approach."

Arm within the normal capture envelope, inside 10 NM and near the glide path angle. A spurious G/S star from a false beam outside it means AP off, APPR off, re-arm inside the zone.

Workbook wording: Discuss what is shown on FMA when the approach is armed

9.1.4.1 KNOW PROCEDURES FOR CAT II/III ILS AUTOLAND LANDING WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

What are the key steps of a CAT II or III autoland?

Both APs, CAT 2 or CAT 3 FMA below 5000 ft, LAND at 350 ft, autoland, degraded guidance strategy.

Descent preparation: choose the lowest achievable minimum, limited by crew qualification, the operating manual, aircraft status and airport status, enter NO DH for CAT III without DH, and brief task sharing, degraded guidance and the airport low visibility procedures. Both APs in APPR, capability displayed below 5000 ft AGL. Above 1000 ft a caution or capability loss means complete the procedure and check the required equipment or go around; below 1000 ft go around; below the 200 ft alert height for CAT 3 DUAL continue unless AUTOLAND lights.

FCOM PRO-NOR-SOP-18-C P 1/32 Approach Using LOC G/S Guidance, Descent Preparation

"For CAT II, CAT III, approaches, review the following items on top of the usual briefing: ■ Task sharing and callouts ■ Management of degraded guidance ■ Low visibility procedures at the airport."

CAT II DH minimum 100 ft, CAT III single 50 ft, CAT III dual no DH with 75 m RVR. Wind limits 35, 10 and 15 kt. Engine-out autoland only in CONF 3 with the procedures done before 1000 ft.

Workbook wording: Know Procedures for CAT II/III ILS Autoland Landing

9.1.4 PERFORM CAT II/III ILS AUTOLAND LANDING WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

What is the autoland sequence?

Both APs, APPR, LAND at 350 ft, FLARE, ROLL OUT, reversers, AP off, then the ACARS Autoland Report.

Fly the ILS with both autopilots engaged after APPR. At 350 ft RA check LAND engaged and announce it; if no LAND mode, autoland is not authorized. The AP flies FLARE and ROLL OUT, the PM checks and announces ground spoilers, reversers and autobrake, and the AP comes off at taxi speed. Every autoland attempt is reported on the ACARS Autoland Report page per FOM 5.6.21.2.

FCOM PRO-NOR-SOP-18-C P 3/32 Final Approach

"AT 350 ft RA LAND mode..... CHECK ENGAGED/ANNOUNCE PF L2 If no LAND mode, autoland is not authorized."

A CAT I autoland is only for practice on a CAT II or III runway with the tower advised, or on a CAT I beam under the FCOM precautions. Check the tower wind against 35, 10 and 15 kt.

Workbook wording: Perform CAT II/III ILS Autoland Landing

8.3.1 PERFORM MANAGED LATERAL/MANAGED VERTICAL APPROACH WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

Fly at least one FLS approach

FLS is the recommended mode for non-precision approaches except RNP AR; select it on PERF APPR and check F-APP capability

FLS, the FMS Landing System, flies F-LOC and F-G/S like an ILS for VOR, NDB, LOC-only and RNAV(GNSS) approaches. It is the default and recommended guidance for a non-precision approach inserted in the F-PLN; RNP AR approaches and cases where FLS is unavailable revert to FINAL APP. The limitation requires F-APP or F-APP+RAW capability on the FMA, with the navaid tuned and monitored in the RAW case.

FCOM PRO-NOR-SOP-16 P 5/10 PERF APPR page

"■ If a non precision approach is inserted in the F-PLN, the default guidance mode is FLS. Check/select the appropriate guidance mode (FLS or FINAL APP). Note: 1. FLS is the recommended guidance mode for the non precision approaches"

Press LS for an FLS approach and check the deviation scales and ident on the PFD. Discontinue if F-APP capability is lost or F-LOC exceeds one dot.

Workbook wording: (Airbus) Perform at least one FLS approach, if equipped

8.3.1 PERFORM MANAGED LATERAL/MANAGED VERTICAL APPROACH

What are the managed approach guidance modes?

LOC G/S for ILS, FLS (F-LOC F-G/S), FINAL APP, then selected LOC FPA, NAV FPA and TRK FPA

The SOP cross-reference table lists guidance by approach type. Fully managed: LOC G/S for ILS and GLS, FLS for LOC-only, VOR, NDB and RNAV(GNSS), and FINAL APP for RNAV(GNSS), VOR, NDB and RNP AR. Selected vertical: LOC FPA or LOC B/C FPA, NAV FPA and TRK FPA where authorized. RNP AR and RNP(VPT) are FINAL APP only.

FCOM PRO-NOR-SOP-18-A P 2/6 Cross-Reference Table

"The FLS (F-LOC F-G/S) is the recommended guidance mode for this type of approach."

Flying reference: HDG-V/S with the FD bars in managed vertical modes, TRK-FPA with the bird in selected vertical modes.

Workbook wording: What are the different levels of Managed (Airbus)/VNAV (Boeing) approaches?

8.3.1 PERFORM MANAGED LATERAL/MANAGED VERTICAL APPROACH

Which modes may be used for a non-precision approach?

FLS or FINAL APP managed; FPA guidance selected where the table authorizes it

For VOR, VOR-DME, NDB and NDB-DME the table allows FLS, FINAL APP, NAV FPA and TRK FPA. RNAV(GNSS) with LNAV minima allows FLS, FINAL APP and NAV FPA; with LNAV/VNAV minima only FLS or FINAL APP. LOC-only and LOC back course use LOC with F-G/S or FPA. LPV and LP minima use SLS. RNP AR is FINAL APP only.

FCOM PRO-NOR-SOP-18-A P 2/6 Cross-Reference Table

"RNP is equivalent to RNAV(GNSS). RNP(AR) is equivalent to RNAV(RNP)."

If NAV accuracy is LOW at 10 000 ft use TRK FPA for the approach. A TOO STEEP PATH message after the FDP rules out FINAL APP.

Workbook wording: What different approach modes(s) are eligible to perform a non-precision approach?

8.3.2 PERFORM MANAGED LATERAL/SELECTED VERTICAL APPROACH (AIRBUS) (OPTIONAL)

Which approaches are flown managed lateral, selected vertical?

LOC-only, LOC back course, VOR, NDB and RNAV with LNAV minima, using LOC FPA or NAV FPA

Managed lateral with selected vertical is LOC FPA, LOC B/C FPA or NAV FPA. The table authorizes it for LOC-only, ILS glide slope out and back course approaches, for VOR, VOR-DME, NDB and NDB-DME, and for RNAV(GNSS) with LNAV minima. It is not authorized for LNAV/VNAV minima, LPV, or RNP AR.

FCOM PRO-NOR-SOP-18-A P 3/6 Flying Reference

"■ In vertical managed modes: HDG-V/S reference associated with the FD crossbars ■ In vertical selected modes: TRK-FPA reference associated with FPD."

FPA needs the bird. Push HDG-V/S to TRK-FPA before the final descent point.

Workbook wording: What approaches would be flown in Managed/Selected?

8.1.1 PERFORM VISUAL APPROACH

Can you fly a visual approach at night?

Yes, unless that airport restricts night visuals

FOM 5.6.6.1 permits visual approaches with 1000 ft ceiling and 3 sm visibility or better, VFR cloud clearance, within 35 nm of the destination, under ATC control, with the airport or preceding aircraft in sight.

FOM 5.6.6.1 Visual Approach or RNAV Visual Flight Procedure (without CVFP)

"Airports that restrict night visual approaches require that an instrument approach procedure be flown to guarantee adequate terrain clearance."

Night restrictions are published in the airport 10-7 (see FOM 5.6.6.3 note). FOM 5.6.6.1 also tells you to consider terrain a threat any time a night visual is planned or even possible.

Workbook wording: Can we fly a visual approach at night?

10.1.2 PERFORM POSTFLIGHT

ACARS post-flight report pages?

All Autoland attempts and all RNP AR procedures

FOM 5.6.21.2 makes the Captain responsible for completing the ACARS Autoland Report page for every Autoland attempt, including the logbook page information. If ACARS/ATSU is inoperative, the Captain makes a logbook entry

stating Autoland successful or unsuccessful with airport, runway and category, and the reason for any failure.

FOM 5.6.21.2 Autoland Reporting - ACARS; FOM 7.1.13.1 RNP AR Required Reporting

"For all Autoland attempts, the Captain is responsible for ensuring that all applicable fields on the ACARS Autoland Report page are filled in, including the logbook page information."

Autoland reporting is every attempt, satisfactory or not. RNP AR reporting covers results of all RNP AR procedures plus any RNP degradation, and excludes visual approaches even if an RNP AR approach was displayed as reference (FOM 7.1.13.1 note).

Workbook wording: ACARS Post-Flight Report (Autoland and RNP-AR Report pages)

8.3.5.27 KNOW CONTINGENCY APPROACH PROCEDURES FOR RNP-AR OPERATIONS FOLLOWING SYSTEM MALFUNCTION, WEATHER DEVIATION OR ATC MANDATE WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

When must you discontinue an RNP AR approach?

No FINAL APP, dual GPS PRIMARY LOST or NAV ACCUR DOWNGRAD, position disagree, or XTK reaching 1 RNP.

Management of degraded navigation for RNAV(RNP): the approach may continue after a single failure of a GPS, FMGS, EFIS DU, MCDU or AP. Discontinue if FINAL APP does not engage, GPS PRIMARY LOST on both NDs, dual NAV ACCUR DOWNGRAD, FM/GPS POSITION DISAGREE, FMS1/FMS2 POS DIFF, dual FMGC or FINAL APP loss, dual AP failure with RNP below 0.3, loss of the GPWS terrain function, or NAV ALT DISCREPANCY. Go around if XTK reaches 1 RNP or V/DEV reaches three quarters of a dot below the path.

FCOM PRO-NOR-SOP-18-C P 18/32 Management of Degraded Navigation

"Discontinue the approach in the following cases: ■ FINAL APP does not engage ■ GPS PRIMARY LOST on both NDs ■ Dual NAV ACCUR DOWNGRAD ■ FM/GPS POSITION DISAGREE ■ FMS1/FMS2 POS DIFF ■ Dual loss of FMGC or dual loss of FINAL APP mode ■ Dual AP failure if the RNP < 0.3 NM"

Both GPS in NAV and both FMS available before the IAF or RNP AR is not permitted. Baro difference between altimeters no more than 100 ft. Brief the degraded navigation chapter as part of the go-around strategy.

Workbook wording: Know Contingency Approach Procedures for RNP-AR Operations Following System Malfunction, Weather Deviation or ATC Mandate

Landing and Rollout ⁽⁹⁾

Airspeed Control

CRITICAL OBSERVABLE ITEM

9.1.1 PERFORM MANUAL LANDING

Is autothrust required for approach and landing?

No. Recommended; required only for CAT III.

The FCTM recommends A/THR for the entire flight and for approaches because of its accurate speed control, with the PF's hand on the levers. It becomes a limitation for CAT III: fail passive and fail operational both require A/THR in selected or managed speed. The QRH severe turbulence procedure also wants A/THR on for the approach.

FCOM LIM-AFS-20 P 1/4 ILS Category III Fail Passive (Single)

"A/THR must be used in selected or managed speed. One autopilot must be engaged in APPR mode, and CAT 3 SINGLE or CAT 3 DUAL must be displayed on the FMA."

The Before Landing check is A/THR in SPEED mode or off; not in THR mode.

Workbook wording: Is the auto-thrust/auto-throttle system required during approach and/or landing?

CRITICAL OBSERVABLE ITEM

9.1.1 PERFORM MANUAL LANDING

What do you weigh before turning autothrust off for the approach?

Unsatisfactory A/THR performance, loss of autotrim, then disconnect by 1000 ft AAL

Use manual thrust if the A/THR performance is unsatisfactory, checked against the ground speed on the ND, or when the autotrim function is lost, to avoid large thrust changes. If you plan manual thrust for landing, disconnect the A/THR at 1000

ft AAL at the latest so the approach is stable through the gate.

FCTM PR-NP-SOP-190-CONF P 4/6 Use of A/THR

"If the A/THR performance is unsatisfactory, the PF should disconnect it and control the thrust manually. If the PF uses manual thrust for landing, the PF should disconnect the A/THR at 1 000 ft AAL at the latest."

Disconnect with the instinctive pushbutton, not the FCU. A thrust lock after an A/THR failure leaves thrust frozen until you move the levers.

Workbook wording: What factors should be considered if deciding to turn off the auto-thrust/auto-throttle system for the approach?

Centerline Drift / Crosswind Control

CRITICAL OBSERVABLE ITEM

9.1.1 PERFORM MANUAL LANDING

Where is the crosswind landing guidance?

FCTM PR-NP-SOP-250 Landing: de-crab in the flare, wings near level, partial crab up to 5 degrees

The FCTM landing chapter gives the technique: rudder to align with the runway in the flare, roll to hold the centerline, and in strong crosswind be ready for small bank into wind. Land with a partial de-crab, up to about 5 degrees of residual crab, rather than excessive bank, to protect the wingtip and nacelle. On the rollout do not hold stick into wind, and be ready to bring reversers to idle if directional control is a problem.

FCTM PR-NP-SOP-250 P 4/12 Landing, Crosswind Conditions

"The aircraft may be landed with a partial de-crab (residual crab angle up to about 5 °) to prevent excessive bank. This technique prevents wingtip (or engine nacelle) strike caused by an excessive bank angle."

Limits: 32 kt certified crosswind takeoff, 45 kt demonstrated landing, and the wet or contaminated table cuts both. CAT II and III autoland is 15 kt.

Workbook wording: Where can you find guidance on landing the aircraft in a crosswind?

Thrust Reverse Usage

CRITICAL OBSERVABLE ITEM

9.1.1 PERFORM MANUAL LANDING

Is max reverse required on landing?

No. REV MAX is standard; REV IDLE is allowed on dry, or wet with MEDIUM TO POOR no-credit assessment.

The selection of REV MAX is the standard practice for landing. On DRY runways the crew may select REV IDLE. On WET runways with GOOD condition REV IDLE is allowed only if the landing distance was assessed at MEDIUM TO POOR with no reverser credit and the result is within the LDA.

FCOM PRO-NOR-SOP-16 P 2/10 Landing Performance

"The selection of REV MAX is the standard practice for landing. However, on DRY runways the flight crew may select REV IDLE."

HA performance policy gives no credit for reverse thrust, so REV IDLE on a dry runway does not change the numbers.

Workbook wording: Is max thrust reverser required upon landing?

CRITICAL OBSERVABLE ITEM

9.1.1 PERFORM MANUAL LANDING

When do you stow the reversers?

Idle reverse at 70 kt, stow at taxi speed before leaving the runway

At 70 kt the PM calls SEVENTY KNOTS and the PF brings all reverser levers to idle, because high reverse at low speed can re-ingest exhaust and stall the engine. At taxi speed, before leaving the runway, stow the reversers. Reverse on taxiways ingests sand and debris and recirculates snow into the inlets.

FCOM PRO-NOR-SOP-19 P 3/6 At 70 Knots, At Taxi Speed

"When the aircraft reaches the taxi speed, and before it leaves the runway, stow the reversers."

Disengage the autobrake with the pedals before 20 kt to avoid brake jerks. High reverse may be held below 70 kt only in an emergency.

Workbook wording: When should the thrust reversers be stowed?

Brakes Usage**CRITICAL OBSERVABLE ITEM****9.1.1 PERFORM MANUAL LANDING****Braking technique with carbon brakes?****Fewer, firmer applications; wear depends on the number of applications, not pressure or duration**

Carbon brake wear depends on the number of brake applications and on brake temperature, not on the pressure applied or the duration. The only way to reduce wear is fewer applications: on a straight taxiway accelerate to 30 kt, one smooth application to 10 kt, no riding the brakes.

FCTM PR-NP-SOP-100 P 1/8 Carbon Brake Wear

"Carbon brake wear depends on the number of brake applications and on brake temperature. It does not depend on the applied pressure, or the duration of the braking."

On landing, autobrake gives one continuous application. If braking manually, one firm application is kinder to carbon than several light ones.

Workbook wording: What braking technique is used with carbon brakes?

CRITICAL OBSERVABLE ITEM**9.1.1 PERFORM MANUAL LANDING****Factored vs unfactored landing data?****Factored adds the SAFO 15 percent margin; unfactored is the actual distance with no margin**

The AeroData landing report gives SAFO Factored Distances with a 15 percent safety margin per SAFO 19001, and Unfactored Distances equal to the actual landing distance without failure and without margin, provided so crews know the real capability in an emergency. FOM 5.6.22.3 sets the A330 model: TALPA, a 7 second air distance, 15 percent on the total distance for max manual and autobrake.

FCOM PER-ARD-ALP P 16/18 Landing Data Report

"The Unfactored Distances are equivalent to the actual landing distances without failure and without a safety margin. Unfactored Distances are provided so that flightcrews can know the actual performance capability of the aircraft in case of emergency conditions."

Except in an emergency do not land where the SAFO Factored Distance exceeds the LDA; dashes replace a factored distance that does not fit.

Workbook wording: What is the difference between factored vs unfactored landing data?

Bounced Landing Recovery**8.5.1 PERFORM GO-AROUND/MISSED APPROACH****Key to a bounced landing recovery without a tail strike?****Hold the pitch attitude; do not increase it; go around from a high bounce**

Light bounce: maintain the pitch attitude and complete the landing with thrust at idle, and do not let pitch increase, especially after a firm touchdown with a high pitch rate. High bounce: maintain the pitch attitude and go around, without trying to avoid the second touchdown, which will be soft enough if the attitude is held.

FCTM PR-NP-SOP-250 P 10/12 Bouncing at Touch Down

"In case of high bounce, maintain the pitch attitude and initiate a go-around. Do not try to avoid a second touch down during the go-around."

The tail strike setup is pitching up to soften the second touchdown. Pitch stays where it is.

Workbook wording: What is a key component during a bounced landing recovery to avoid a tail strike?

9.1.1.14 PERFORM FCOM PROCEDURE(S) FOLLOWING LANDING WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

What are the after landing procedures?**Spoilers disarmed, lights, radar off, start selector NORM, flaps up, TCAS standby, transponder, APU, anti-ice, brake temps, checklist**

After Landing flow: ground spoilers disarmed once the runway is vacated; land and wing lights off, strobes AUTO, nose light TAXI; WXR and PWS off and both display selectors off; ENG START selector NORM; flaps 0 unless icing or slush, then wait for engine shutdown and a ground crew check; TCAS STBY; transponder per airport; APU as required; engine anti-ice as required; brake temperatures monitored; After Landing checklist once clear of the runway.

FCOM PRO-NOR-SOP-21 P 1/4 After Landing

"GROUND SPOILERS When the runway is vacated: GND

SPLRS.....DISARM PF"

Brake temperature splits drive maintenance action: more than 150 C between two brakes on a gear with one at or above 600 C or at or below 60 C, 200 C between gear averages, or any brake over 800 C.

Workbook wording: What are the FCOM procedures after landing?

MEL and Maintenance (17)**ETOPS Verification Flight**

16.1.1.7.6 ETOPS VERIFICATION FLIGHT PROCEDURE

Must the verification request be in the logbook before departure?**Yes**

The Flight Crew is notified of an open item on the MIC sheet. Maintenance places this entry in the Aircraft Maintenance Logbook: AIRCRAFT RELEASED TO ETOPS SERVICE IAW WITH HA ETOPS MAINTENANCE MANUAL FOR VERIFICATION FLIGHT DUE TO REPAIR OF _____.

FOM 6.5.10 ETOPS Verification Flight (HA)

"AIRCRAFT RELEASED TO ETOPS SERVICE IAW WITH HA ETOPS MAINTENANCE MANUAL FOR VERIFICATION FLIGHT DUE TO REPAIR OF"

Workbook wording: Does the verification flight item request have to be in the maintenance logbook prior to departure?

16.1.1.7.6 ETOPS VERIFICATION FLIGHT PROCEDURE

Must Maintenance talk to the crew before an EVF?**Yes, verbally**

Maintenance will verbally notify the Flight Crew and place the release entry in the Aircraft Maintenance Logbook. The notification is therefore threefold: the MIC sheet open item, the verbal notification, and the logbook entry.

FOM 6.5.10 ETOPS Verification Flight (HA)

"Maintenance will also verbally notify the Flight Crew"

Workbook wording: Is Maintenance required to communicate with the flight crew prior to departing on an ETOPS verification flight?

16.1.1.7.6 ETOPS VERIFICATION FLIGHT PROCEDURE

After talking to Maintenance, must the PIC call Dispatch?**Yes, a verbal acknowledgement**

That closes the loop between Maintenance, the crew and Dispatch before departure.

FOM 6.5.10 ETOPS Verification Flight (HA)

"The Flight Crew must verbally acknowledge the EVF requirement with Dispatch."

Workbook wording: Does the PIC have to communicate with dispatch after speaking with Maintenance?

ETOPS APU Verification Flight**16.1.1.7.6 ETOPS VERIFICATION FLIGHT PROCEDURE****What logbook entry is required after the verification flight?****Verification flight completed satisfactorily or unsatisfactorily**

The Captain must make an entry in the Aircraft Maintenance Logbook detailing the results: VERIFICATION FLIGHT FOR ____ COMPLETED SATISFACTORILY or UNSATISFACTORILY. If the test was unsuccessful the Captain must list any discrepancies. For the APU specifically, a successful entry must state: VERIFICATION FLIGHT FOR APU COMPLETED, APU OPERATES NORMALLY.

FOM 6.5.10 ETOPS Verification Flight (HA)

"VERIFICATION FLIGHT FOR ____ COMPLETED "SATISFACTORILY" or "UNSATISFACTORILY"

Captain writes this, not the FO and not Maintenance. The APU wording is distinct from the generic component wording, so know both.

Workbook wording: What maintenance logbook entry (specific verbiage) must be entered after the verification flight?

Company Reporting**CRITICAL OBSERVABLE ITEM****5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE****How do you report a discrepancy before landing?****ACARS or radio to Dispatch, who relays to Maintenance**

Report through Dispatch using ACARS or radio. FOM 11.1.2 requires contacting Dispatch for abnormalities as safety of flight permits, and if time does not permit in flight, after gate arrival. Reporting early lets Maintenance position parts and personnel at the gate rather than starting the troubleshooting clock on block-in.

FOM 11.1.2 Abnormal Operations Dispatch Notification

"If time does not permit contact with Dispatch while in flight, contact shall be made after gate arrival."

That is also the answer to the advantages question in the maintenance logbook section: enroute notification converts ground time from diagnosis into repair.

Workbook wording: How can you report a discrepancy to Maintenance in advance, prior to landing?

SATCOM / HF Procedures**CRITICAL OBSERVABLE ITEM****5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE****Who do you call on SATCOM for maintenance or a guest issue?****Dispatch**

Inflight, the Dispatcher is the primary contact and shares operational control with the Captain (FOM 7.1.1.2). ACARS messages sent to Maintenance Control go automatically to the Dispatcher, and Maintenance does not normally communicate directly with an airborne aircraft because their diagnostic guidance differs from operational guidance for crews.

FOM 7.1.1.2 In-Flight Operations

"The Dispatcher is the primary contact for requests and notifications, both managing information flow and sharing operational control with the Captain."

Maintenance will not normally talk directly to an airborne aircraft; the FODO provides the link to Maintenance and Engineering. For a medical event, MedLink is the SATCOM call (FOM 7.1.7).

Workbook wording: When we make a SATCOM call for Maintenance or guest issue, who do we call?

CRITICAL OBSERVABLE ITEM**1.2.1.2.5.3 KNOW MAINTENANCE LOGBOOK PROCEDURES****Where are the logbook review procedures?****Maintenance Logbook Preflight Review, in the FOM maintenance chapter.**

Hawaiian procedures sit under Maintenance Logbook Procedures, Preflight Review. Every pilot reviews at least the last 10 pilot entered discrepancies, not counting No Items or Info to Maintenance entries. Alaska has its own preflight review paragraph a few pages earlier in the same chapter.

FOM 12.4.12.1 Maintenance Logbook Preflight Review

"Prior to the first flight on a particular aircraft, all Flight Crew shall review the Aircraft Maintenance Logbook."

Deferred items get a second look. Check the MEL, CDL or SDL for restrictions, operational procedures and performance penalties, and confirm every one is on the release.

Workbook wording: Where in the FOM can you find the maintenance logbook review/checking procedures?

CRITICAL OBSERVABLE ITEM

1.2.1.2.5.3 KNOW MAINTENANCE LOGBOOK PROCEDURES

How long is the PDSC good for?

Four hours from the Airworthiness Release signature.

The clock starts when the Airworthiness Release block is signed in the M258 logbook, not at scheduled departure. If four hours pass and boarding has not started, the check is invalid and Maintenance must redo it. If boarding started in time you may continue, but tell Maintenance Control.

FOM 6.5.4 ETOPS Pre-Departure Service Check, PDSC Policy

"An ETOPS PDSC Airworthiness Release will be valid for four (4) hours from the time the Airworthiness Release block was signed in the M258 Aircraft Maintenance Logbook."

An aircraft swap always needs a new check. The check belongs to that tail and that flight number and cannot follow the airplane to another flight.

CRITICAL OBSERVABLE ITEM

1.2.1.2.5.3 KNOW MAINTENANCE LOGBOOK PROCEDURES

White HAL 46-1333 versus yellow 46-1333b placard?

No published source located, ask the check airman

The FOM describes HAL 46-1333 as the Project Operation and Maintenance Information Form placed in the logbook to notify the crew of special installations or engineering modifications, and the APU in-flight start request uses the same form as a placard card. The yellow 46-1333b variant is MEL introduction material, not in the FOM.

FOM 12.4.12.7 HAL Project Operation and Maintenance Information Form

"When required, a Project Operation and Maintenance Information Form (HAL 46-1333) will be placed in the Aircraft Maintenance Logbook (HAL M258) to notify Flight Crew of special installations or engineering modifications that have been accomplished on an aircraft."

No published source located, ask the check airman. The MEL general section carries the placard color rules.

Workbook wording: What is the difference between a white (HAL 46-1333) vs yellow (HAL 46-1333b) MEL placard?

CRITICAL OBSERVABLE ITEM

1.2.1.2.5.3 KNOW MAINTENANCE LOGBOOK PROCEDURES

What are the four repair intervals?

Categories A, B, C, and D.

Repair categories run A through D. Category A carries its own limit written into the MEL item itself, while B, C and D are fixed day limits published in the MEL general pages. All four are dispatch limits, not limits on the flight once it is under way.

FOM 8.5.4.1 Dispatch Release Sample and Description

"The repair categories of A, B, C, or D in an MEL are dispatch time limits not flight operational time limits."

Dispatch happens at pushback or engine start, whichever comes first. Start inside the limit and you may finish that flight, but you may not continue past the next termination point.

Workbook wording: What are the 4 different repair intervals

CRITICAL OBSERVABLE ITEM**1.2.1.2.5.3 KNOW MAINTENANCE LOGBOOK PROCEDURES****What is an (O) procedure in the MEL?****A crew operational procedure required with that item inoperative.**

An (O) means the flight crew has something to do to operate with that item deferred, such as a limit, a workaround or an extra check. Read it during the logbook review, before you accept the airplane. An (M) is the maintenance side, done by mechanics.

FOM 12.4.12.1 Maintenance Logbook Preflight Review

"all Flight Crew shall review the aircraft MEL/CDL/SDL for applicability, restrictions, MEL Operational Procedures (O), and performance decrements."

The full wording lives in the MEL introduction. The FOM requires you to review deferred items for applicability, restrictions, operational procedures and performance decrements.

Workbook wording: What is an "(O)" procedure listed in an MEL?

CRITICAL OBSERVABLE ITEM**1.2.1.2.5.3 KNOW MAINTENANCE LOGBOOK PROCEDURES****Where is the crew placarding procedure?****Maintenance chapter of the FOM, and Crew Placarding in the MEL.**

The FOM paragraph sets the conditions: a discrepancy after pushback but before takeoff, with Dispatch and Maintenance Control both concurring, and no return to the gate needed. The step by step placarding text is in the MEL under Crew Placarding. Remote placarding follows right after it.

FOM 12.4.12.9 Crew Placarding Procedure

"If the item is Crew Placardable in the MEL/CDL/SDL, follow the appropriate procedures as described in the MEL/CDL/SDL under Crew Placarding and a return to gate (BTB) is not required."

Maintenance Control normally sends the whole item by ACARS. Print it, check the blocks, do the placarding and any procedures, and note the item on the release.

Workbook wording: Where can you find the Crew Placarding Procedure?

Maintenance Communication Procedure**CRITICAL OBSERVABLE ITEM****1.2.1.2.5.3 KNOW MAINTENANCE LOGBOOK PROCEDURES****EVF communication requirements with Maintenance?****Call or ACARS Dispatch 30-60 minutes after takeoff, before EEP**

FOM 6.5.10 (HA): the crew is notified of an open item on the MIC sheet and Maintenance also verbally notifies the crew. The logbook entry reads AIRCRAFT RELEASED TO ETOPS SERVICE IAW WITH HA ETOPS MAINTENANCE MANUAL FOR VERIFICATION FLIGHT DUE TO REPAIR OF _____. The crew verbally acknowledges the EVF requirement with Dispatch.

FOM 6.5.10 ETOPS Verification Flight (HA)

"Between 30 and 60 minutes after takeoff and prior to reaching the EEP, the Flight Crew will call or send an ACARS message to Dispatch to report the status of the affected component or system."

Gotcha: the in-flight report goes to Dispatch, not to Maintenance. Maintenance's role is the up-front verbal notification plus the logbook entry, and the crew verbally acknowledges the EVF requirement with Dispatch. The report AND Dispatch's acknowledgment must both be complete before the EEP.

Workbook wording: What are the ETOPS verification flight communication requirements regarding contacting Maintenance prior to departure?

CRITICAL OBSERVABLE ITEM**1.2.1.2.5.3 KNOW MAINTENANCE LOGBOOK PROCEDURES****Why tell Maintenance enroute rather than at the gate?****The arrival station can have Maintenance ready before you block in.**

Airborne, put the discrepancy in the logbook and pass it to Maintenance and Dispatch by the ACARS maintenance report page or company radio. Telling them early is what lets the inbound station plan for the airplane. Waiting until block in gives them no lead time.

FOM 12.4.5.3 After Takeoff

“Communicate defects via ACARS MAINT REPORT page (primary) or Company radio. Ensure the inbound station is aware of the defect.”

MEL and crew placarding procedures are not used in flight. Enroute you report and coordinate. The deferral or repair happens on the ground.

Workbook wording: What are the advantages of notifying MX of a discrepancy enroute (time permitting) vs after blocking in?

APU In-Flight Start Program (APU component currency)

CRITICAL OBSERVABLE ITEM

16.1.1.7 PERFORM ETOPS PROCEDURES

What release items matter for the APU in-flight start program?

The APU Administrative Control Item and any conflicting MEL items

Per FOM 12.4.12.5, a Maintenance request for an APU in-flight start is issued as an Administrative Control Item (ACI) found in Section 05 of the fleet MEL, and that ACI is reflected on the Dispatch Release.

FOM 12.4.12.5 Maintenance Requests / FOM 6.5.6 APU In-Flight Start Program (HA)

“These requests will be in the form of an Administrative Control Item (ACI) as found in Section 05 of the fleet MEL. The associated ACI will be applied to the aircraft and will be reflected on the Dispatch Release.”

The ACI comes from Section 05 of the fleet MEL. Crew is also notified by an open item on the MIC sheet and a placard card (form HAL-46-1333) in the AML per FOM 6.5.6.

Workbook wording: What items on the dispatch release must you take note of when planning an APU In-Flight Start Program flight?

CRITICAL OBSERVABLE ITEM

16.1.1.7 PERFORM ETOPS PROCEDURES

What goes in the logbook after the flight?

No Items, Info to Maintenance, or a discrepancy, plus signature.

If you are done with the airplane, the Captain closes out the logbook. Record No Items, Info to Maintenance, or the mechanical discrepancy, then sign and add your employee number on the proper lines. Info to Maintenance is information only, so never use it to preface a real discrepancy.

FOM 12.4.12.4 Post-Flight Logbook Closeout

“This can be accomplished by recording “No Items,” “Info to Maintenance,” or a mechanical discrepancy as appropriate and entering their signature and employee number on the appropriate lines.”

Date entries by current GMT wherever you are, one discrepancy per block, and leave the Action Taken column alone.

Workbook wording: What items must be entered into the maintenance logbook after flight completion?

CRITICAL OBSERVABLE ITEM

16.1.1.7 PERFORM ETOPS PROCEDURES

What A/C Currency Items are listed in the “Administrative Control Items” (ACI’s)?

Max thrust takeoff, autoland, and APU in-flight start

FOM 12.4.12.5: Maintenance may request a max thrust takeoff, and on the A330 an autoland or an APU in-flight start. Each request is an Administrative Control Item from Section 05 of the fleet MEL, applied to the aircraft and reflected on the Dispatch Release, and the crew follows the ACI procedure in the MEL.

FOM 12.4.12.5 Maintenance Requests

“• (787/A321/A330) Autoland, APU in-flight start”

A completed ACI may still show on a later release that was generated before you finished it.

Abnormals and QRH (3)

Non-Normal Start Operations

1.3.4.1 COMPLETE NORMAL ENGINE START FCOM PROCEDURE

Is there a QRH procedure for a tailpipe fire?

Yes. ENGINE TAILPIPE FIRE, QRH 17.09A

ENG MASTER of the affected engine OFF, establish air bleed pressure, beacon ON, then when N3 is below 30 percent ENG START selector to CRANK and ENG MAN START pushbutton ON to motor the engine. When the fire stops, MAN START OFF and selector NORM. The caution says external fire agents are corrosive, so use them only if the procedure does not stop the fire.

QRH 17.09A ENGINE TAILPIPE FIRE

"ENGINE TAILPIPE FIRE Ident.: ABN-17-00010574.0004001 / 16 MAY 23 Applicable to: ALL CAUTION External fire agents can cause severe corrosive damage."

A tailpipe fire is a fuel fire in the exhaust with no fire warning. Do not pull the fire push button; you need bleed air to crank it out.

Workbook wording: Is there a QRH procedure for a TAILPIPE FIRE?

1.3.4.1 COMPLETE NORMAL ENGINE START FCOM PROCEDURE

Is there a QRH procedure for an engine fire on the ground?

No. It is the ECAM procedure ENG FIRE (ON GROUND) in FCOM PRO-ABN-ENG.

The QRH abnormal section carries only the non-ECAM procedures. An engine fire on the ground is handled by ECAM as ENG 1(2) FIRE (ON GROUND), listed in the FCOM abnormal chapter alongside ENG 1(2) FIRE (IN FLIGHT) and the FIRE DET FAULT variants. The tailpipe fire is the QRH item, at 17.09A.

FCOM PRO-ABN-PLP-TOC P 7/16 ENG procedures list

"ENG 1(2) FIRE (ON GROUND) X"

Ground fire means the fire push button, both agents and the evacuation decision. Tailpipe fire means crank it with the master off. Do not mix the two.

Workbook wording: Is there a QRH procedure for an ENG FIRE on the ground?

Oceanic Contingency Procedures

CRITICAL OBSERVABLE ITEM

13.1.1.1.1 KNOW ENGINE FAILURE IN CRUISE PROCEDURE WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

Review the PRC immediate actions for an engine failure in cruise

MCT and A/THR off together, pull green dot, 5 NM offset oceanic, pull FL200, start APU, then ECAM

As soon as the failure is recognized: simultaneously set MCT and disconnect autothrust with the instinctive disconnect button, pull speed to green dot, for oceanic operations turn at least 30 degrees to establish a 5 NM parallel offset, pull FL200 until the final driftdown altitude is known, start the APU. Exterior lights on, monitor TCAS, declare MAYDAY, then accomplish ECAM actions. Later reset REC MAX EO in the FCU and adjust speed for the strategy.

PRC p1 Engine Failure During Cruise Procedure

"SIMULTANEOUSLY, SET MCT AND DISCONNECT AUTOTHRUST Use the Instinctive Disconnect P/B. PULL SPEED - GREEN DOT"

The crew must not decelerate below green dot; at high level close to the weight limit the speed decays fast, so do not delay the descent.

Workbook wording: Review immediate action items (787 back of QRC, Airbus in PRC) for engine failure in cruise

CRM and PM Duties (16)

CRITICAL OBSERVABLE ITEM

12.3.5 DEMONSTRATE AND APPLY PRINCIPLES OF TEAM EFFECTIVENESS WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

Demonstrate team effectiveness

Captain sets the tone, uses every crewmember, mentors with credibility, and solicits feedback

Graded CRM dimension on the OE grade sheet. The standard is the FOM CRM and TEM chapter: Communication, Workload Management, Leadership and Flight Discipline as core skills, with TEM as the framework for preventing, trapping and mitigating threats and errors.

FOM 10.1.1.3 Leadership Command, Mentoring Teamwork

"The Captain sets an approachable and responsive tone for the Flight Deck to optimize team performance by acknowledging the importance of and utilizing all applicable crewmembers in task accomplishment."

Assess the crew elements, proficiency, experience and recency, and modify roles as the situation dictates. On an augmented crew that includes the relief pilot's briefing before the rest break.

Workbook wording: Demonstrate and Apply Principles of Team Effectiveness

CRITICAL OBSERVABLE ITEM

12.1.8 DEMONSTRATE SITUATIONAL AWARENESS AND MANAGEMENT OF INFORMATION WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

Demonstrate situational awareness

Continuously monitor, state and remain aware of conditions; verbalize the plan for actual, expected and contingent situations

Graded CRM dimension on the OE grade sheet. The standard is the FOM CRM and TEM chapter: Communication, Workload Management, Leadership and Flight Discipline as core skills, with TEM as the framework for preventing, trapping and mitigating threats and errors.

FOM 10.1.1.2 Workload Management, Situational Assessment

"Continuously monitor, state, and remain aware of conditions that affect operations."

The FOM standard is stated aloud: verbalizing the assessment is how the PM knows you have it. Vigilance scales with the workload condition.

Workbook wording: Demonstrate Situational Awareness and Management of Information

CRITICAL OBSERVABLE ITEM

12.1.3 DEMONSTRATE EFFECTIVE COMMUNICATION (INCLUDING BRIEFINGS) WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

Demonstrate effective communication and briefings

Open interactive communication, seek information, scalable assertiveness: Ask, Say, Command, Assume Control

Graded CRM dimension on the OE grade sheet. The standard is the FOM CRM and TEM chapter: Communication, Workload Management, Leadership and Flight Discipline as core skills, with TEM as the framework for preventing, trapping and mitigating threats and errors.

FOM 10.1.1.1 Communication

"The Captain is responsible for setting the overall tone and stating minimal acceptable performance standards for safe operations when not specifically defined in the SOP."

Briefings review threats and management strategies and assign specific PM duties. Do not mitigate your speech in a high-risk situation.

Workbook wording: Demonstrate Effective Communication (Including Briefings)

CRITICAL OBSERVABLE ITEM

12.3.2 DEMONSTRATE AND APPLY LEADERSHIP CONCEPTS WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

Demonstrate leadership

Accept PIC authority, balance command with crew participation, decide and state the decision, resolve conflicts on what is right

Graded CRM dimension on the OE grade sheet. The standard is the FOM CRM and TEM chapter: Communication, Workload Management, Leadership and Flight Discipline as core skills, with TEM as the framework for preventing, trapping and mitigating threats and errors.

FOM 10.1.1.3 Leadership Command, Decision Making

"After technical and/or CRM input is considered, the Captain makes a decision and clearly states it to the crew."

Immediate decisions when time or operational restrictions dictate; otherwise methodical threat, risk, course of action. Resolutions are command, collaborate or accommodate.

Workbook wording: Demonstrate and Apply Leadership Concepts

CRITICAL OBSERVABLE ITEM

12.1.7 DEMONSTRATE PROBLEM SOLVING AND DECISION MAKING INCLUDING THREAT ERROR MANAGEMENT WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

Demonstrate TEM and decision making

Anticipate, recognize, recover: identify threats, build strategies, trap errors with SOPs and checklists, mitigate any undesired aircraft state

Graded CRM dimension on the OE grade sheet. The standard is the FOM CRM and TEM chapter: Communication, Workload Management, Leadership and Flight Discipline as core skills, with TEM as the framework for preventing, trapping and mitigating threats and errors.

FOM 10.1.2.3 Anticipate/Recognize/Recover

"Anticipation is at the heart of TEM. It is essentially a two-step process: (1) identify relevant threats and (2) develop management strategies for each utilizing various resources."

See it, Say it obligates the PF to respond. If the PF does not react to a UAS the PM commands the correction in one or two words, and as a last resort takes control.

Workbook wording: Demonstrate Problem Solving and Decision Making including Threat Error Management

CRITICAL OBSERVABLE ITEM

12.1.9 DEMONSTRATE WORKLOAD MANAGEMENT WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

Demonstrate workload management

Fly the aircraft first, prioritize by risk, manage time, use and manage automation, report overload

Graded CRM dimension on the OE grade sheet. The standard is the FOM CRM and TEM chapter: Communication, Workload Management, Leadership and Flight Discipline as core skills, with TEM as the framework for preventing, trapping and mitigating threats and errors.

FOM 10.1.1.2 Workload Management, Prioritize Tasks

"■ Fly the aircraft first. ■ When the aircraft is stable and safe, navigate while properly analyzing the situation. ■ Prioritize operational tasks based upon risk/criticality to safety of flight."

Automation is a workload tool: use it in high-density traffic, drop it when it raises workload or lowers awareness, and when hand flying have the PM make the automation inputs.

Workbook wording: Demonstrate Workload Management

11.1.1 PERFORM PM DUTIES DURING TAXI WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

PM duties during taxi

Airport diagram out and followed, ATC and company comms, flight control check calls, taxi flow items, runway crossing dual acknowledgment

Task sharing is defined phase by phase in FCOM PRO-NOR-TSK and in each SOP chapter with the PF, PM, PF-PM and BOTH labels. PF or PM means a single crewmember does it, PF-PM means both do it but not necessarily together, BOTH means at the same time because coordination is needed. The FOM adds the PM's flight discipline standard: monitor the flight path first and never let secondary tasks interfere.

FCOM PRO-NOR-TSK P 1/26 Tasksharing, General Information

"PF or PM is used for individual actions. The actions should be completed by a single crew member, the PF or the PM. This rule also applies for CM1 or CM2."

FOM 10.1.1.4: the PM assesses and communicates threats, holds the shared mental model, and uses Ask, Say, Command to restore safety margins.

Workbook wording: Perform PM Duties During Taxi

11.1.2 PERFORM PM DUTIES DURING TAKEOFF WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL**PM duties during takeoff****THRUST SET, ONE HUNDRED KNOTS, V1 monitor, ROTATE, POSITIVE RATE, gear, PFD and engine monitoring**

Task sharing is defined phase by phase in FCOM PRO-NOR-TSK and in each SOP chapter with the PF, PM, PF-PM and BOTH labels. PF or PM means a single crewmember does it, PF-PM means both do it but not necessarily together, BOTH means at the same time because coordination is needed. The FOM adds the PM's flight discipline standard: monitor the flight path first and never let secondary tasks interfere.

FCOM PRO-NOR-TSK P 1/26 Tasksharing, General Information

"PF or PM is used for individual actions. The actions should be completed by a single crew member, the PF or the PM. This rule also applies for CM1 or CM2."

FOM 10.1.1.4: the PM assesses and communicates threats, holds the shared mental model, and uses Ask, Say, Command to restore safety margins.

Workbook wording: Perform PM Duties during Takeoff

11.1.3 PERFORM PM DUTIES DURING DEPARTURE AND CLIMB WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL**PM duties during departure and climb****Flap retraction on schedule, after takeoff flow, altitude and speed deviation callouts, FMA CHECKED responses**

Task sharing is defined phase by phase in FCOM PRO-NOR-TSK and in each SOP chapter with the PF, PM, PF-PM and BOTH labels. PF or PM means a single crewmember does it, PF-PM means both do it but not necessarily together, BOTH means at the same time because coordination is needed. The FOM adds the PM's flight discipline standard: monitor the flight path first and never let secondary tasks interfere.

FCOM PRO-NOR-TSK P 1/26 Tasksharing, General Information

"PF or PM is used for individual actions. The actions should be completed by a single crew member, the PF or the PM. This rule also applies for CM1 or CM2."

FOM 10.1.1.4: the PM assesses and communicates threats, holds the shared mental model, and uses Ask, Say, Command to restore safety margins.

Workbook wording: Perform PM Duties During Departure and Climb

11.1.4 PERFORM PM DUTIES DURING CRUISE WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL**PM duties during cruise****Waypoint checks, fuel score, position reports, weather and dispatch communication, altimeter crosschecks**

Task sharing is defined phase by phase in FCOM PRO-NOR-TSK and in each SOP chapter with the PF, PM, PF-PM and BOTH labels. PF or PM means a single crewmember does it, PF-PM means both do it but not necessarily together, BOTH means at the same time because coordination is needed. The FOM adds the PM's flight discipline standard: monitor the flight path first and never let secondary tasks interfere.

FCOM PRO-NOR-TSK P 1/26 Tasksharing, General Information

"PF or PM is used for individual actions. The actions should be completed by a single crew member, the PF or the PM. This rule also applies for CM1 or CM2."

FOM 10.1.1.4: the PM assesses and communicates threats, holds the shared mental model, and uses Ask, Say, Command to restore safety margins.

Workbook wording: Perform PM Duties During Cruise

11.1.5 PERFORM PM DUTIES DURING DESCENT AND ARRIVAL WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL**PM duties during descent and arrival****Weather and landing information, ECAM status, landing performance, 10 000 ft flow, transition level QNH**

Task sharing is defined phase by phase in FCOM PRO-NOR-TSK and in each SOP chapter with the PF, PM, PF-PM and BOTH labels. PF or PM means a single crewmember does it, PF-PM means both do it but not necessarily together, BOTH means at the same time because coordination is needed. The FOM adds the PM's flight discipline standard: monitor the flight path first and never let secondary tasks interfere.

FCOM PRO-NOR-TSK P 1/26 Tasksharing, General Information

"PF or PM is used for individual actions. The actions should be completed by a single crew member, the PF or the PM. This rule also applies for CM1 or CM2."

FOM 10.1.1.4: the PM assesses and communicates threats, holds the shared mental model, and uses Ask, Say, Command to restore safety margins.

Workbook wording: Perform PM Duties During Descent and Arrival

11.1.5.2 MONITOR AND RESPOND TO ATC/COMPANY COMMUNICATIONS WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL**Monitor and respond to ATC and company communications****PM handles ATC and company on VHF 1 and 2, dual acknowledgment of runway clearances, written complex taxi instructions**

Task sharing is defined phase by phase in FCOM PRO-NOR-TSK and in each SOP chapter with the PF, PM, PF-PM and BOTH labels. PF or PM means a single crewmember does it, PF-PM means both do it but not necessarily together, BOTH means at the same time because coordination is needed. The FOM adds the PM's flight discipline standard: monitor the flight path first and never let secondary tasks interfere.

FCOM PRO-NOR-TSK P 1/26 Tasksharing, General Information

"PF or PM is used for individual actions. The actions should be completed by a single crew member, the PF or the PM. This rule also applies for CM1 or CM2."

FOM 10.1.1.4: the PM assesses and communicates threats, holds the shared mental model, and uses Ask, Say, Command to restore safety margins.

Workbook wording: Monitor and Respond to ATC/Company Communications

11.1.7 PERFORM PM DUTIES DURING APPROACH WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL**PM duties during approach****Flap and gear selections on order, LOC and GLIDESLOPE half dot calls, ONE HUNDRED ABOVE, MINIMUM, stabilized gate calls**

Task sharing is defined phase by phase in FCOM PRO-NOR-TSK and in each SOP chapter with the PF, PM, PF-PM and BOTH labels. PF or PM means a single crewmember does it, PF-PM means both do it but not necessarily together, BOTH means at the same time because coordination is needed. The FOM adds the PM's flight discipline standard: monitor the flight path first and never let secondary tasks interfere.

FCOM PRO-NOR-TSK P 1/26 Tasksharing, General Information

"PF or PM is used for individual actions. The actions should be completed by a single crew member, the PF or the PM. This rule also applies for CM1 or CM2."

FOM 10.1.1.4: the PM assesses and communicates threats, holds the shared mental model, and uses Ask, Say, Command to restore safety margins.

Workbook wording: Perform PM Duties During Approach

11.1.9 PERFORM PM DUTIES DURING LANDING WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL**PM duties during landing****SPOILERS, REVERSERS, AUTO BRK, DECEL, SEVENTY KNOTS announced from the ECAM and PFD**

Task sharing is defined phase by phase in FCOM PRO-NOR-TSK and in each SOP chapter with the PF, PM, PF-PM and BOTH labels. PF or PM means a single crewmember does it, PF-PM means both do it but not necessarily together, BOTH means at the same time because coordination is needed. The FOM adds the PM's flight discipline standard: monitor the flight path first and never let secondary tasks interfere.

FCOM PRO-NOR-TSK P 1/26 Tasksharing, General Information

"PF or PM is used for individual actions. The actions should be completed by a single crew member, the PF or the PM. This rule also applies for CM1 or CM2."

FOM 10.1.1.4: the PM assesses and communicates threats, holds the shared mental model, and uses Ask, Say, Command to restore safety margins.

Workbook wording: Perform PM Duties During Landing

11.1.8 PERFORM PM DUTIES DURING GO-AROUND/MISSED APPROACH WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL**PM duties during go-around****FLAPS on order, POSITIVE RATE, gear up, FMA and altitude monitoring, ATC notification**

Task sharing is defined phase by phase in FCOM PRO-NOR-TSK and in each SOP chapter with the PF, PM, PF-PM and BOTH labels. PF or PM means a single crewmember does it, PF-PM means both do it but not necessarily together, BOTH means at the same time because coordination is needed. The FOM adds the PM's flight discipline standard: monitor the flight path first and never let secondary tasks interfere.

FCOM PRO-NOR-TSK P 1/26 Tasksharing, General Information

"PF or PM is used for individual actions. The actions should be completed by a single crew member, the PF or the PM. This rule also applies for CM1 or CM2."

FOM 10.1.1.4: the PM assesses and communicates threats, holds the shared mental model, and uses Ask, Say, Command to restore safety margins.

Workbook wording: Perform PM Duties During Go-Around/Missed Approach

11.1.10 PERFORM PM DUTIES FOLLOWING LANDING WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL**PM duties after landing****After landing flow: radar off, start selector NORM, flaps up, TCAS standby, APU, anti-ice, brake temperatures**

Task sharing is defined phase by phase in FCOM PRO-NOR-TSK and in each SOP chapter with the PF, PM, PF-PM and BOTH labels. PF or PM means a single crewmember does it, PF-PM means both do it but not necessarily together, BOTH means at the same time because coordination is needed. The FOM adds the PM's flight discipline standard: monitor the flight path first and never let secondary tasks interfere.

FCOM PRO-NOR-TSK P 1/26 Tasksharing, General Information

"PF or PM is used for individual actions. The actions should be completed by a single crew member, the PF or the PM. This rule also applies for CM1 or CM2."

FOM 10.1.1.4: the PM assesses and communicates threats, holds the shared mental model, and uses Ask, Say, Command to restore safety margins.

Workbook wording: Perform PM Duties following Landing

General Operations (2)

16.1.2.1.3 KNOW ACARS PROCEDURES

Where is the spare ACARS paper on the flight deck?

No published source located, ask the check airman

The FOM authorizes the crew to replace ACARS printer paper as a consumable and, for HA, wants an Info to Maintenance logbook entry when you do so the spare gets restocked. It does not say where the spare is stowed on the A330.

FOM 12.2.6 Consumables Replacement - Flight Deck

"(HA) If ACARS printer paper is replaced, the Flight Crew should enter an "Info to Maintenance" logbook entry to ensure spare printer paper is restocked."

No published source located, ask the check airman

Workbook wording: Where is the spare ACARS paper located on the flight deck?

16.1.2.1.3 KNOW ACARS PROCEDURES

Where are the instructions for changing the ACARS paper?

No published source located, ask the check airman

The FOM covers the policy, crew may replace expended consumables, and the HA logbook note, but no loading instructions are in the FOM, FCOM or PRC extracts.

FOM 12.2.6 Consumables Replacement - Flight Deck

"Consumable items such as light bulbs, ACARS printer paper, tissue boxes, and trash bags are provided to the Flight Crew and may be replaced by the Flight Crew when they are expended during normal operations."

No published source located, ask the check airman. The printer usually carries a loading diagram inside the door.

Workbook wording: Where are the instructions on how to change the ACARS paper?